Artificial Neural Networks & Deep Learning

Lecture 1

Waqas Sultani

Associate Professor, Department of AI
Information Technology University, Lahore

https://waqassultani.github.io/







Who Am I? – Education

- Ph.D. Computer Vision, 2017

  - Center for Research in Computer Vision , University of Central Florida, USA
  - Advisor: Dr. Mubarak Shah
  - 6th Best Computer Vision Research Group in US



- MS. Computer Engineering, 2010

  - Perception and Intelligence Lab , Seoul National Unveristy, South Korea
  - Advisor: Jin Young Choi
  - South Korea's No.1 Unversity

- BS, Electrical Engineering, 2006

  - University of Engineering and Technology, Taxila, Pakistan

Work Experience

I have more than 14 years of extensive hands-on experience working on videos and images

- Principal Machine Learning Scientist
  - Vehcile Tracking and Speed estimation
  - Traffic Signals Detection

- Associate Professor, Computer Science
  - Head of MedAI Research Group
  - Supervised around 10 MS students and Co-Supervising 2 PhD Students

- Xerox, Webster, NewYork
  - Worked on Developing Elementray School students monitoriing system
  - 2 US patents granted

- Advanced Engineering Organization, Pakistan
  - System Design Engineer.

Who We Are?

- **Intelligent Machine Learning Lab @ ITU**
  - Computer Vision
  - Machine Learning / Deep Learning
  - Medical Vision
  - Remote Sensing

- **Two Research groups**
  - CVML and MedAI

- **Recent publications in top venues by lab members**
  - NeurIPS, CVPR, ECCV, AAAI, ICRA, MICCAI, WACV, ISPRS-JPRS etc..







Members

Dr. Waqas Sultani , Dr. Mohsen Ali

Google Research Scholar Award 2021

Google Research Scholar Award **2023**

Donors / Collaborators / Secured Grants (PI & co-PI)

Members

PhD students : 3

MS students : 10

Undergraduate Students : 10

Recent Publications

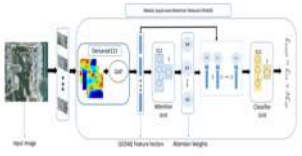
- CVPR 2021, 2022, 2206
- ECCV 2020
- MICCAI, 2020, 2024
- NeurIPS 2021, 2022, 2023
- AAAI 2023
- ICRA 2023

Research Areas

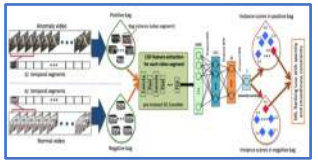
Weakly Supervised Classification

UAV Analysis

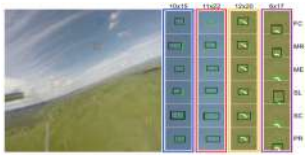
Medical Image Analysis



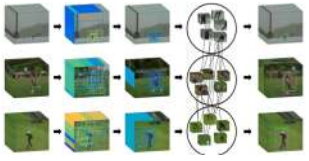
JPR 2020



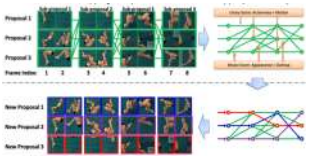
CVPR 2018



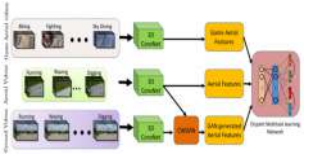
CVPR 2021, ICRA 2023



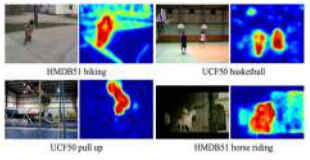
CVIU 2017



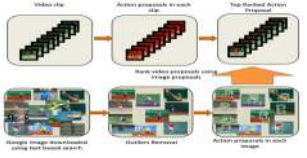
CVIU 2017



CVIU 2021



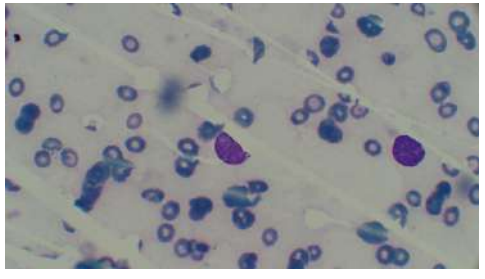
CVPR 2014



CVPR 2016

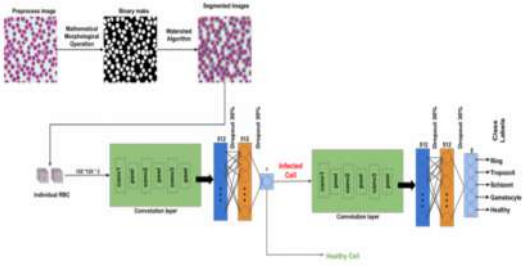
Research Areas

Medical Image Analysis



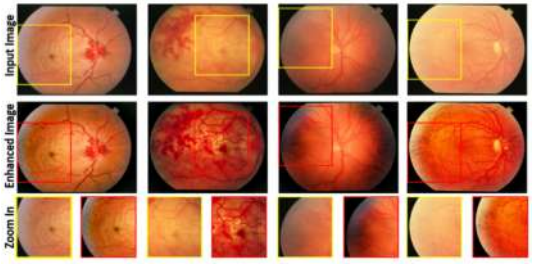
Google Research

Research Grant, 2023

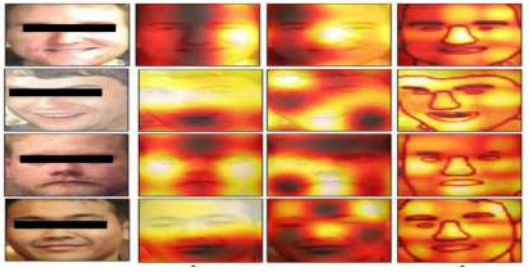


FaceBook AI

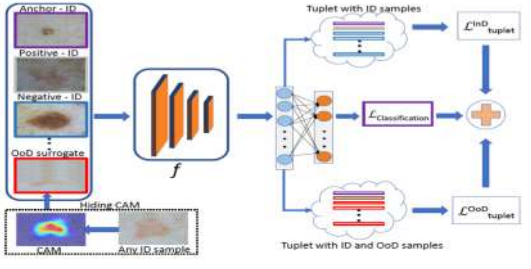
CV4GC Research Grant, 2019



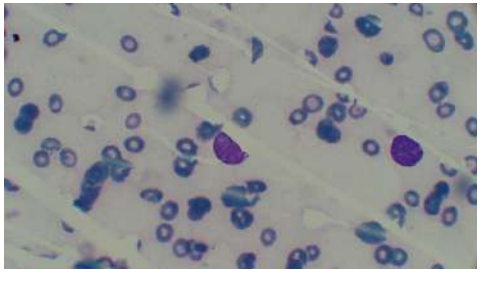
MICCAI 2020, CMB 2022



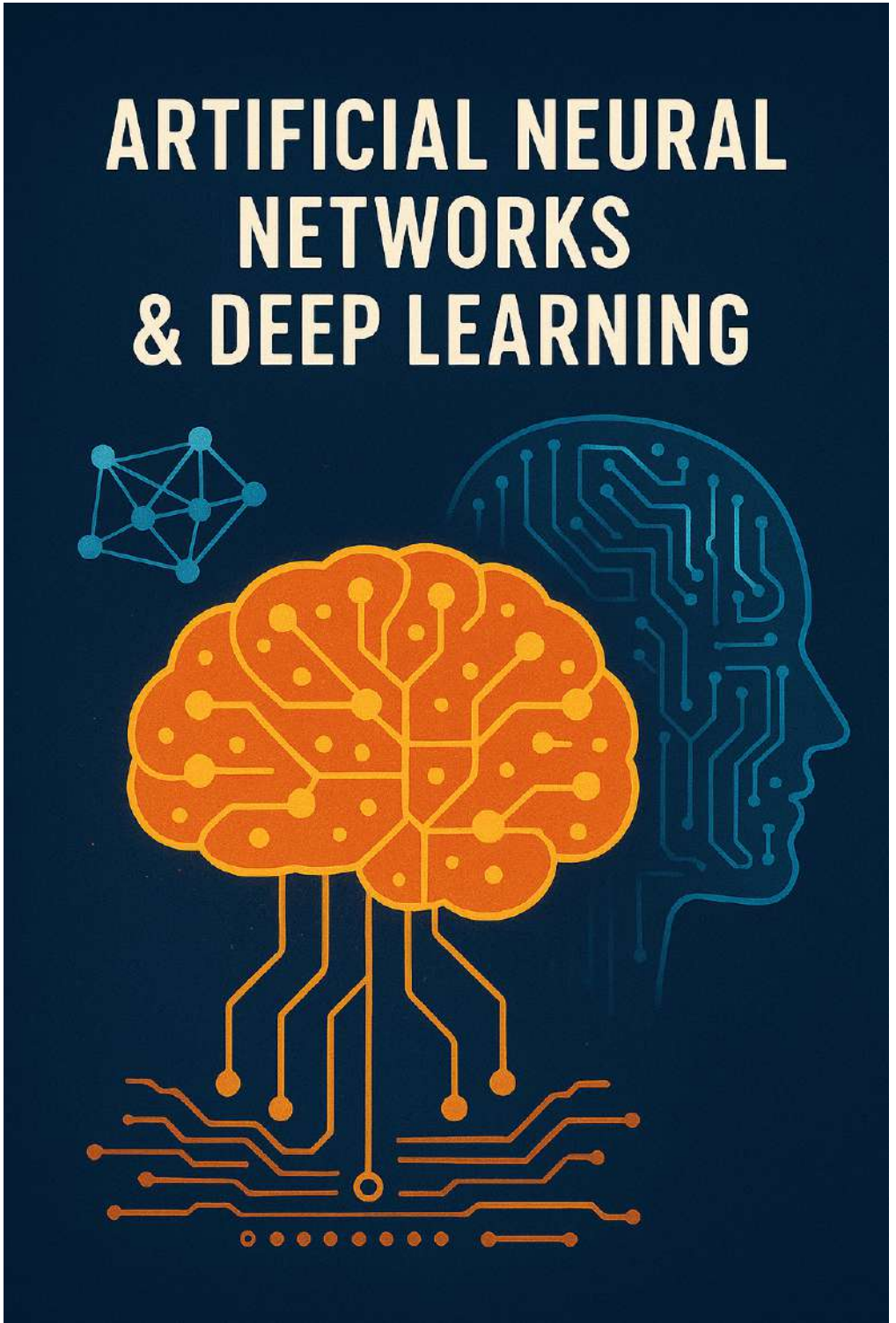
CMB 2021



OOD, 2021



MICCAI, 2023



- **Office hours**
  - **Appointment through email**

    Email: waqas.sultani@itu.edu.pk,
    Subject: ANN_DL_

Email of TAs:

1) Abdul Rehman: phdcs23002@itu.edu.pk
2) Bushra: msds25051@itu.edu.pk
3) Hamza Khalid: msds-26008@itu.edu.pk

Course Information

Course work:

Assessement	Description	Weight(%)
Quizzes	10 +	12%
Assignment	3 to 4	12
Labs	15	20
Project	1	6
Mid-term Exam	1	20%
Final Exam	1	30%

You are allowed to verbally discuss the assignments or use chatgpt etc. However, you must write the answer completely on your own.

Do not share written solutions or partial solutions with other students.

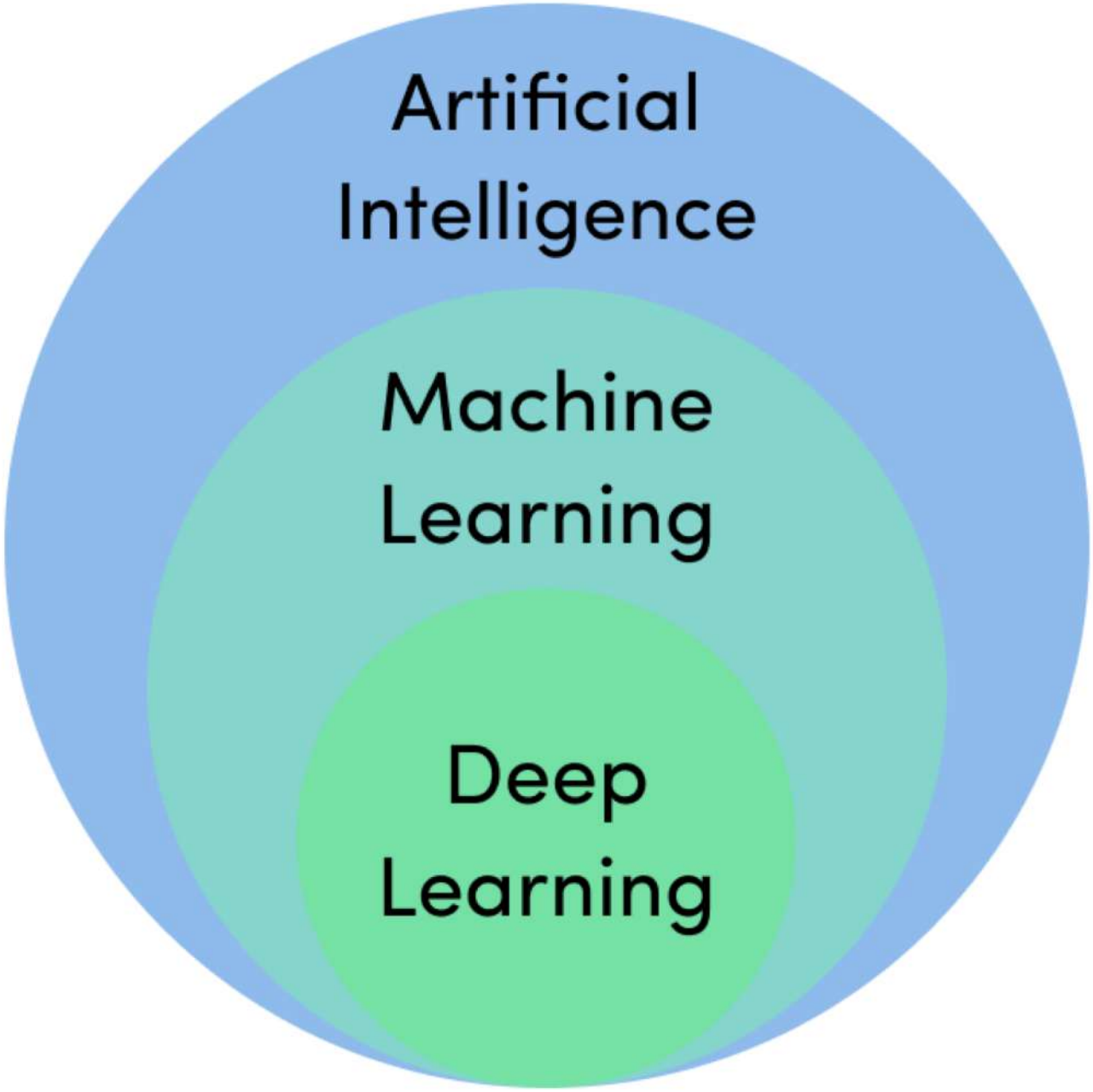
Text Book

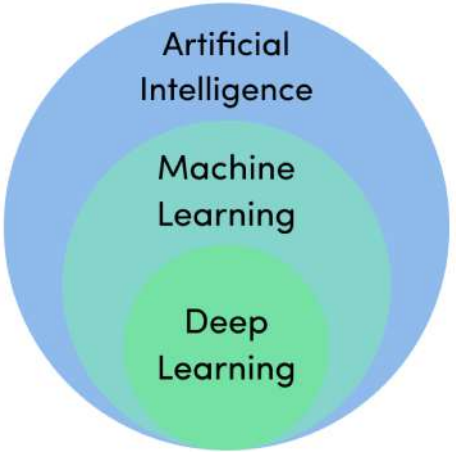
- Deep Learning by Ian Goodfellow Link

Recommended Online Books

- http://deeplearning.net/
- http://neuralnetworksanddeeplearning.com/
- **Courses**
- Machine Learning, Oxford – Phil Blunsom | Matthew Fellows Link
- Convolutional Neural Networks for Visual Recognition, Stanford (cs231n) Link
- A curated list of courses (Recommended) Link
- Deep Learning for Computer Vision Michigan Link
- Deep Learning MIT Link
- Deep Learning for Natural Language Processing, Stanford Link

Deep Learning --- General Introduction


Artificial
Intelligence
Machine
Learning
Deep
Learning

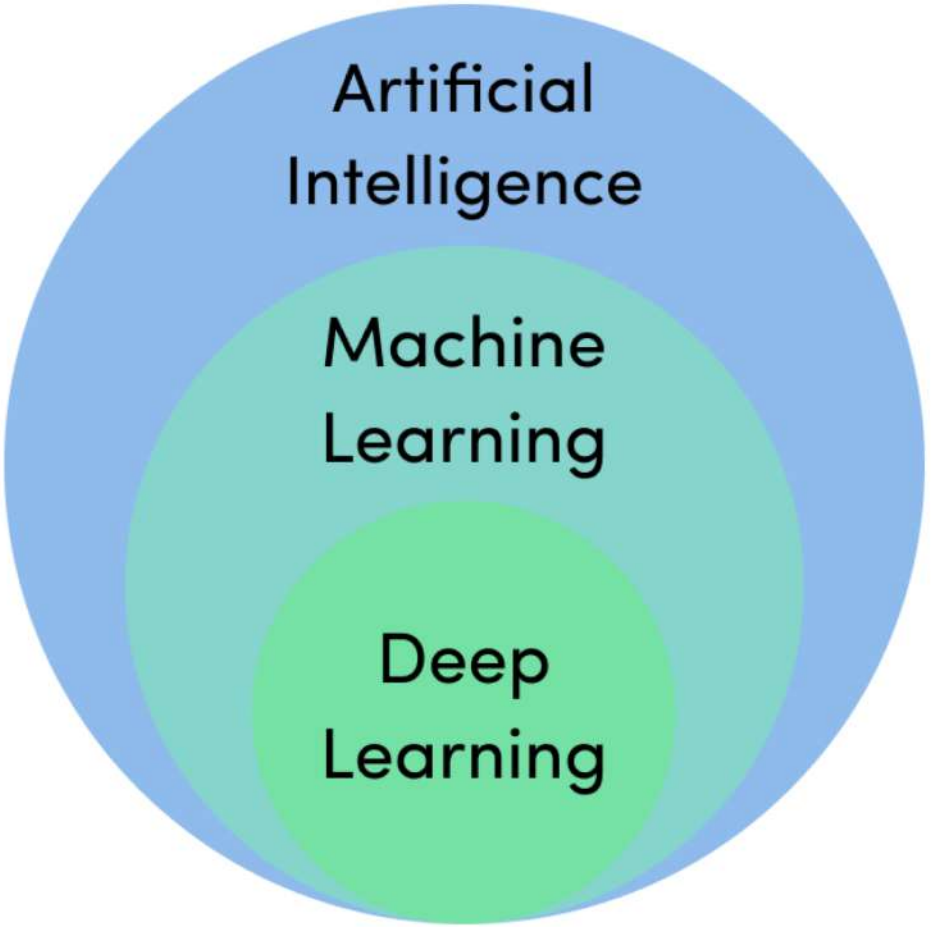


AI

•A branch of computer science dealing with the simulation of intelligent behaviour in computers.

•The capability of a machine to imitate intelligent human behaviour.

•A computer system able to perform tasks that normally require human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages.

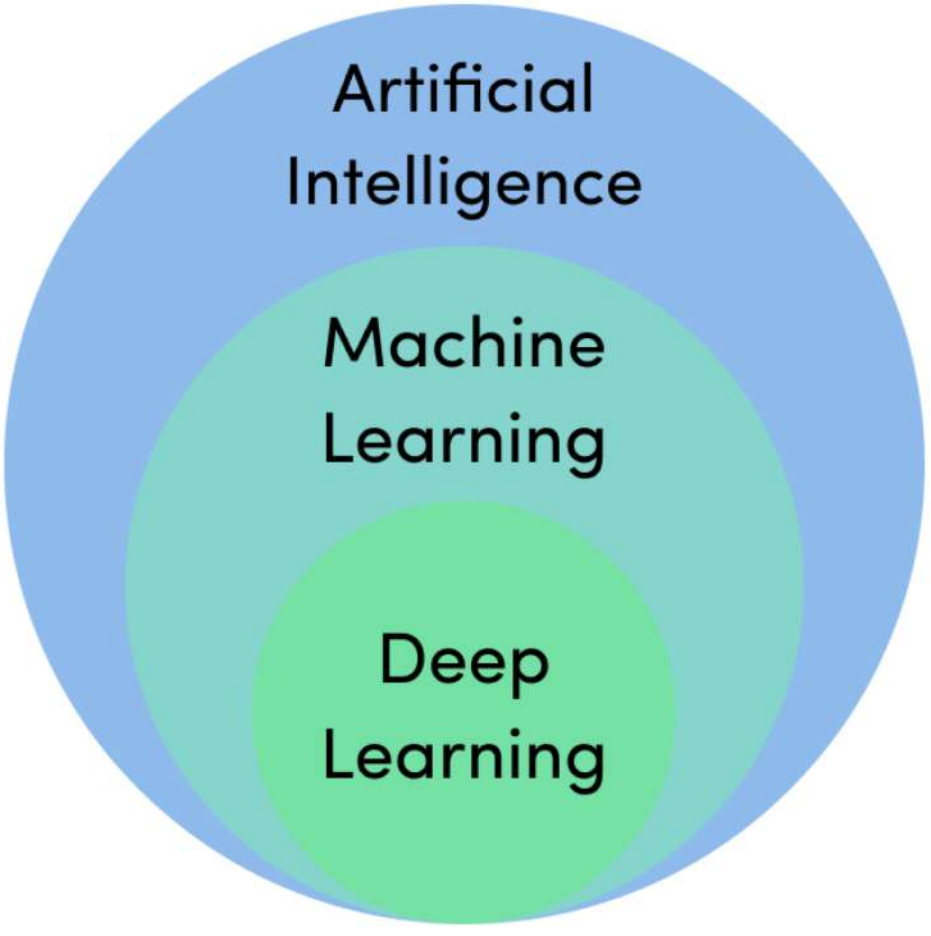


AI

- A branch of computer science dealing with the simulation of intelligent behaviour in computers.
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- A computer system able to perform tasks that normally require human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages.

ML

- A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P if its performance at tasks in T, as measured by P, improves with experience E. –Tom Mitchell

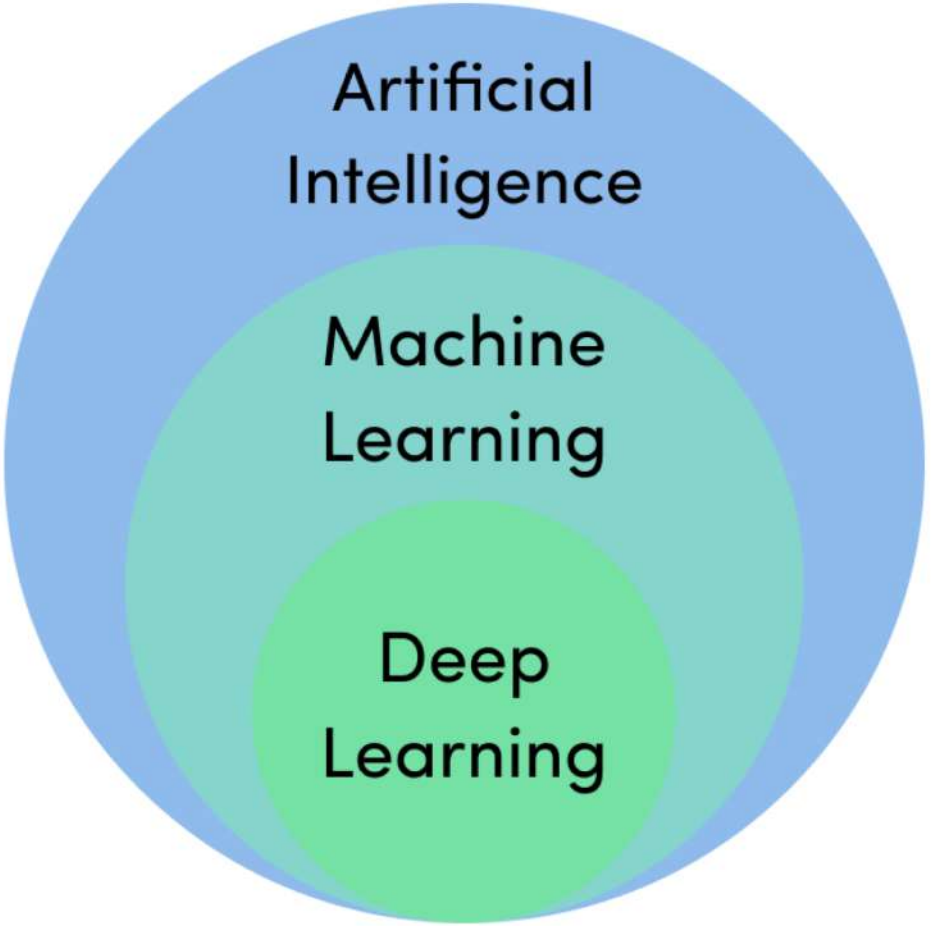


AI (is NOT only ML)

•Machine learning is a subset of AI. That is, all machine learning counts as AI, but not all AI counts as machine learning.

•For example, symbolic logic – rules engines, expert systems and knowledge graphs – could all be described as AI, and none of them are machine learning.

•One aspect that separates machine *learning* from the knowledge graphs and expert systems is its ability to modify itself when exposed to more data; i.e. machine learning is *dynamic* and does not require human intervention to make certain changes.



Deep Learning

•Deep learning is a subset of machine learning. Usually, when people use the term deep learning, they are referring to deep artificial neural networks.

What is Deep Learning?

Artificial Intelligence

Any technique that enables computers to mimic human behavior



Machine Learning

Ability to learn without explicitly being programmed

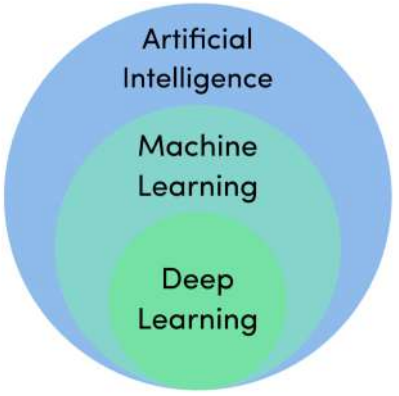


Deep Learning

Extract patterns from data using neural networks



Teaching computers how to **learn a task** directly from **raw data**

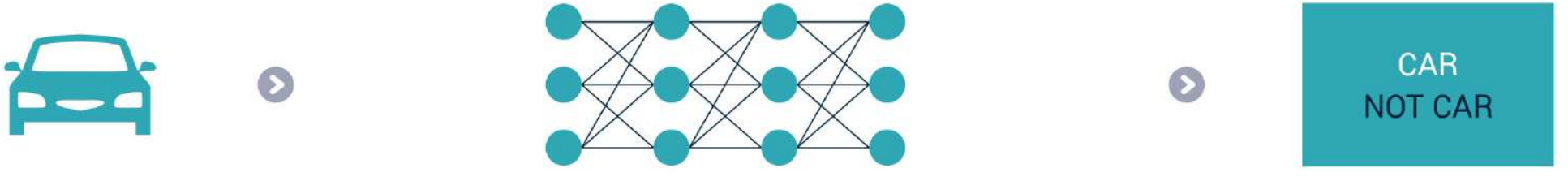


Machine Learning



Input Feature extraction Classification Output

Deep Learning



Input Feature extraction + Classification Output

Why Now?

1952 Stochastic Gradient Descent

1958 Perceptron
- Learnable Weights

1986 Backpropagation
- Multi-Layer Perceptron

1995 Deep Convolutional NN
- Digit Recognition

Neural Networks date back decades, so why the resurgence?

1. Big Data
- Larger Datasets
- Easier Collection & Storage

IMAGENET





2. Hardware
- Graphics Processing Units (GPUs)
- Massively Parallelizable



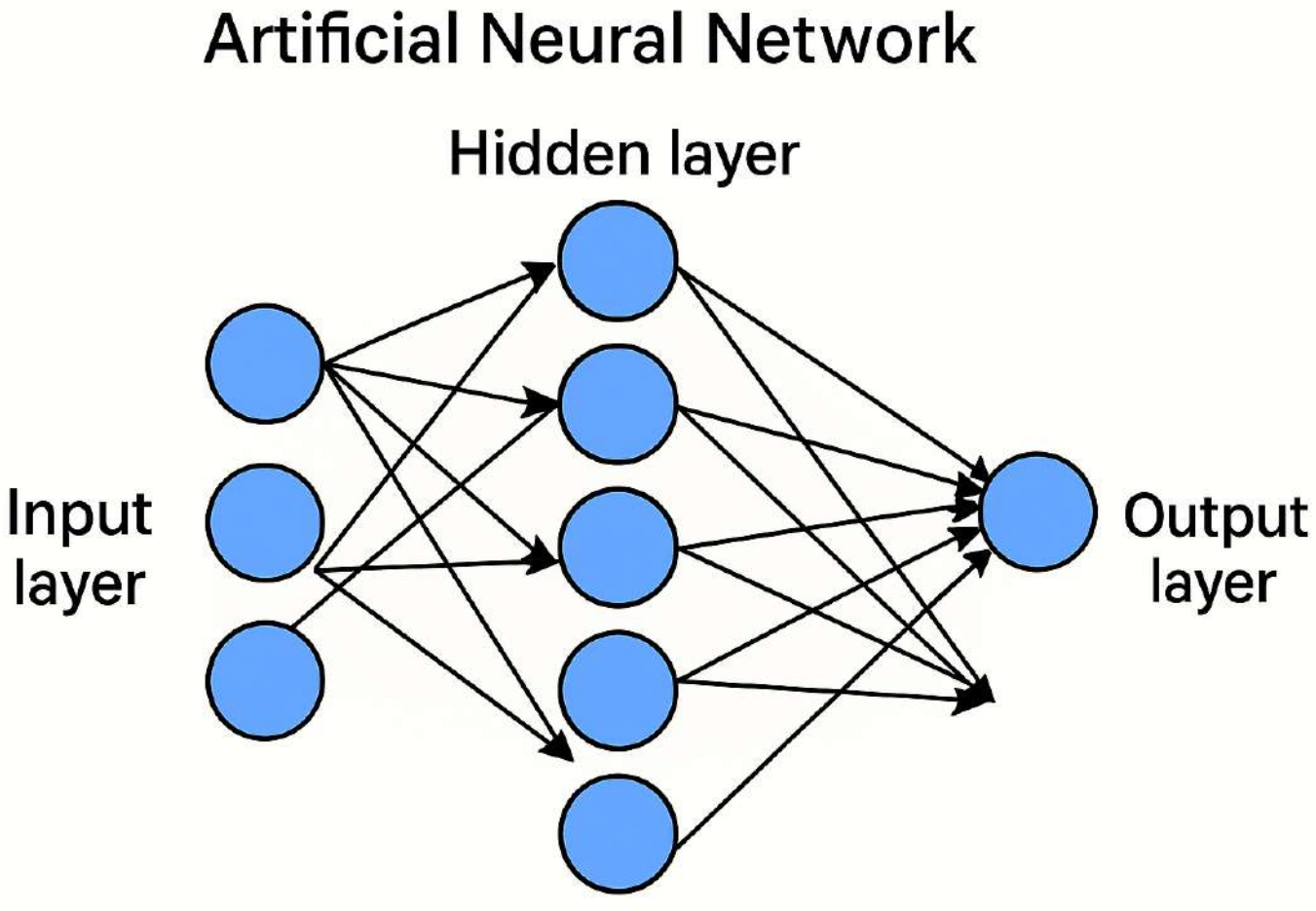
3. Software
- Improved Techniques
- New Models
- Toolboxes

TensorFlow

PYTORCH

Caffe2

Chainer


Artificial Neural Network
Hidden layer
Input layer
Output layer

Research

Research Evaluations: Top venue Journals and conferences

- Getting PhD scholarships in good universities
- Understand the better way to solve the problem

- Conference
- Journal
- Patent

How do we formulate learning problems?

Src: Sergey Levine

Machine Learning

- **Machine learning** is the subfield of computer science that gives computers the ability to learn without being explicitly programmed (Arthur Samuel, 1959)
- What we do in Machine Learning?
  - Making predictions or
    - decisions from Data.

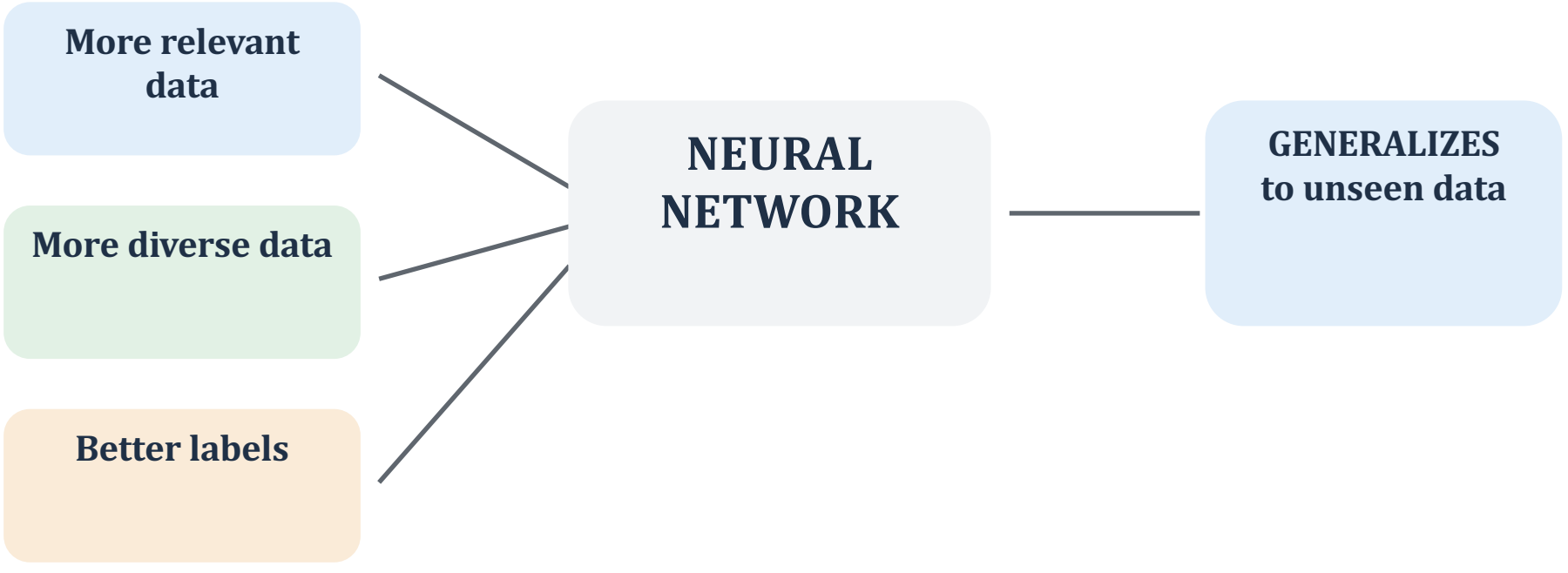
	Supervised Learning	*Unsupervised Learning*
Discrete	classification or categorization	clustering
Continuous	regression	dimensionality reduction

Data, Labels & Dataset Splits



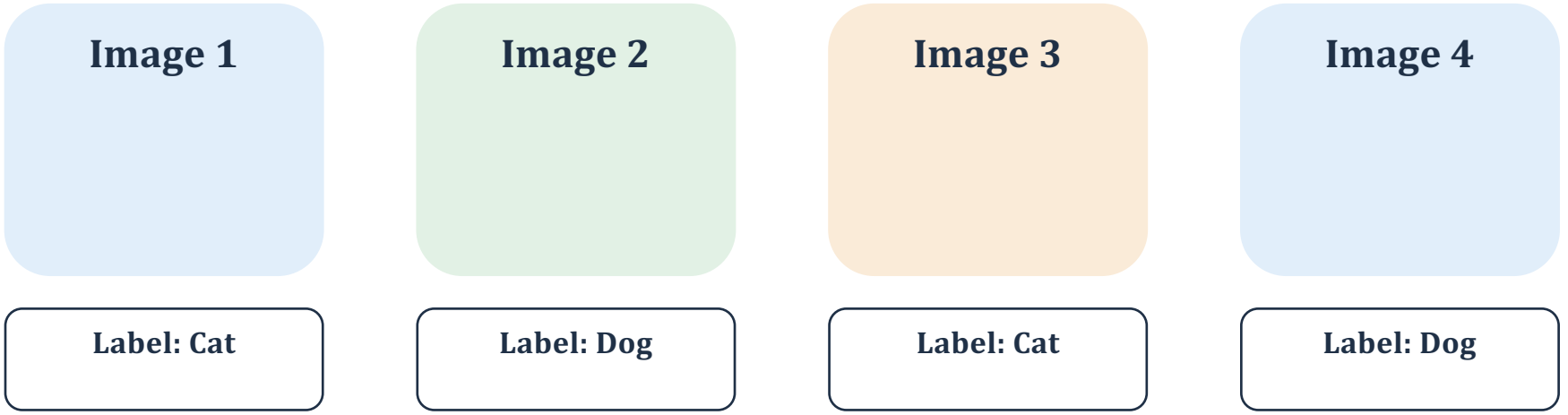
A powerful neural network is only as good as the data it learns from.

Why Does Data Matter?



Poor data → poor learning, even with a sophisticated architecture.

What Is a Labeled Example?



Dataset = Inputs (X) + Input Labels (Y), Output(X) + Output Labels (Y)

DATA PREPARATION

Data Labeling

1

What Is Data Labeling?

The process of tagging raw data , images, text, audio, video , with the correct answer, so a supervised model can learn the mapping from input to output.

2

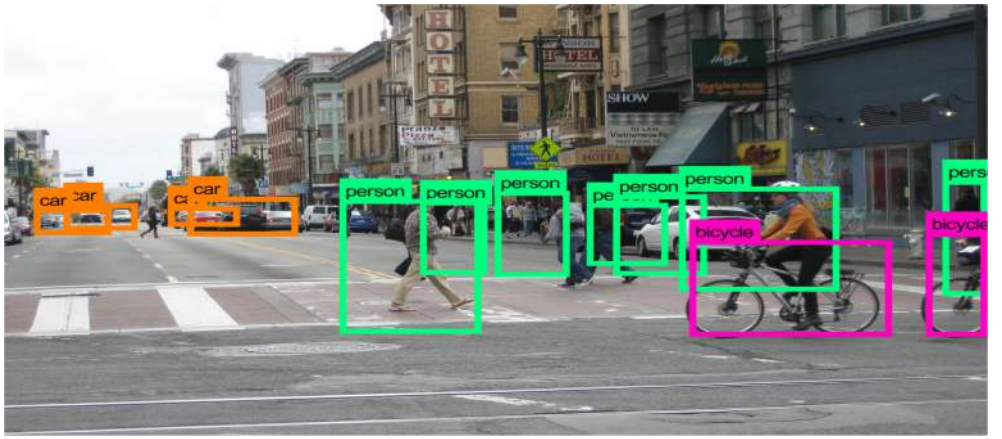
Why It Matters

A model is only as good as its labels. Supervised learning performance is capped by label accuracy , noisy labels mean a noisy model, no matter the architecture.

3

Common Label Types

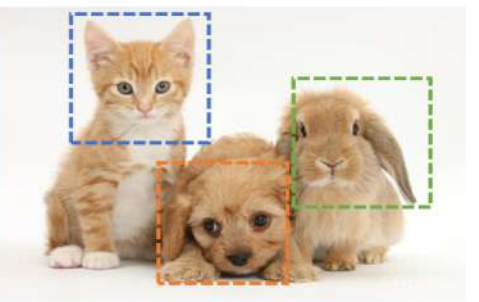
Class tags (cat/dog), bounding boxes, segmentation masks, transcripts, and sentiment scores — the label format follows directly from the task.





Single-label classification

Dog



Multi-label classification

Cat, Dog, Rabbit

DATA PREPARATION

Annotation

1 What Is Annotation?

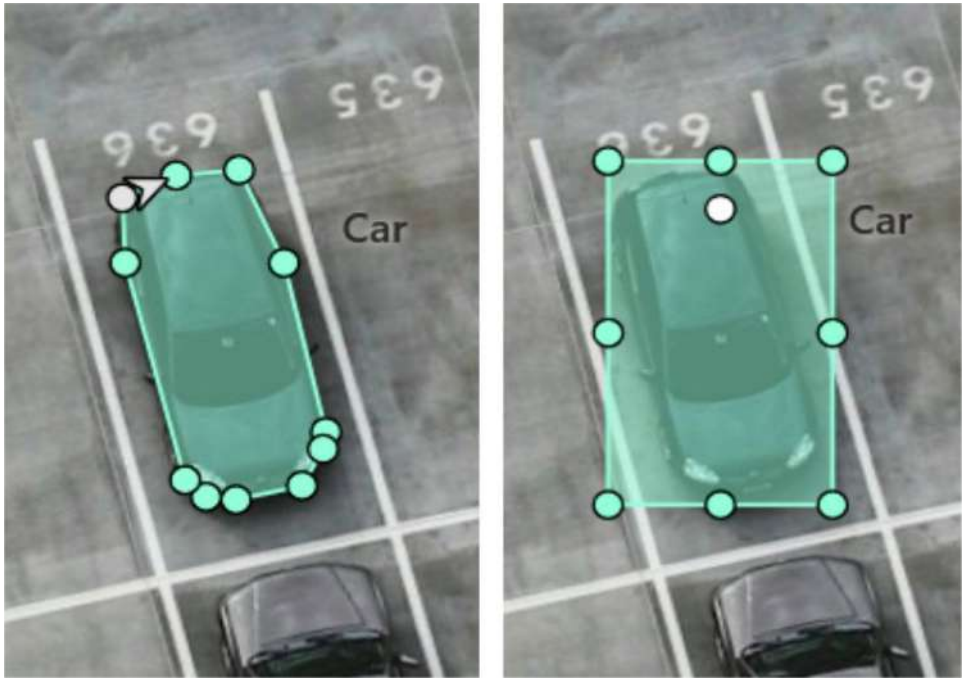
The hands-on task — done by humans, tools, or a human-in-the-loop pipeline — of attaching labels and metadata onto raw data. Labeling is the goal; annotation is the workflow.

2 Common Annotation Types

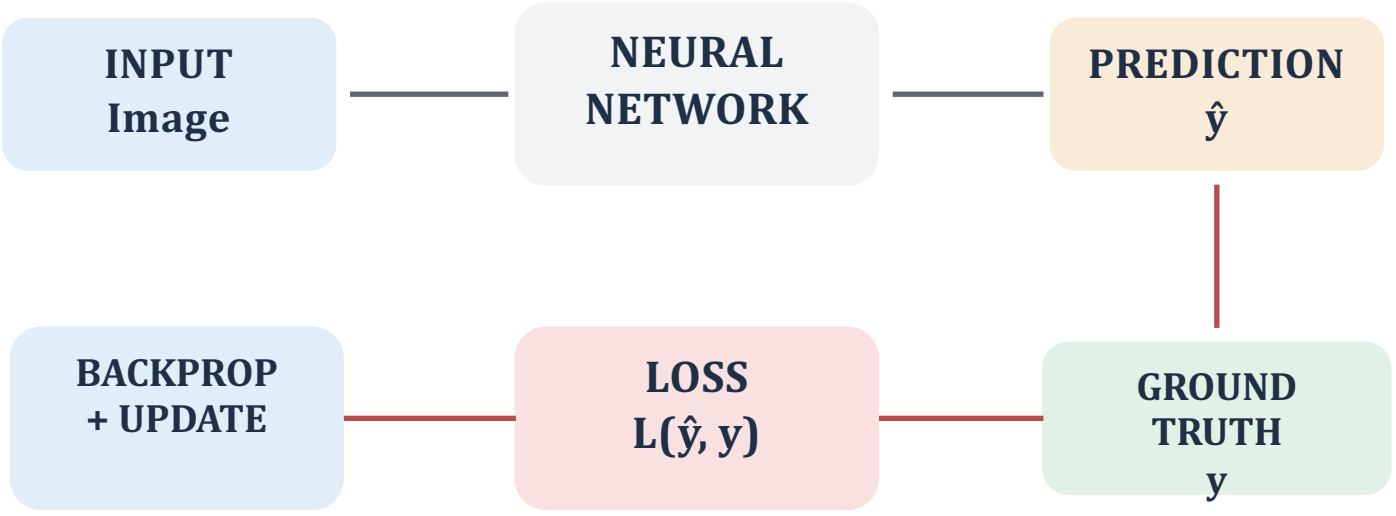
Bounding boxes & polygons (vision), segmentation masks, key-points, named-entity & POS tags (text), and timestamps (audio/video).

3 Popular Tools

CVAT, LabelImg, Label Studio, Prodigy, and Amazon SageMaker Ground Truth are widely used for team-scale annotation projects.



Why Are Labels Important?



Without a useful target, the network cannot directly learn the intended task.

What Makes Good Training Data?

Quantity alone is not enough.

Representative

Real-world conditions

Diverse

Lighting, angle, scale, background

Correctly labeled

Low annotation error

Balanced

Avoid extreme class imbalance

Independent

Avoid near-duplicates

Example: training only on clean white backgrounds may teach the model the background, not the object.

Training, Validation & Test Sets

Each split has a different job.



TRAINING
Learn weights & biases

VALIDATION
Choose / tune settings

TEST
Final unbiased evaluation

Typical split: 70/15/15 — but the right split depends on the problem and dataset size.

Parameters vs. Hyperparameters

A crucial distinction for understanding training.

PARAMETERS — learned from training data

- Weights
- Biases

HYPERPARAMETERS — chosen by us

- Learning rate
- Batch size
- Number of layers / units
- Dropout, weight decay, epochs

Training data → learn parameters | Validation data → select hyperparameters

A Common Mistake: Data Leakage

The test set must remain genuinely unseen.

TRAINING

TEST

Near-duplicate images, video frames, or the same patient/sample can leak information across splits.

Person / Object X

Same X

Result: artificially high test accuracy

For Images & Video: Split by the Right Unit

Random image-level splitting is not always safe.

Random image split

- Frames from the same video can appear in train and test
- Multiple images of the same physical sample can cross the split

✓ Group-wise split

- Keep all frames from one video in one split
- Keep all images from one patient / sample in one split

Goal: test generalization to genuinely new examples

The Golden Rule

Remember the role of each dataset.

TRAIN	VALIDATE	TEST
Learn	Decide	Judge
Weights & biases	Hyperparameters / model selection	Final generalization

Never optimize your model for the test set.

A Simple Analogy: Learning to Drive

TRAINING
Practice with instructor

VALIDATION
Practice test

TEST
Actual driving test

You can practice repeatedly during training.
You use validation to decide what to improve.
You should not keep looking at the actual test while preparing.

ML: Training → Learning | Validation → Decision | Test → Final evaluation

MODEL DEVELOPMENT DATA SPLITS

Validation Data

1

Definition

A held-out slice of the dataset the model never trains on directly. Used during training to check how well the model generalizes.

2

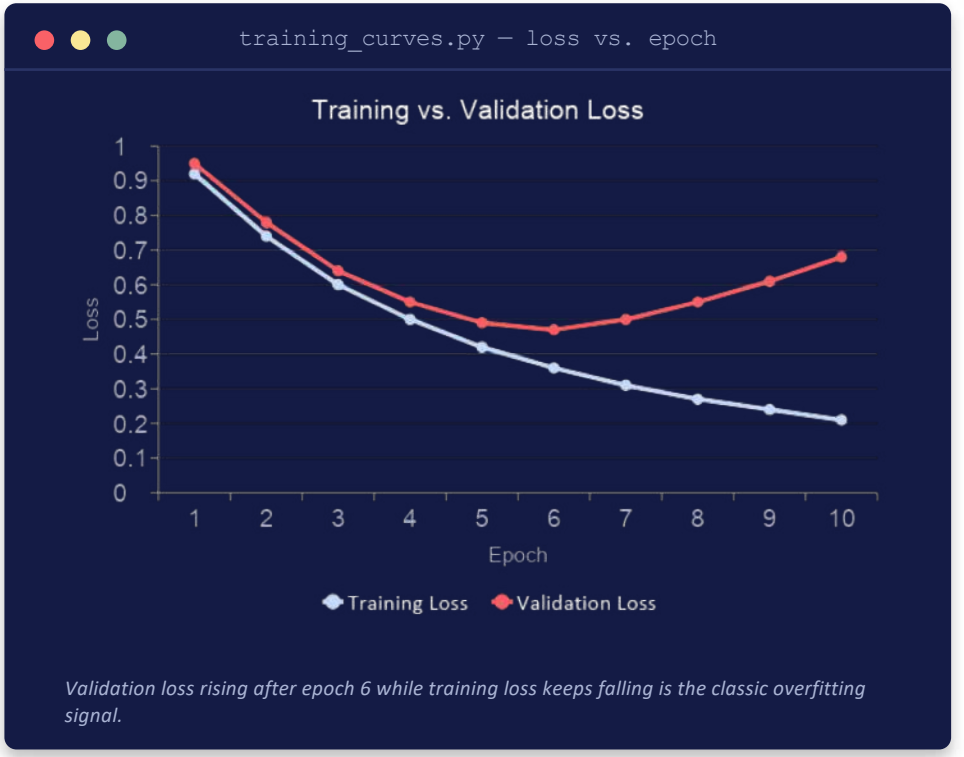
Typical Split

Typically 10-20% of the dataset; small datasets often use k-fold cross-validation instead of a single fixed split.

3

Key Use

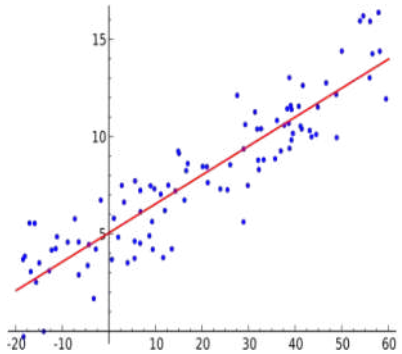
Drives hyperparameter tuning, model selection, and early stopping — training halts once validation loss stops improving, guarding against overfitting.



Validation loss rising after epoch 6 while training loss keeps falling is the classic overfitting signal.

Supervised learning

Given: $\mathcal{D} = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$ learn $f_\theta(x) \approx y$



predict y from x



$f_\theta(x) = y$ [object label]

The overwhelming majority of machine learning that is used in industry is supervised learning

- Encompasses all prediction/recognition models trained from ground truth data
- Multi-billion $/year industry!
- Simple basic principles

Example supervised learning problems

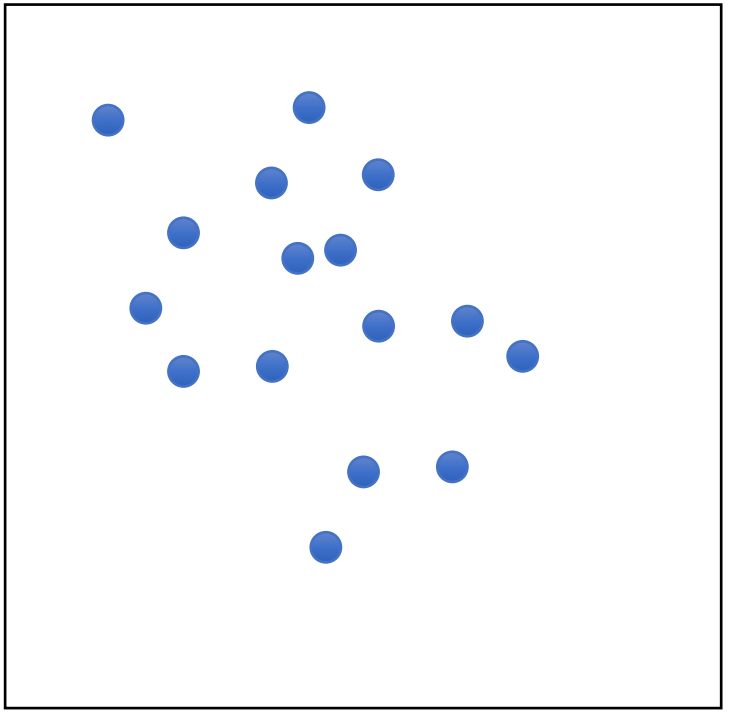
Given: $\mathcal{D} = \{(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)\}$ learn $f_\theta(x) \approx y$

Predict…	**Based on…**
category of object	image
sentence in French	sentence in English
presence of disease	X-ray image
text of a phrase	audio utterance
y	x

Input (X)	Output (Y)	Application
email	spam? (0/1)	spam filtering
audio	text transcripts	speech recognition
English	Spanish	machine translation
ad, user info	click? (0/1)	online advertising
image, radar info	position of other cars	self-driving car
image of phone	defect? (0/1)	visual inspection

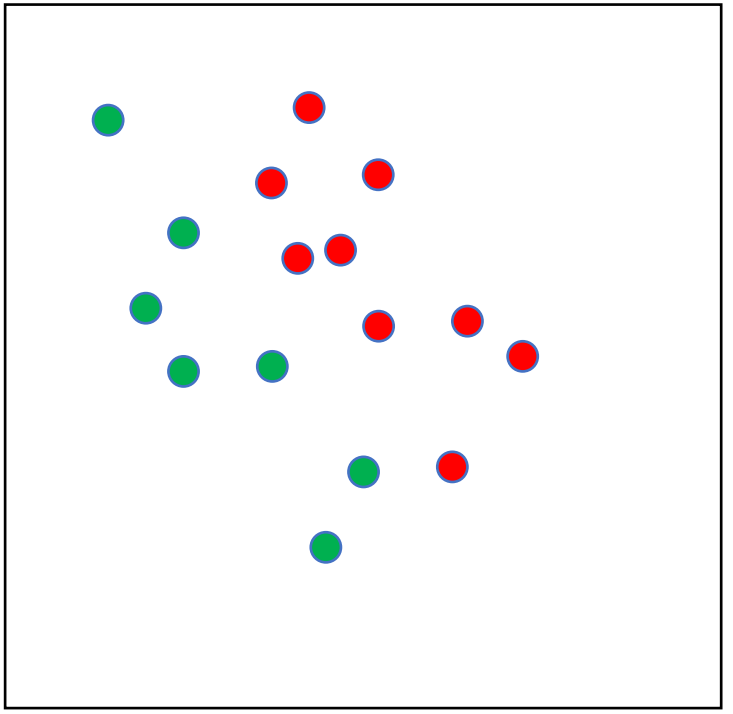
Supervised Learning

- Data consists
  - Input-output pairs
- Input
  - data points
  - features
- Output
  - labels
  - targets
  -



Supervised Learning

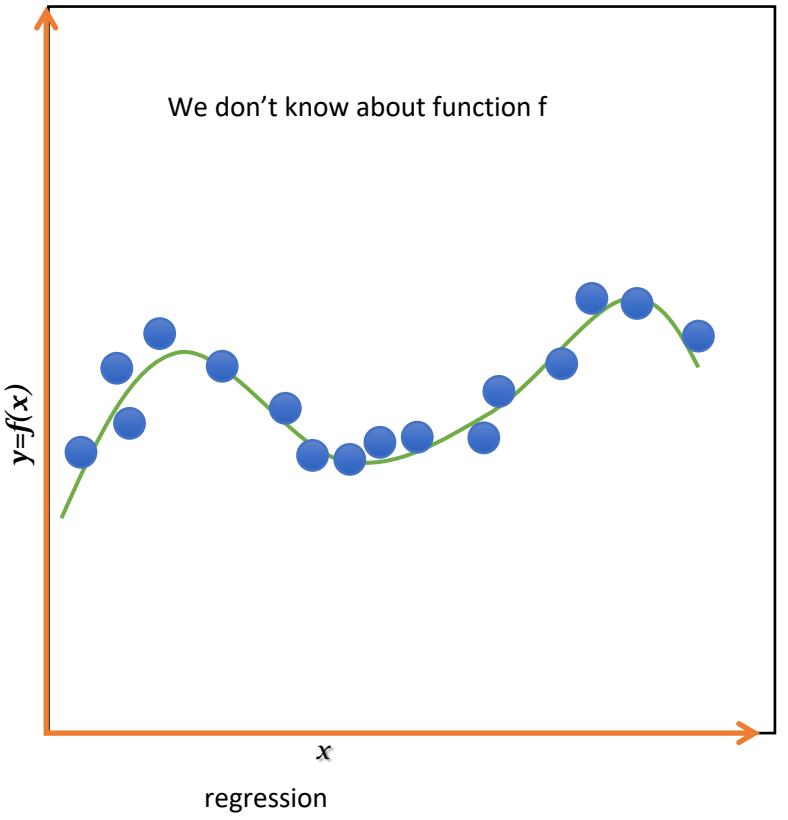
- Data consists
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  -



Classification

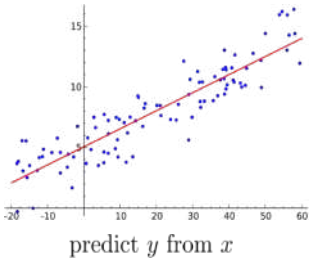
Supervised Learning

- Data consists
  - Input-output pairs
- Input
  - data points
  - features
- Output
  - labels
  - targets



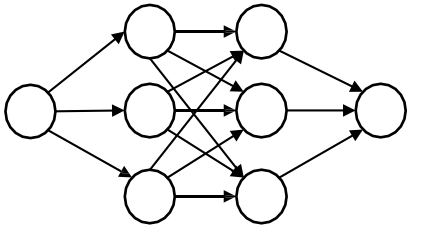
Supervised learning

Given: $\mathcal{D} = \{(x_1, y_1), (x_2, y_2), ..., (x_n, y_n)\}$ learn $f_\theta(x) \approx y$





$f_\theta(x) = y$ [object label]



Questions to answer:

how do we represent $f_\theta(x)$?

$$f_\theta(x) = \theta_1 x_1 + \theta_2 x_2 + \theta_3$$

$$f_\theta(x) = \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$

how do we measure difference between $f_\theta(x_i)$ and y_i?

$$||f_\theta(x_i) - y_i||^2$$

$$\delta(f_\theta(x_i) \neq y_i)$$

probability?

how do we find the best setting of θ?

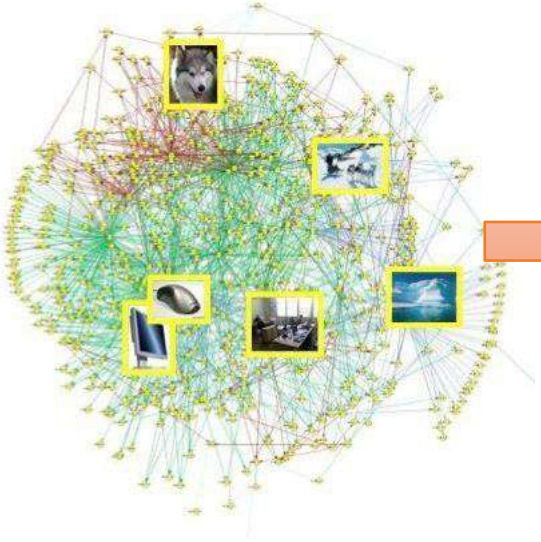
gradient descent

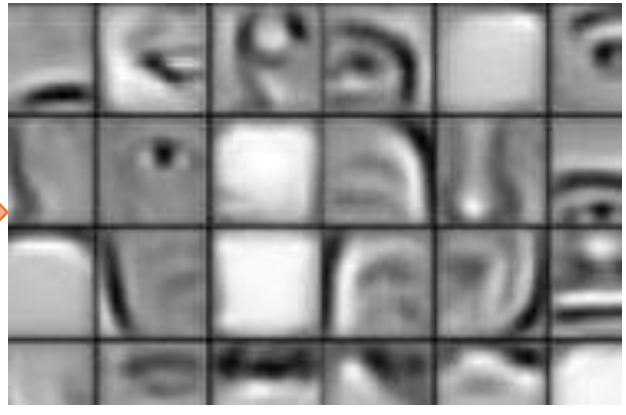
random search least squares

Src: Sergey Levine

Unsupervised learning

Unlabeled Data





what does that mean?

representation

generative modeling:







GANs
VAEs
pixel RNN, etc.

Src: Sergey Levine

Examples/Applications of ANN & DNN


Bus Stop

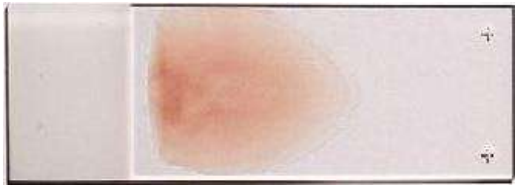

Green (0.9
Green (0.8
J.RASHID & SONS
Jubilee

Automatic Malaria Detection

- There are more than 3,000 species of mosquitoes

- Out of 400 species of the Anopheles mosquito only 60 species carry plasmodium, a parasite that causes malaria

- According to the World Health Organization, only in 2018, 228 million cases of malaria occurred worldwide and there were 405,000 deaths globally due to malaria

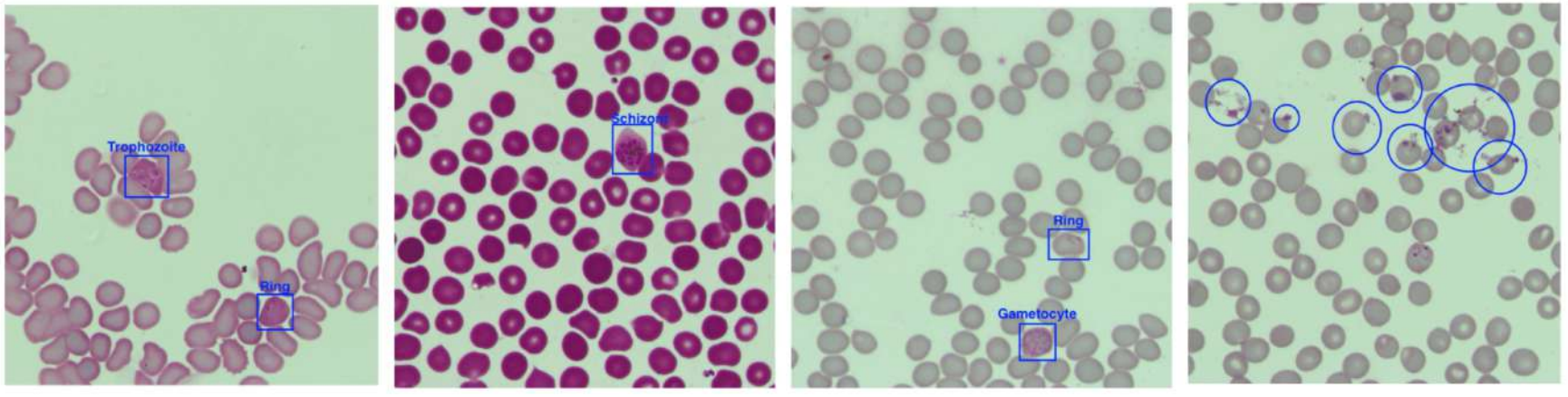




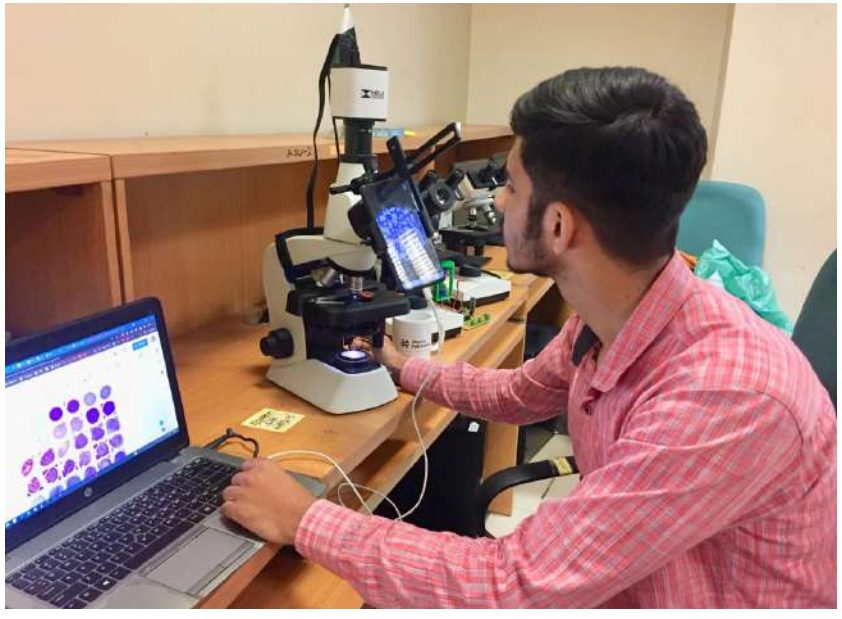
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https://arxiv.org/abs/2102.08708

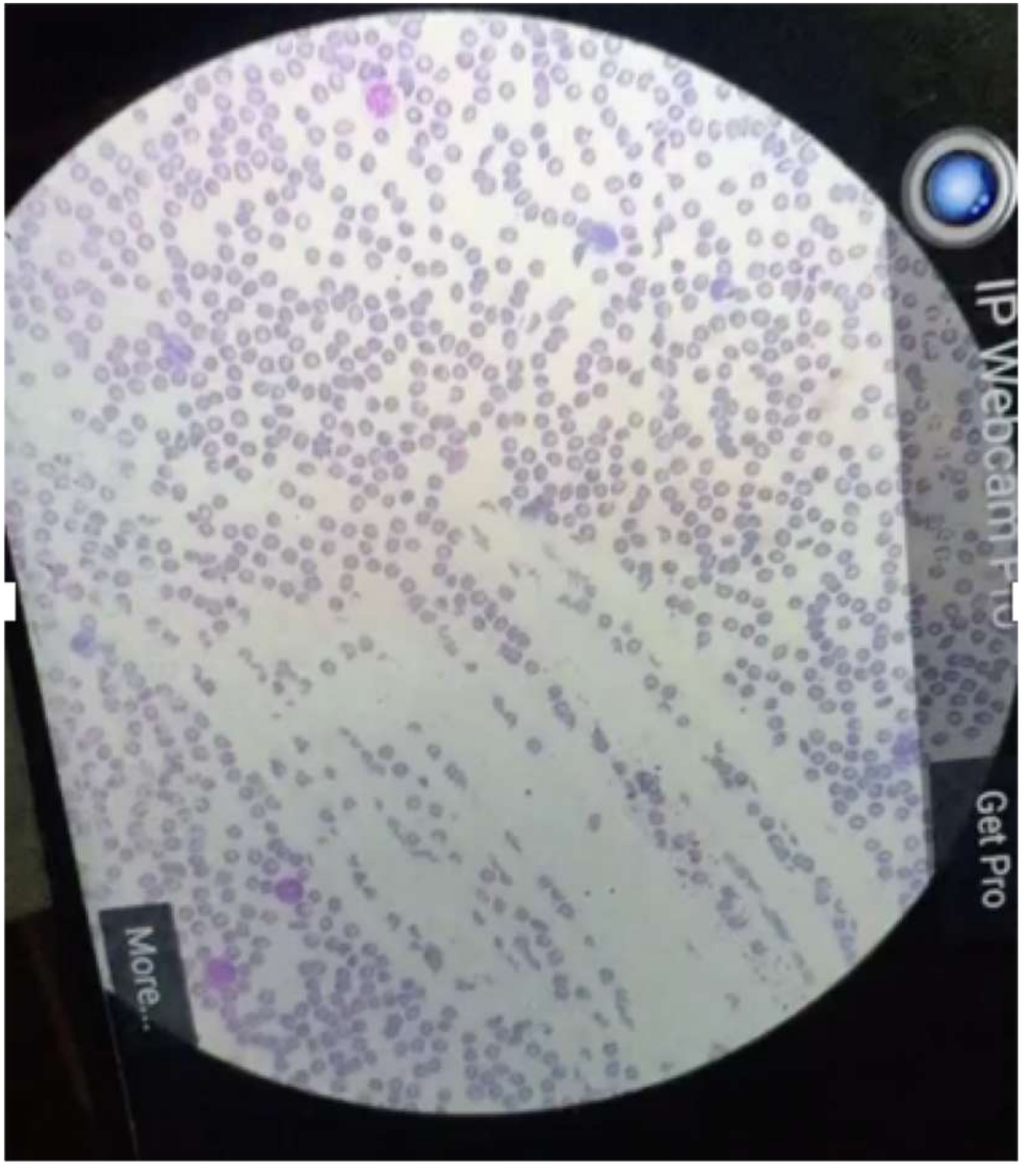




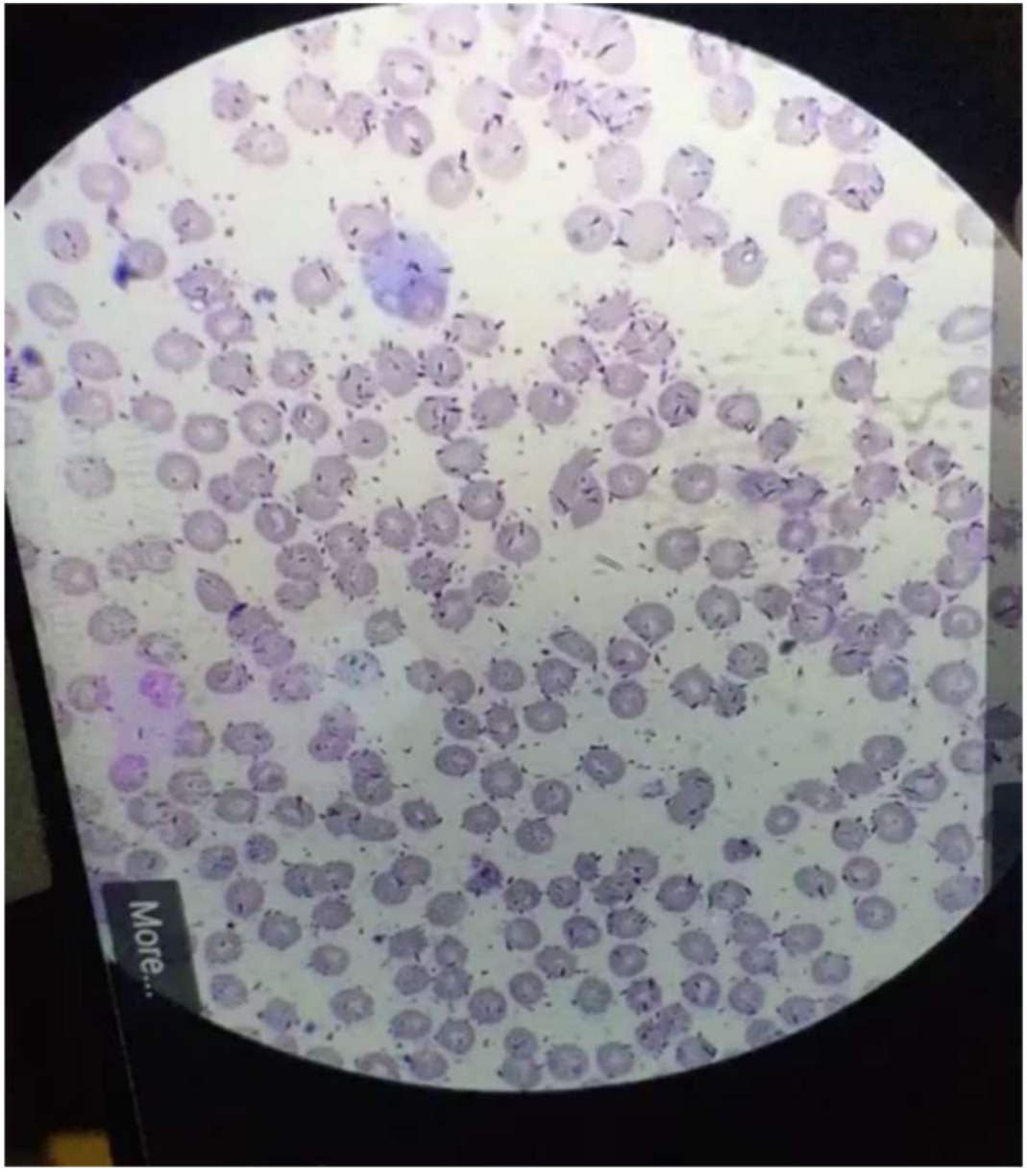




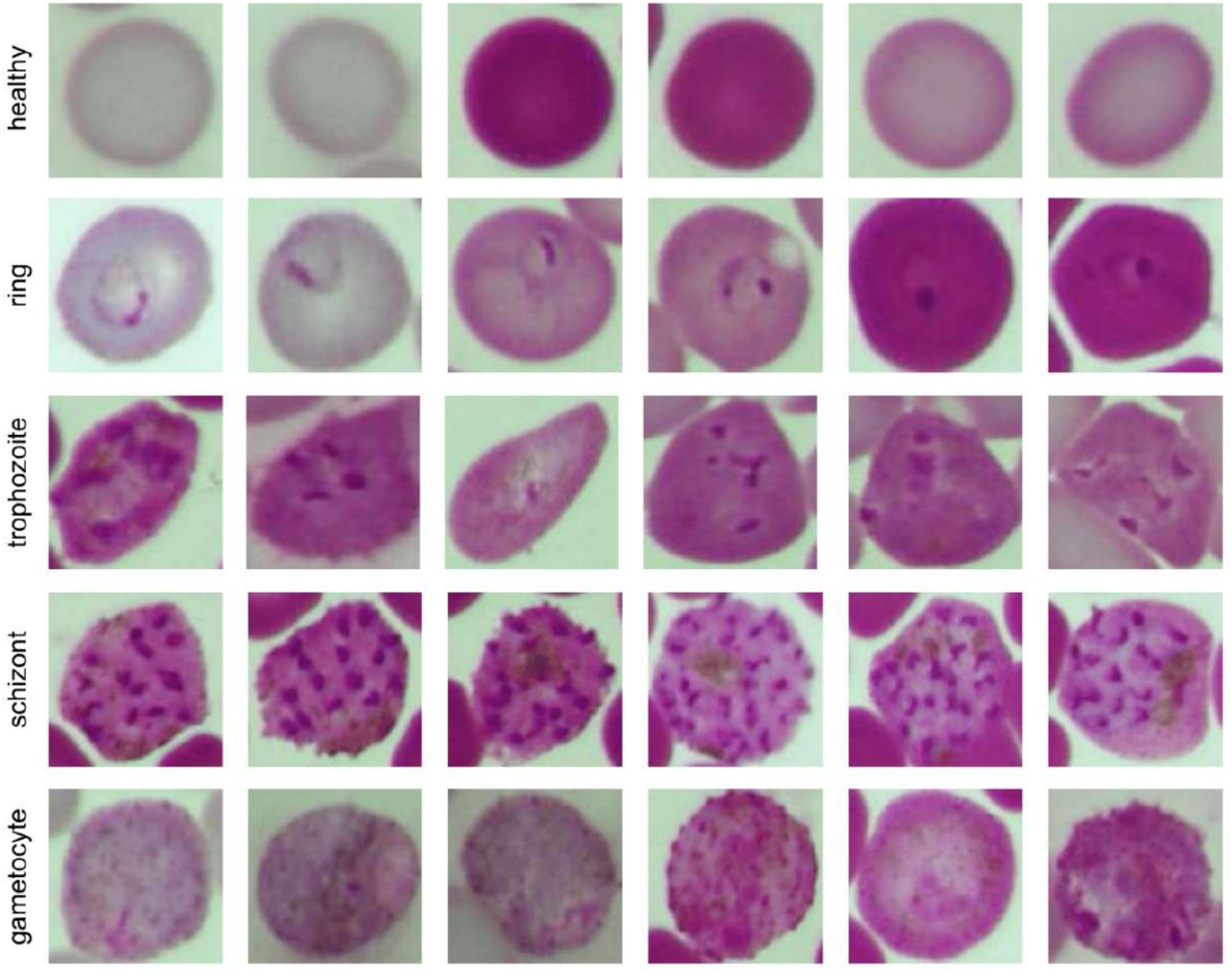
100x



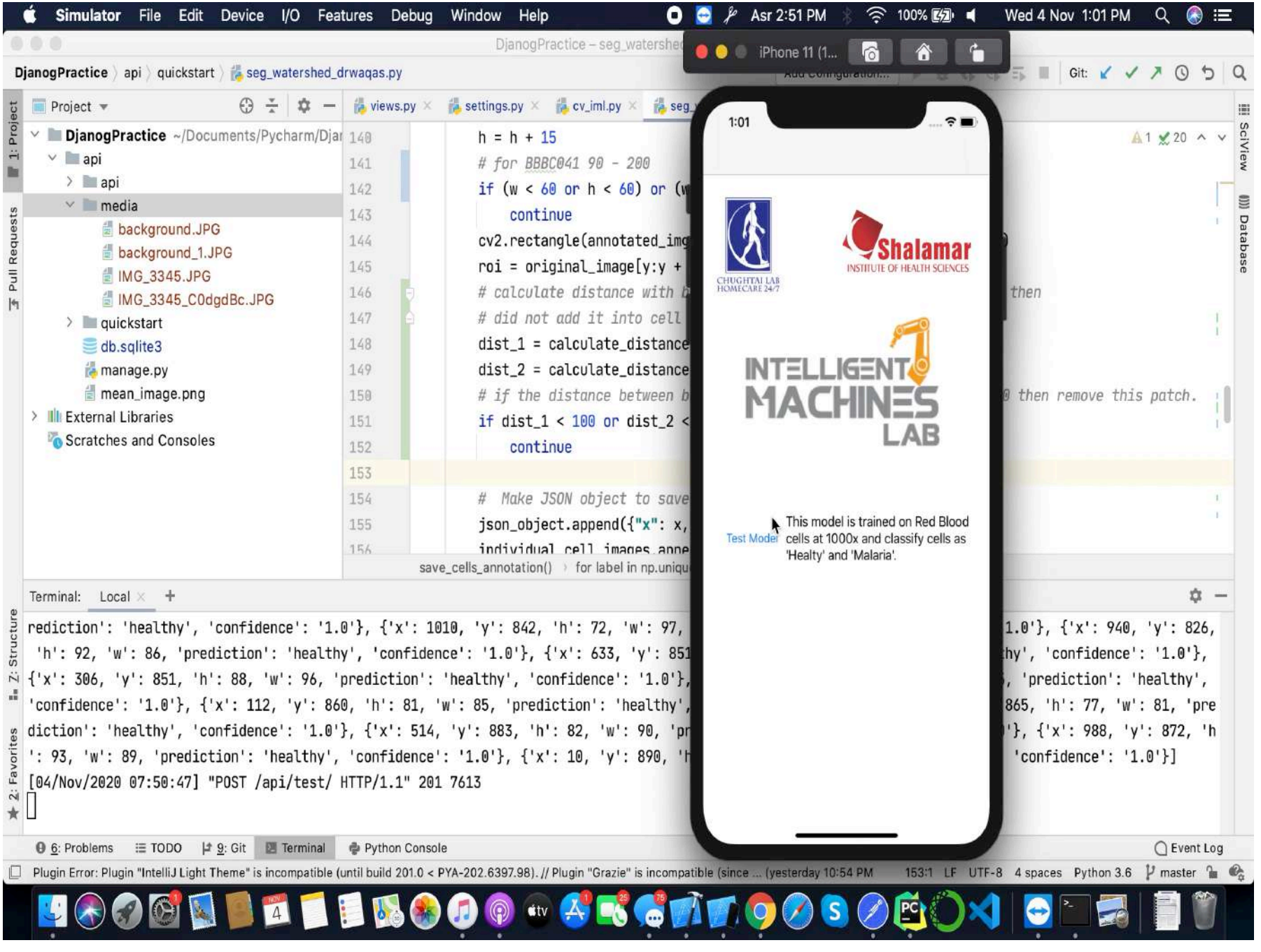
400x

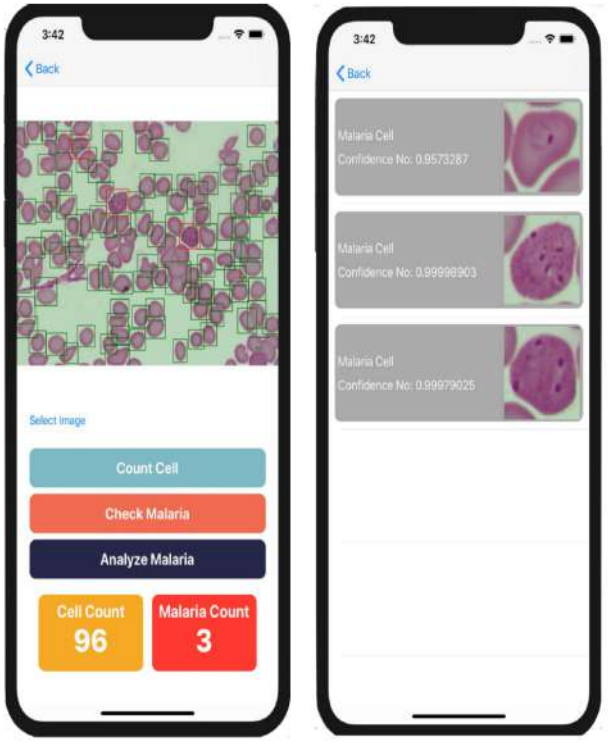


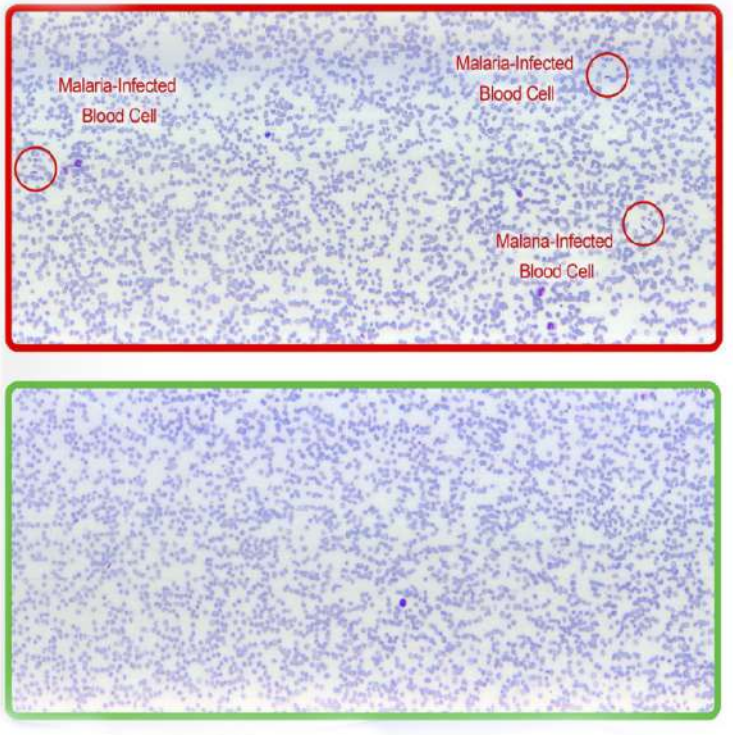
1000x

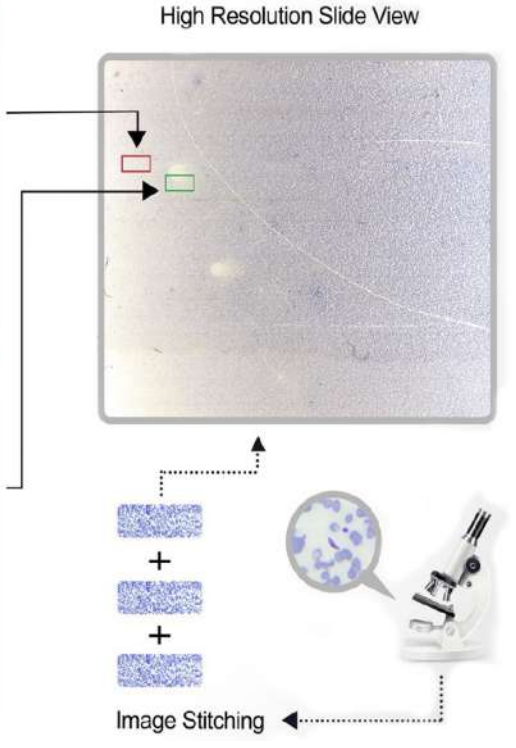


https://arxiv.org/abs/2102.08708


Simulator File Edit Device I/O Features Debug Window Help
Asr 2:51 PM 100% Wed 4 Nov 1:01 PM
DjanogPractice – seg_watershed
iPhone 11 (1...
DjanogPractice api quickstart seg_watershed_drwaqas.py
Project
DjanogPractice ~/Documents/Pycharm/Djar
api
api
media
background.JPG
background_1.JPG
IMG_3345.JPG
IMG_3345_C0dgdBc.JPG
quickstart
db.sqlite3
manage.py
mean_image.png
External Libraries
Scratches and Consoles
views.py
settings.py
cv_iml.py
h = h + 15
for BBBC041 90 - 200
if (w < 60 or h < 60) or (w
continue
cv2.rectangle(annotated_img
roi = original_image[y:y +
calculate distance with b
did not add it into cell
dist_1 = calculate_distance
dist_2 = calculate_distance
if the distance between b
if dist_1 < 100 or dist_2 <
continue
Make JSON object to save
json_object.append({"x": x,
individual_cell_images.appe
save_cells_annotation() for label in np.uniqu
Terminal: Local
rediction': 'healthy', 'confidence': '1.0'}, {'x': 1010, 'y': 842, 'h': 72, 'w': 97,
'h': 92, 'w': 86, 'prediction': 'healthy', 'confidence': '1.0'}, {'x': 633, 'y': 851
{'x': 306, 'y': 851, 'h': 88, 'w': 96, 'prediction': 'healthy', 'confidence': '1.0'},
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diction': 'healthy', 'confidence': '1.0'}, {'x': 514, 'y': 883, 'h': 82, 'w': 90, 'pr
': 93, 'w': 89, 'prediction': 'healthy', 'confidence': '1.0'}, {'x': 10, 'y': 890, 'h
[04/Nov/2020 07:50:47] "POST /api/test/ HTTP/1.1" 201 7613
1.0'}, {'x': 940, 'y': 826,
hy', 'confidence': '1.0'},
, 'prediction': 'healthy',
865, 'h': 77, 'w': 81, 'pre
'}, {'x': 988, 'y': 872, 'h
'confidence': '1.0'}]
6: Problems TODO 9: Git Terminal Python Console Event Log
Plugin Error: Plugin "IntelliJ Light Theme" is incompatible (until build 201.0 < PYA-202.6397.98). // Plugin "Grazie" is incompatible (since ... (yesterday 10:54 PM 153:1 LF UTF-8 4 spaces Python 3.6 master
1:01
CHUGHTAI LAB HOMECARE 24/7
Shalamar INSTITUTE OF HEALTH SCIENCES
INTELLIGENT MACHINES LAB
Test Model
This model is trained on Red Blood cells at 1000x and classify cells as 'Healty' and 'Malaria'.

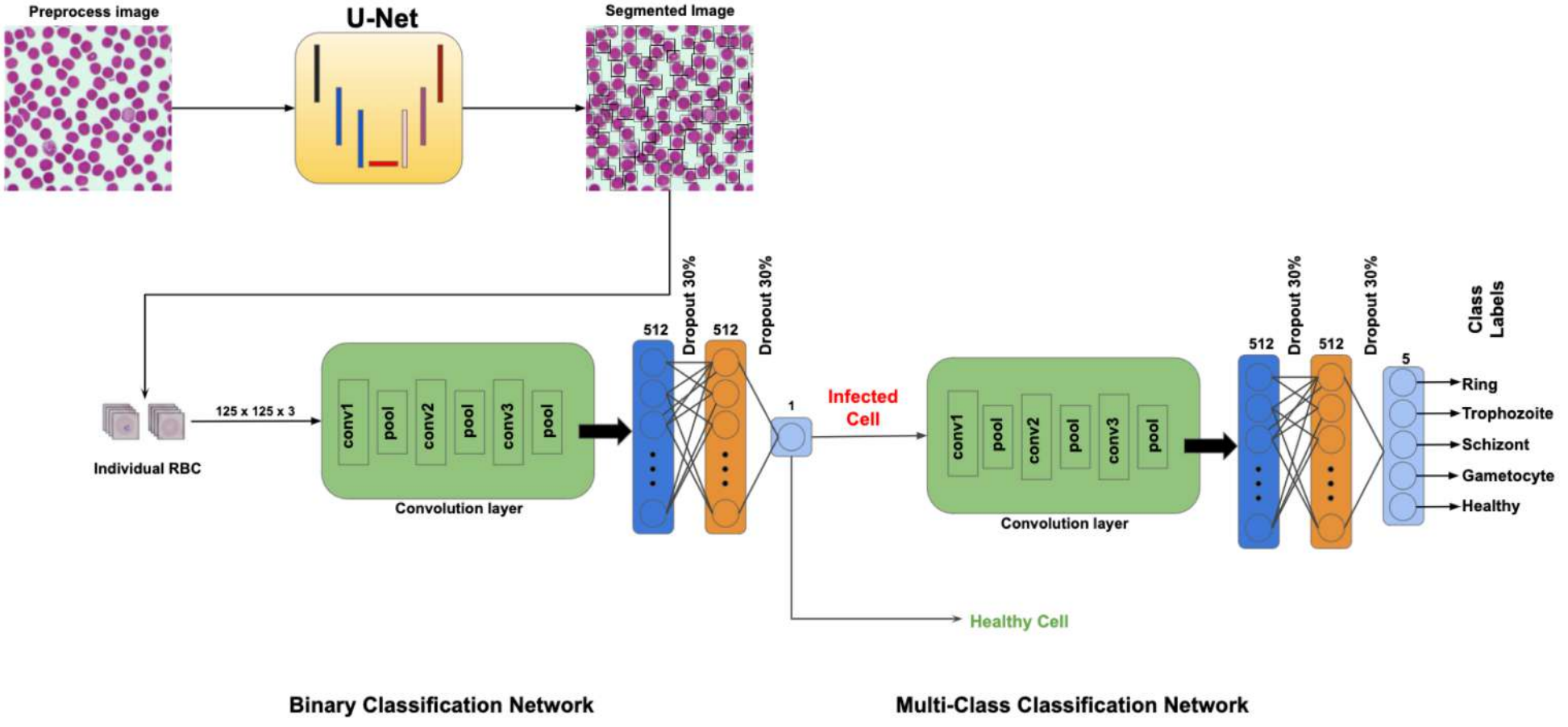






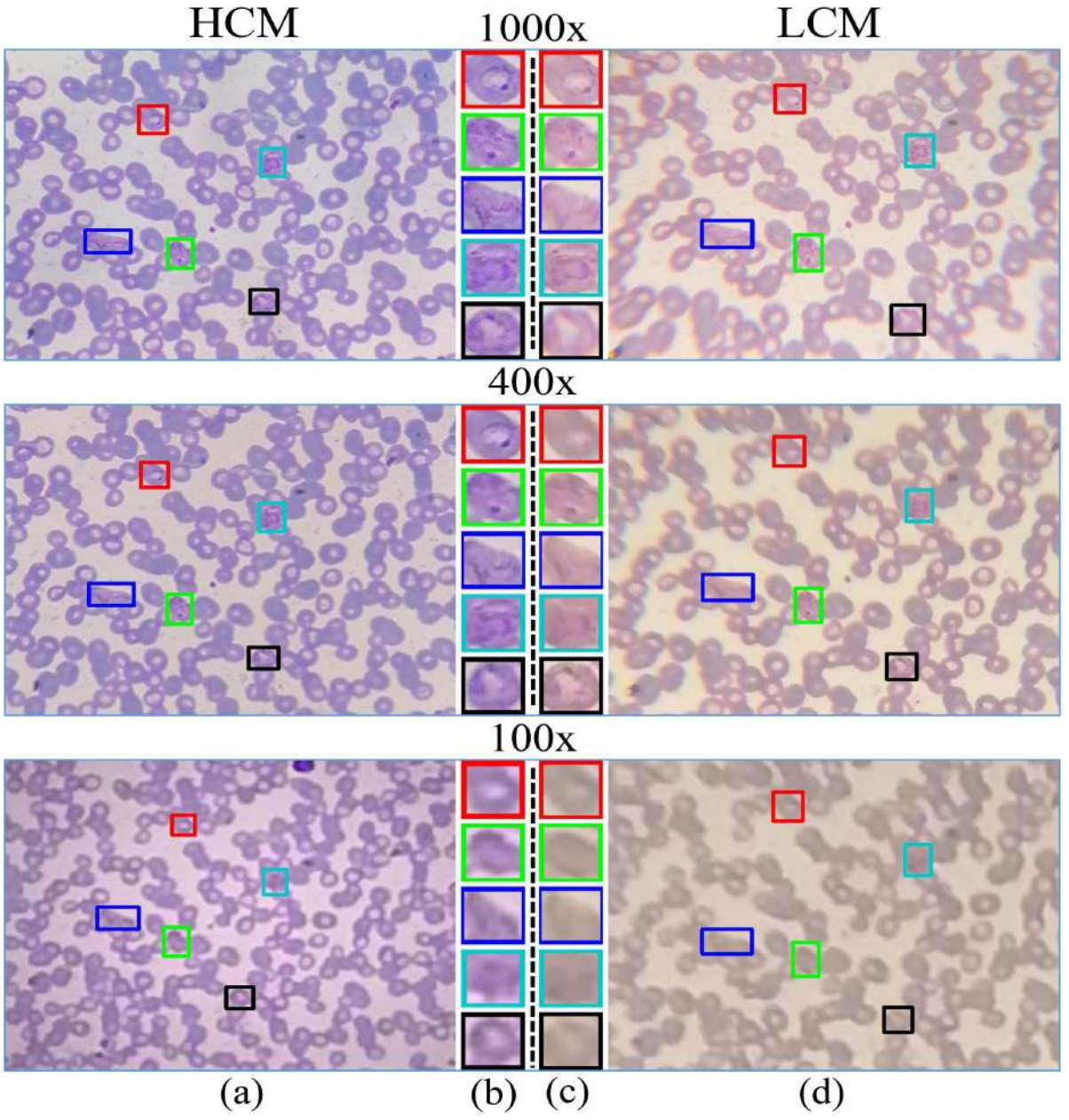
Published in Neural Computing and Application 2021

https://arxiv.org/abs/2102.08708



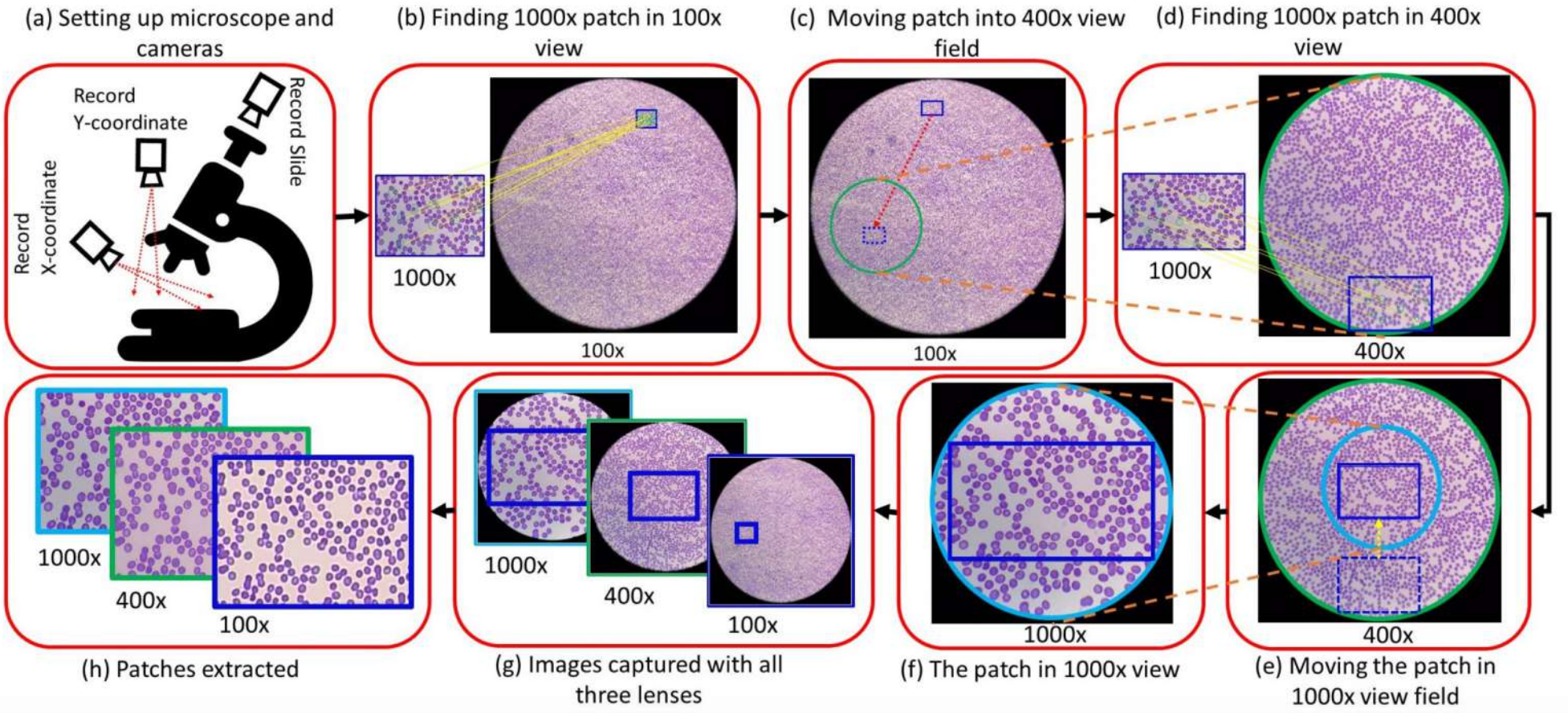
Published in Neural Computing and Application 2021

https://arxiv.org/abs/2102.08708



Published in CVPR 2022

https://openaccess.thecvf.com/content/CVPR2022/papers/Sultani_Towards_Low-Cost_and_Efficient_Malaria_Detection_CVPR_2022_paper.pdf

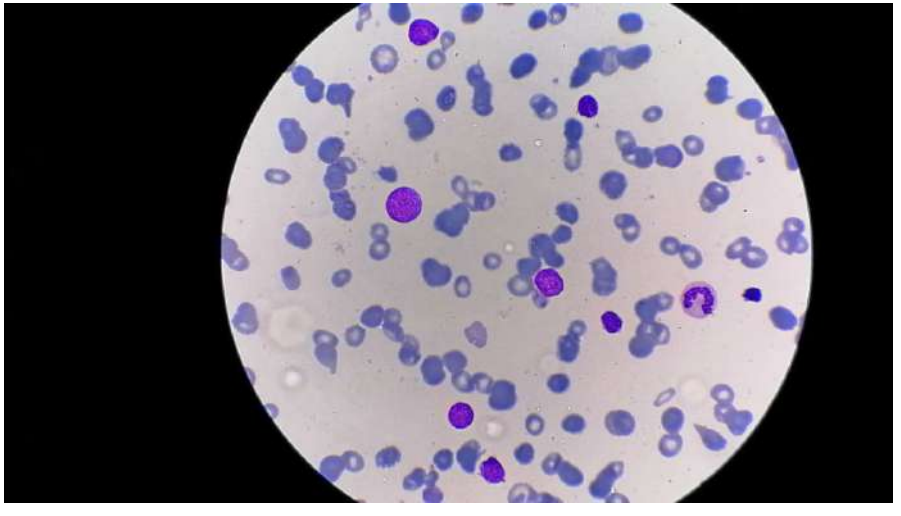


Published in CVPR 2022

https://openaccess.thecvf.com/content/CVPR2022/papers/Sultani_Towards_Low-Cost_and_Efficient_Malaria_Detection_CVPR_2022_paper.pdf

Image Resolution: 1000x

High Cost



Low Cost

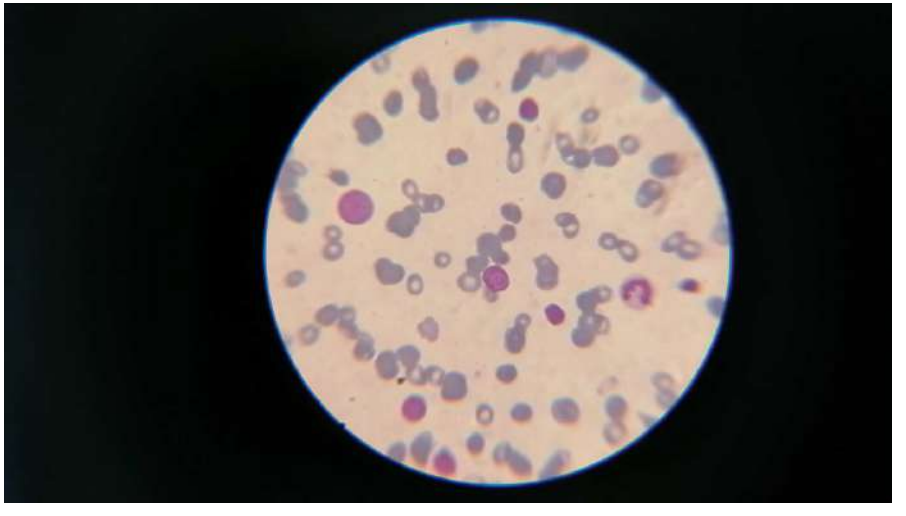
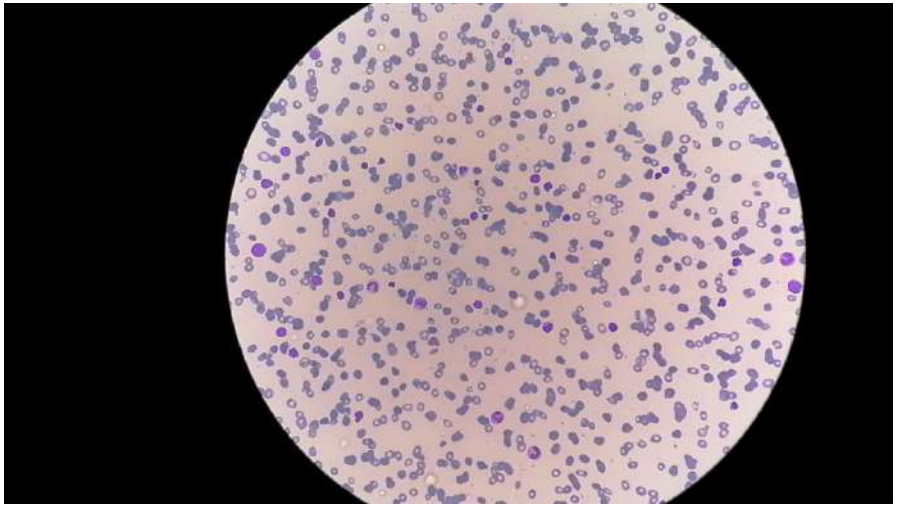
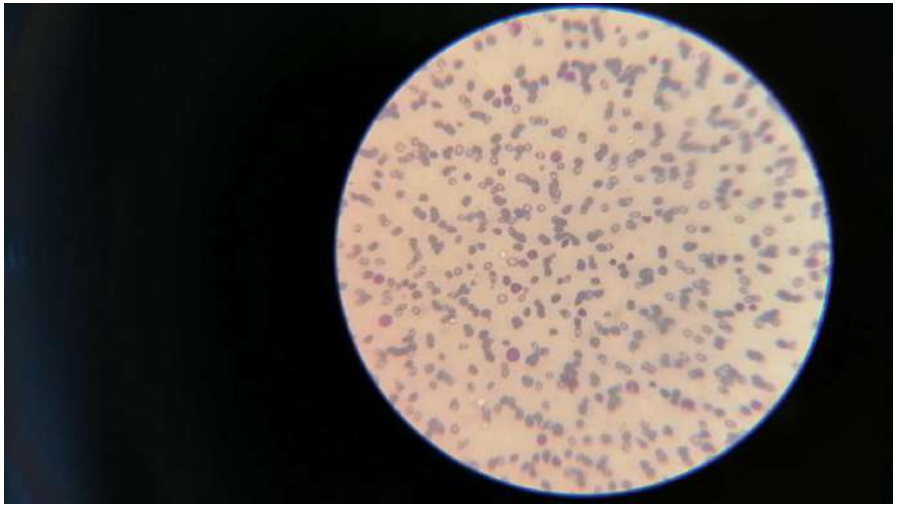


Image Resolution: 400x

High Cost

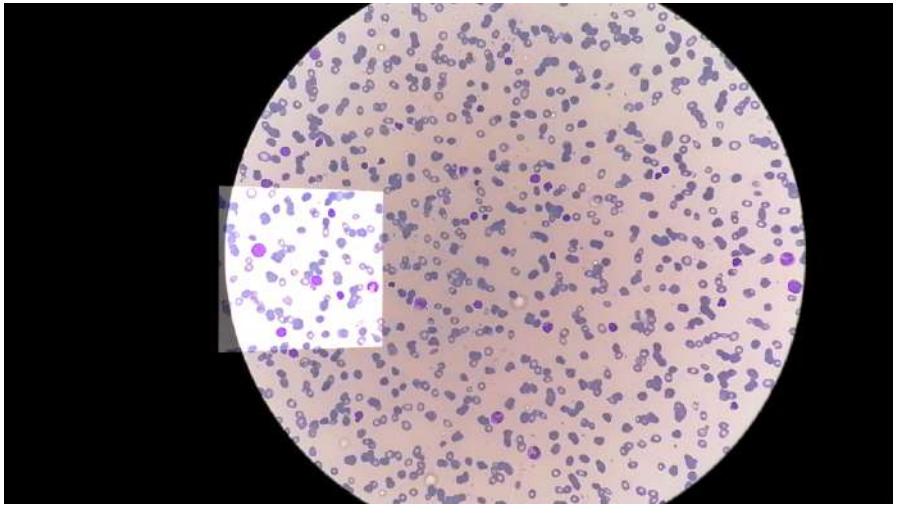


Low Cost



Matching: 1000x to 400x

High Cost



Low Cost

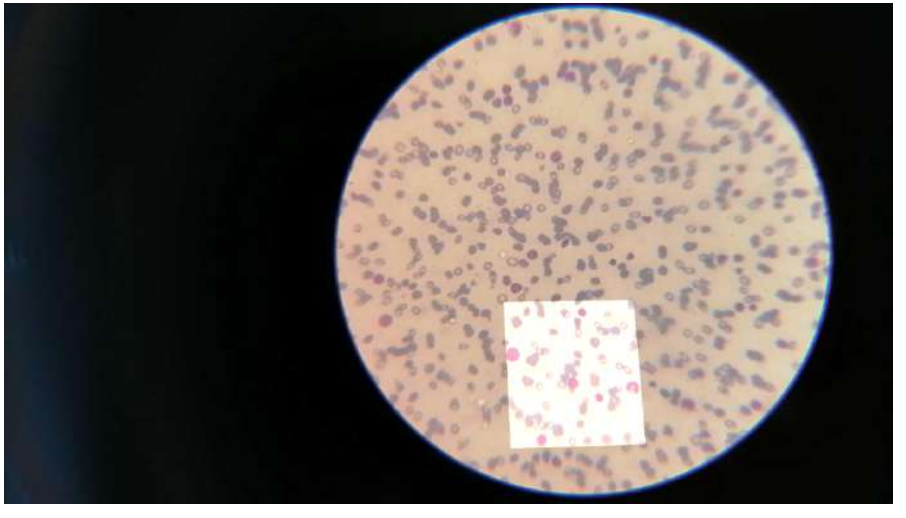
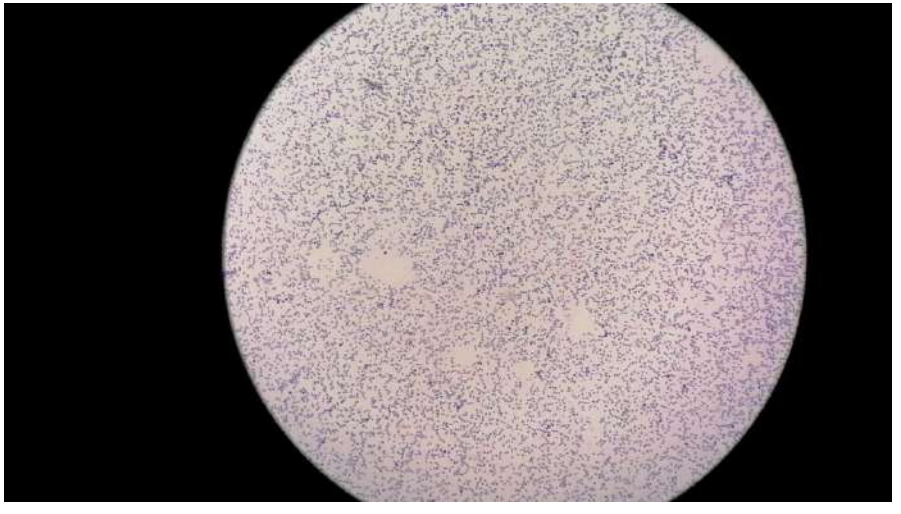
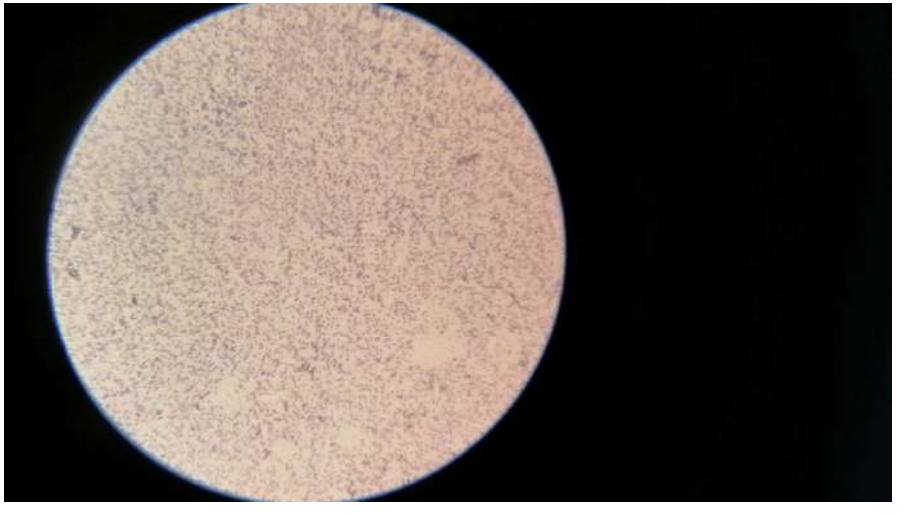


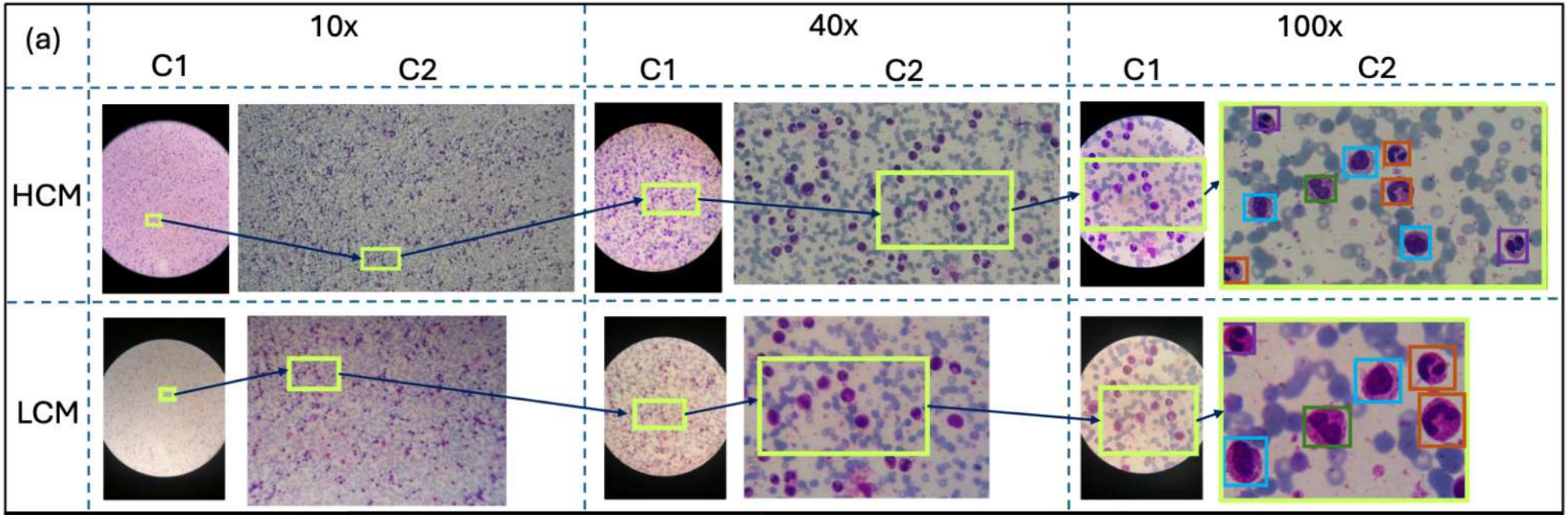
Image Resolution: 100x

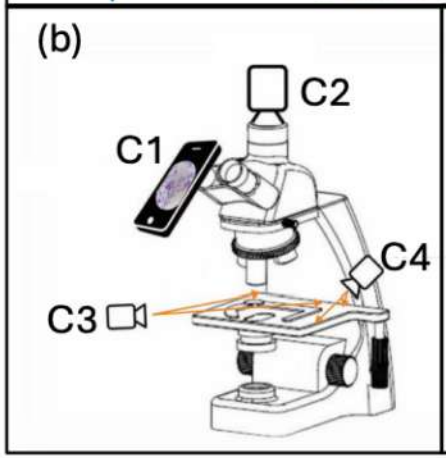
High Cost

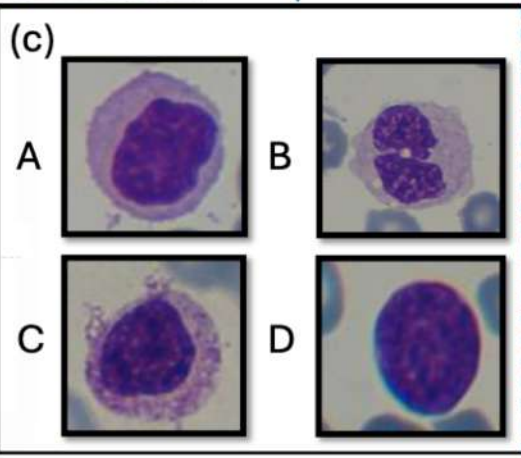


Low Cost



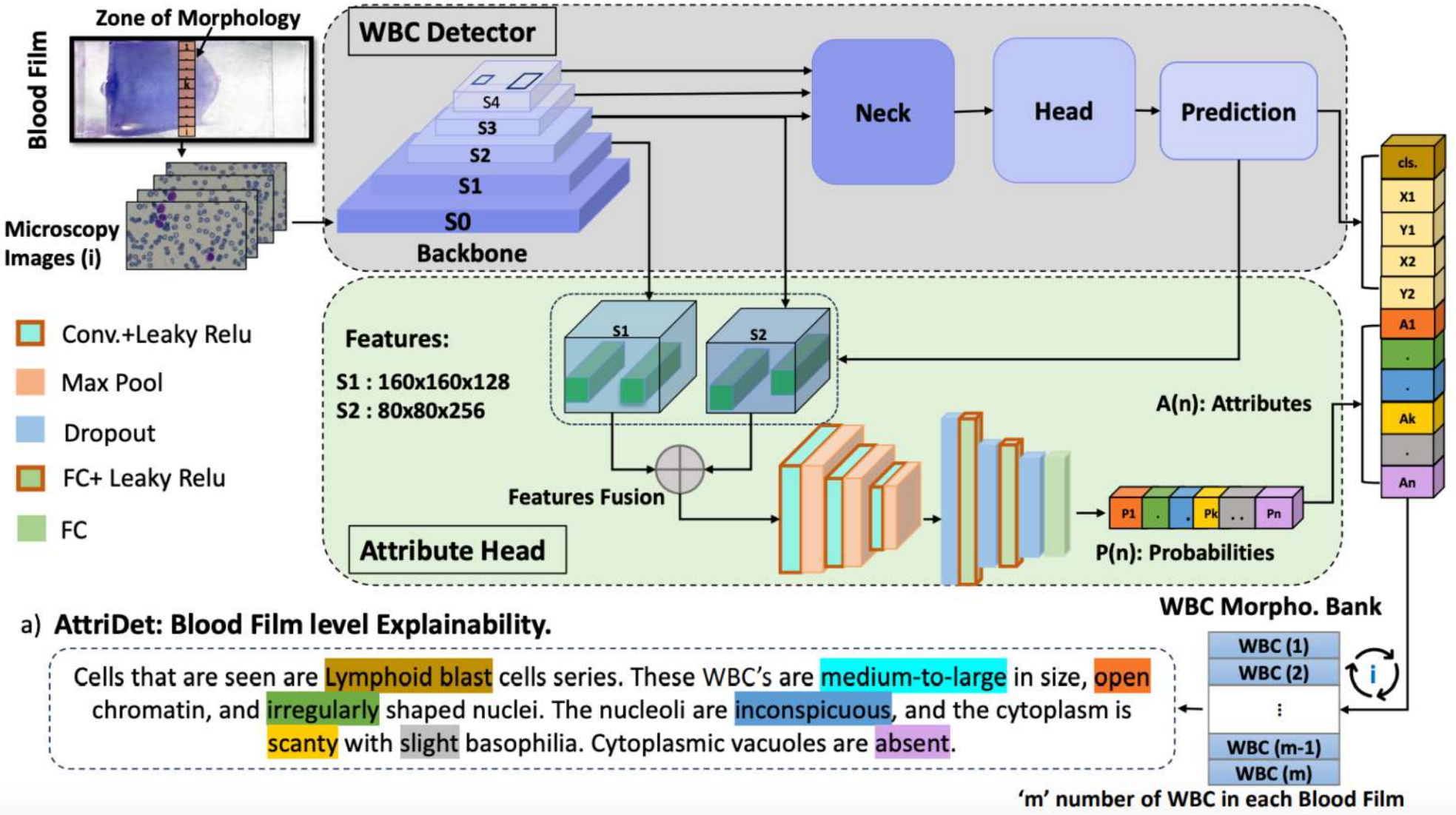






WBC	A	B	C	D
Type	Monoblast	Monocyte	Myelocyte	Lymphoblast
Size	Medium	Medium	Medium	Medium
Nuclear Chromatin	Open	Coarse	Coarse	Open
Nuclear shape	Irregular	Irregular	Regular	Regular
Nucleolus	Prominent	Inconspicuous	Inconspicuous	Inconspicuous
Cytoplasm	Abundant	Abundant	Abundant	Scanty
Cytoplasmic basophilia	Moderate	Moderate	Moderate	Slight
Cytoplasmic vacuoles	Absent	Prominent	Absent	Absent

Published in MICCA 2024:
A Large-scale Multi Domain Leukemia Dataset for the White Blood Cells Detection with Morphological Attributes for Explainability



a) **AttriDet: Blood Film level Explainability.**

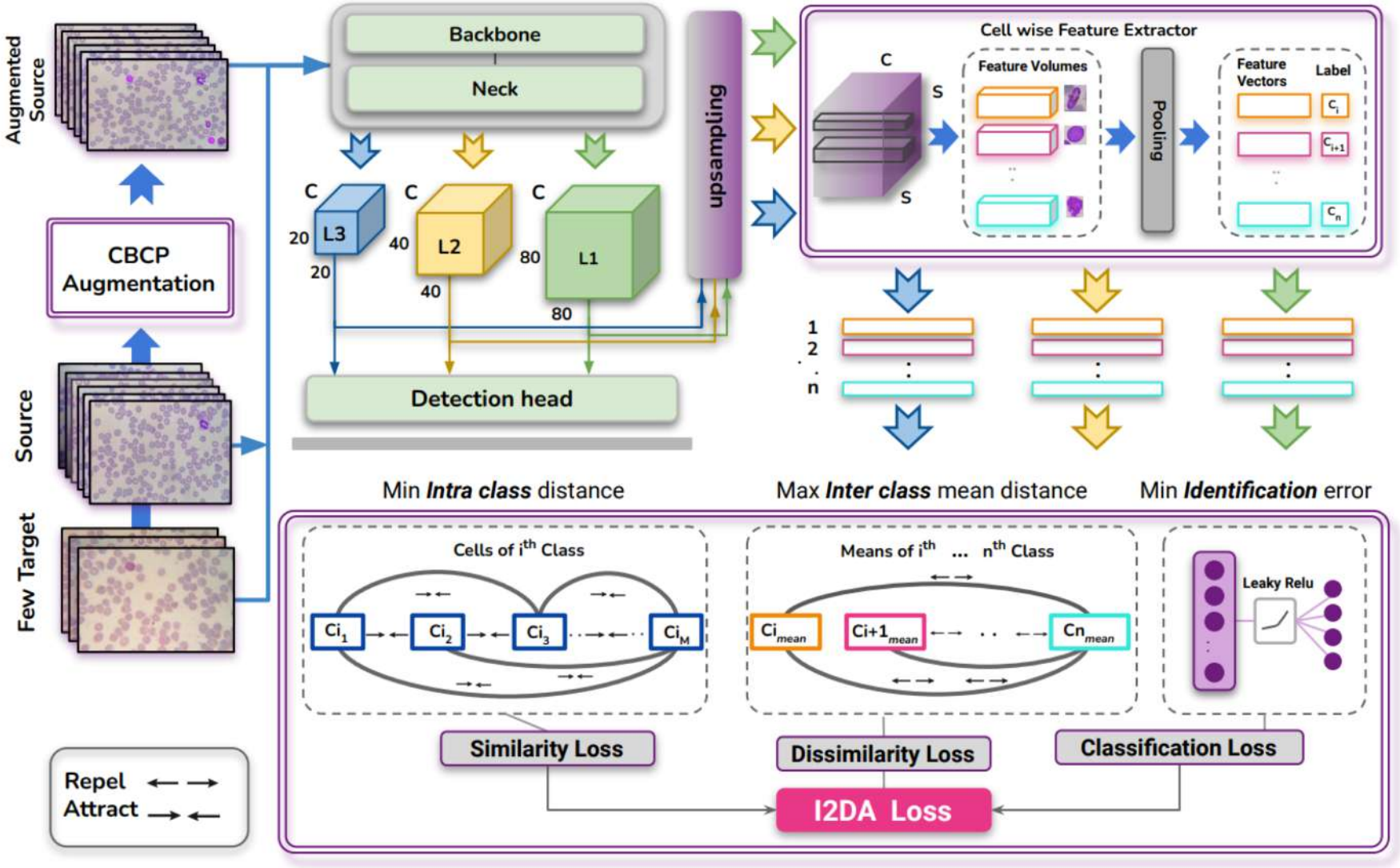
Cells that are seen are Lymphoid blast cells series. These WBC's are medium-to-large in size, open chromatin, and irregularly shaped nuclei. The nucleoli are inconspicuous, and the cytoplasm is scanty with slight basophilia. Cytoplasmic vacuoles are absent.

Published in MICCA 2024I:
A Large-scale Multi Domain Leukemia Dataset for the White Blood Cells Detection with Morphological Attributes for Explainability

Leveraging Sparse Annotations for Leukemia Diagnosis on the Large Leukemia Dataset

Abdul Rehman+, Talha Meraj+, Aiman Mahmood Minhas-, Ayisha Imran-,
Mohsen Ali+, Waqas Sultani+, Mubarak Shah*

+ Intelligent Machines Lab, Information Technology University, Lahore, Pakistan
- Department of Hematology, Chughtai Labs, Lahore, Pakistan
* Center for Research in Computer Vision, University of Central Florida, United State



Few-Shot Domain Adaptive Object Detection for Microscopic Images, MICCAI 2024

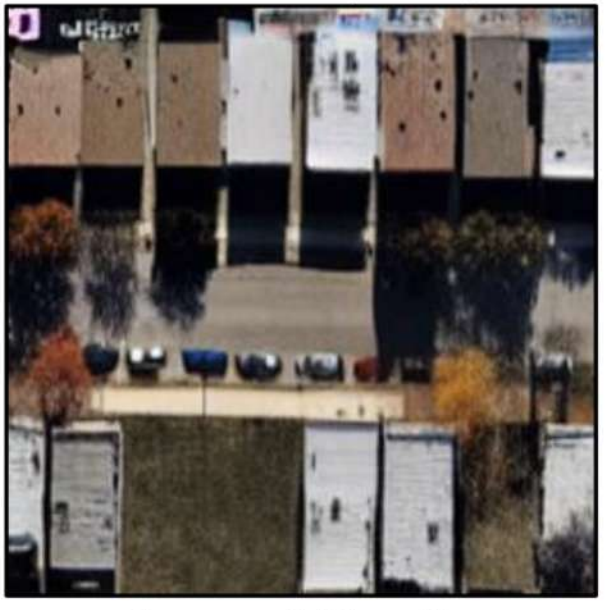
Cross-View Meets Diffusion: Aerial Image Synthesis with Geometry and Text Guidance

WACV 2025

Input Text Prompt

Realistic Chicago aerial satellite top view image with high quality details with buildings and roads in Chicago that probably has the following objects and characteristics: *street green dumpster, treelined block urban, buildings sides street, parked cars sides, urban area threestory.*

GPG2A (ours)

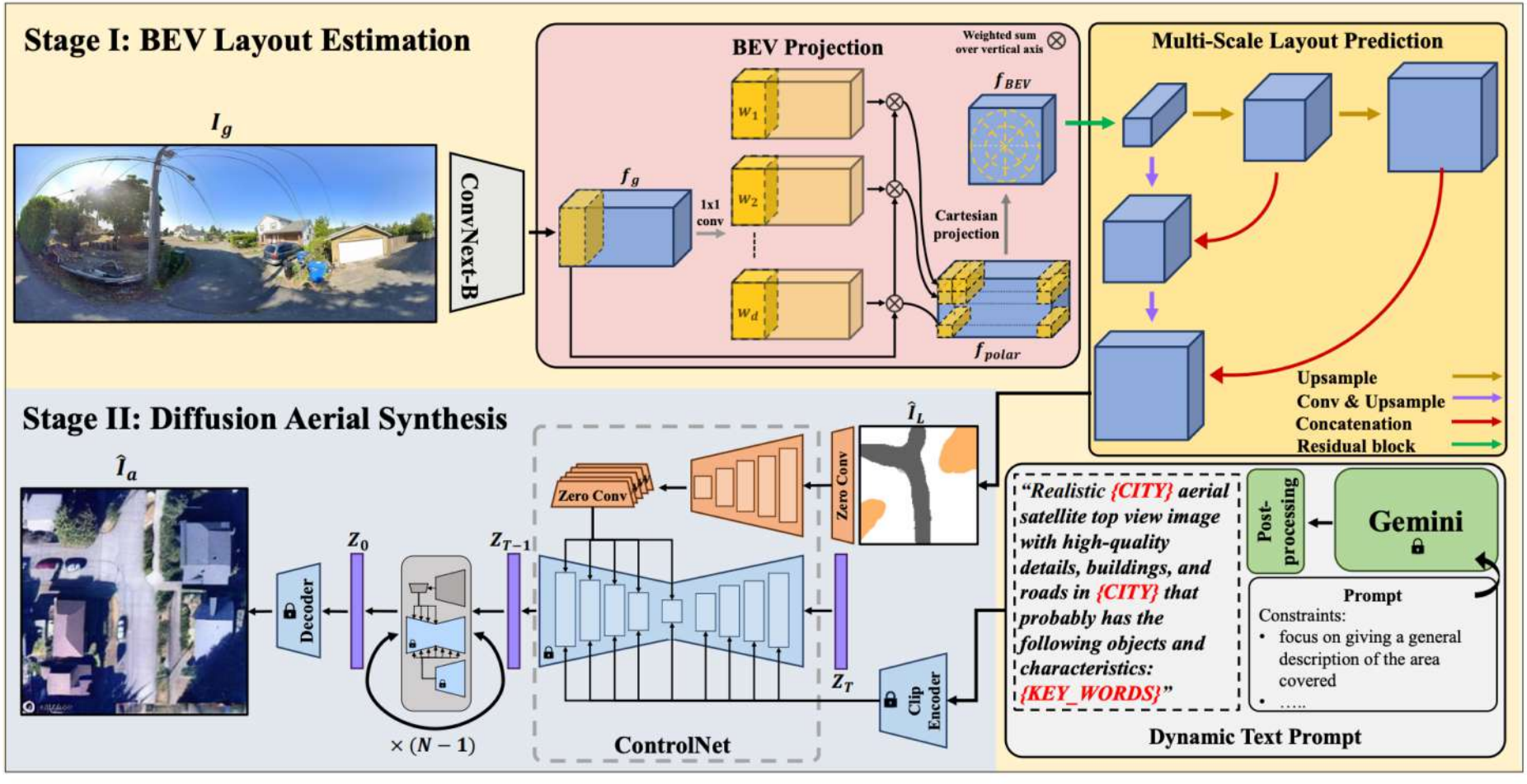


Input Ground Image

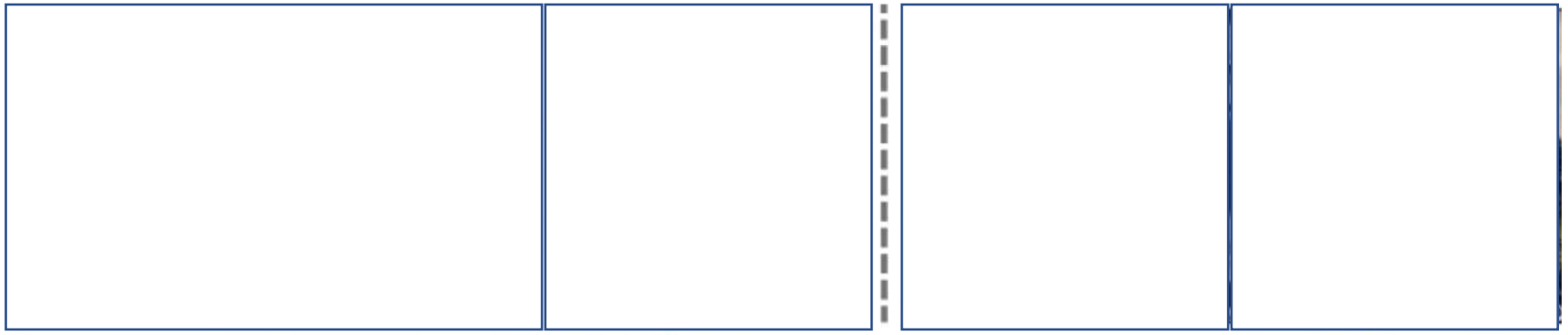


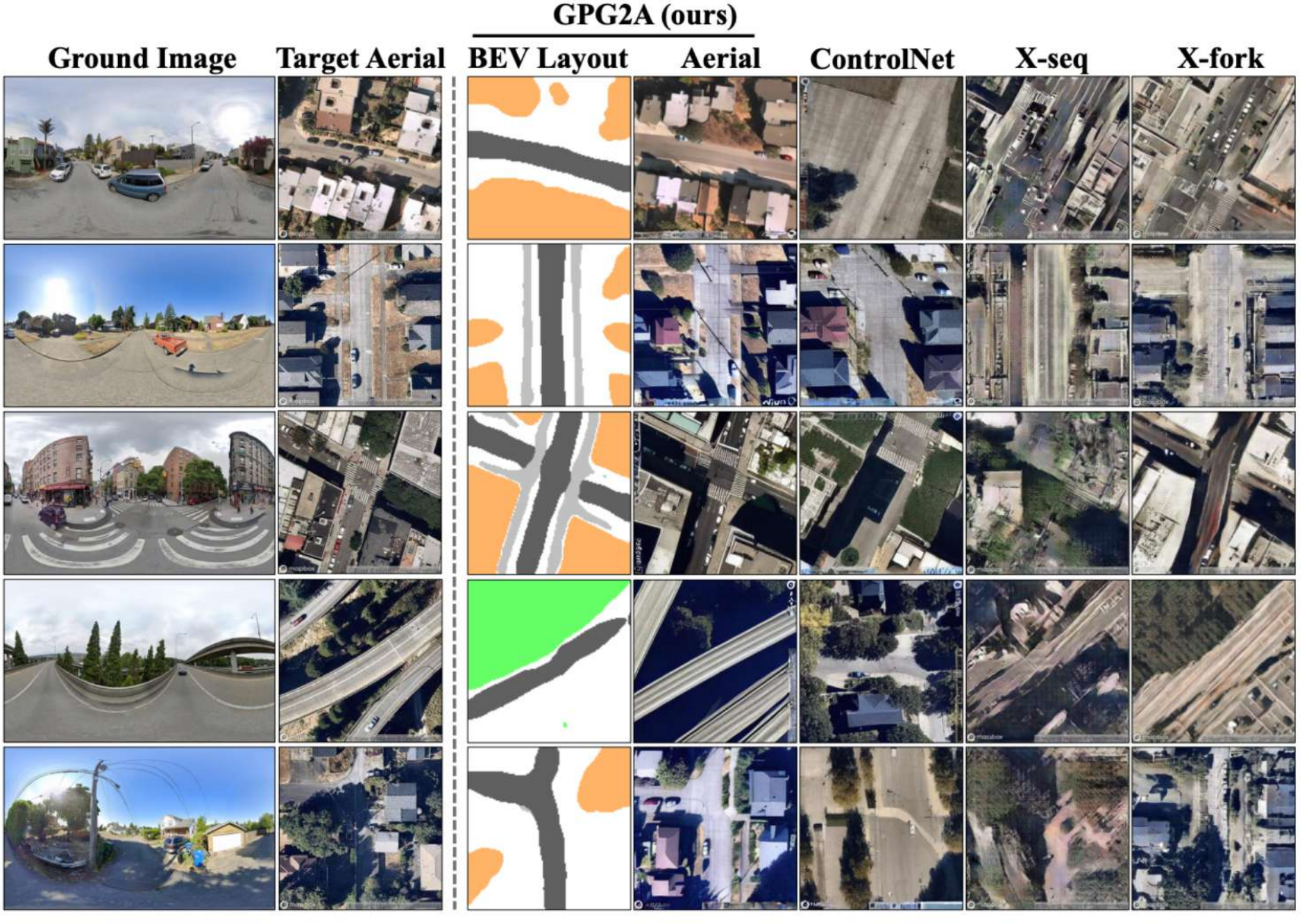
Ground Truth




Stage I: BEV Layout Estimation
I_g
ConvNext-B
BEV Projection
Weighted sum over vertical axis
f_g
1x1 conv
w_1
w_2
w_d
f_{BEV}
Cartesian projection
f_{polar}
Multi-Scale Layout Prediction
Upsample
Conv & Upsample
Concatenation
Residual block
Stage II: Diffusion Aerial Synthesis
$\hat{I}_a$
Decoder
Z_0
Z_{T-1}
$\times (N-1)$
Zero Conv
Zero Conv
$\hat{I}_L$
Z_T
Clip Encoder
ControlNet
"Realistic {CITY} aerial satellite top view image with high-quality details, buildings, and roads in {CITY} that probably has the following objects and characteristics: {KEY_WORDS}"
Post-processing
Gemini
Prompt
Constraints:
• focus on giving a general description of the area covered
• .....
Dynamic Text Prompt

Cross-View Meets Diffusion: Aerial Image Synthesis with Geometry and Text Guidance




GPG2A (ours)
Ground Image
Target Aerial
BEV Layout
Aerial
ControlNet
X-seq
X-fork

Out of Distribuition Detection in Malaria and Skin Cancer Images

Published in Biomedical Signal Processing 2022

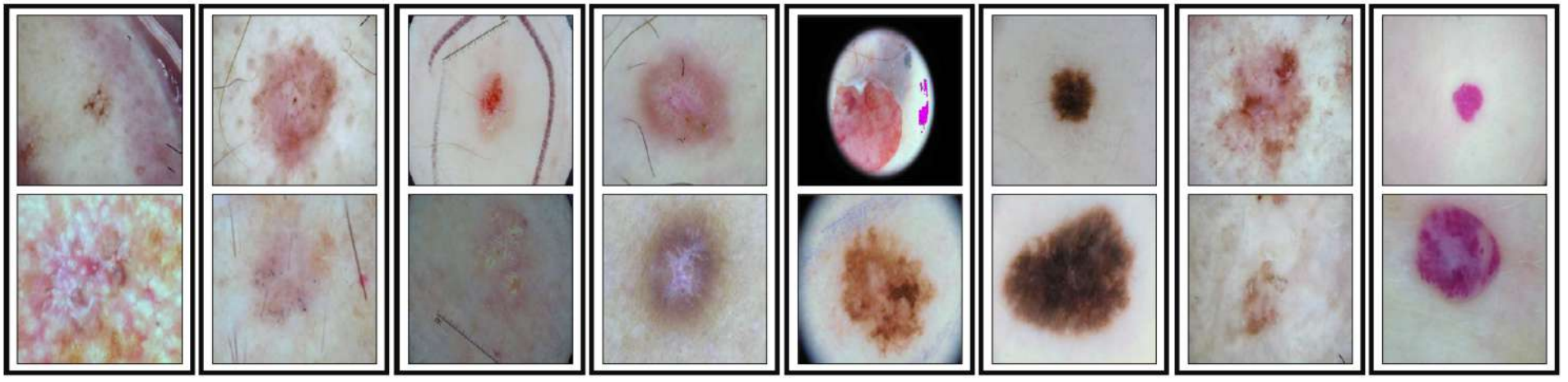
https://arxiv.org/pdf/2111.01505.pdf

- Skin cancer is one of the most common types of cancer. There are around 1.19 million new cases of skin cancer only in 2020.

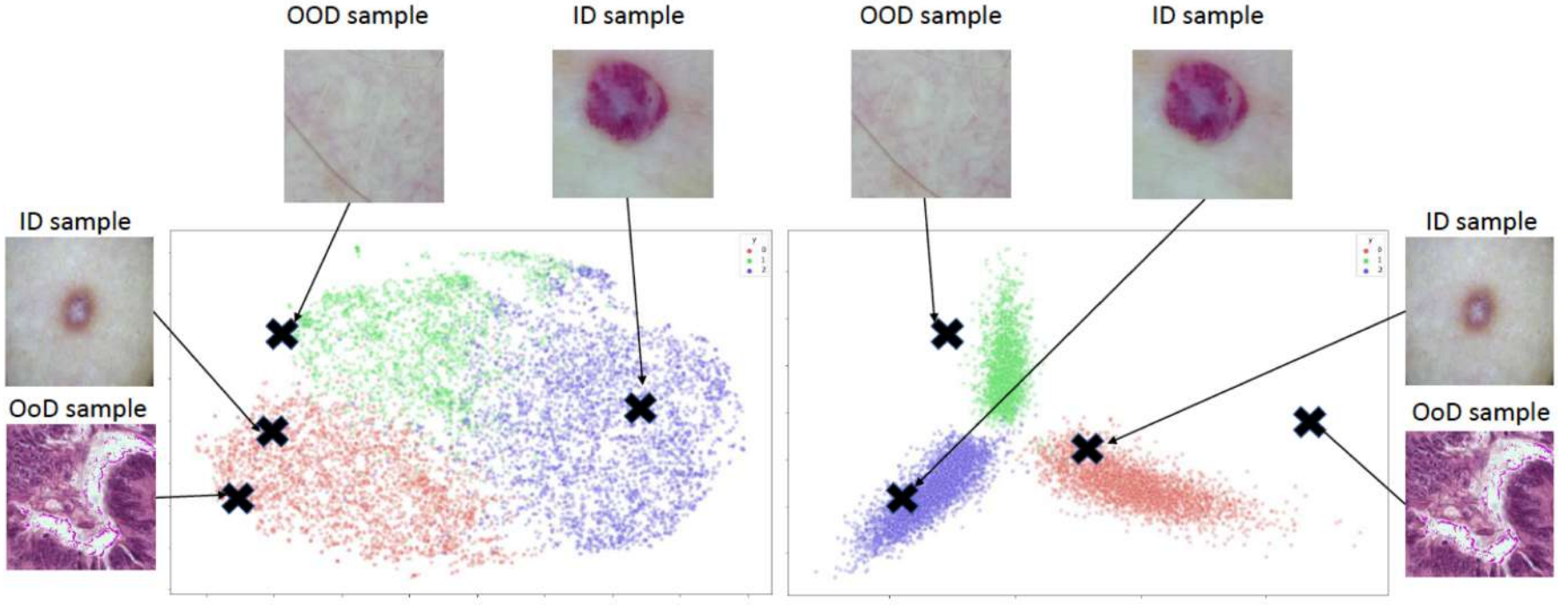
- An early diagnosis, including the detection of cancer and its correct classification, has been correlated to a high rate of overall survival

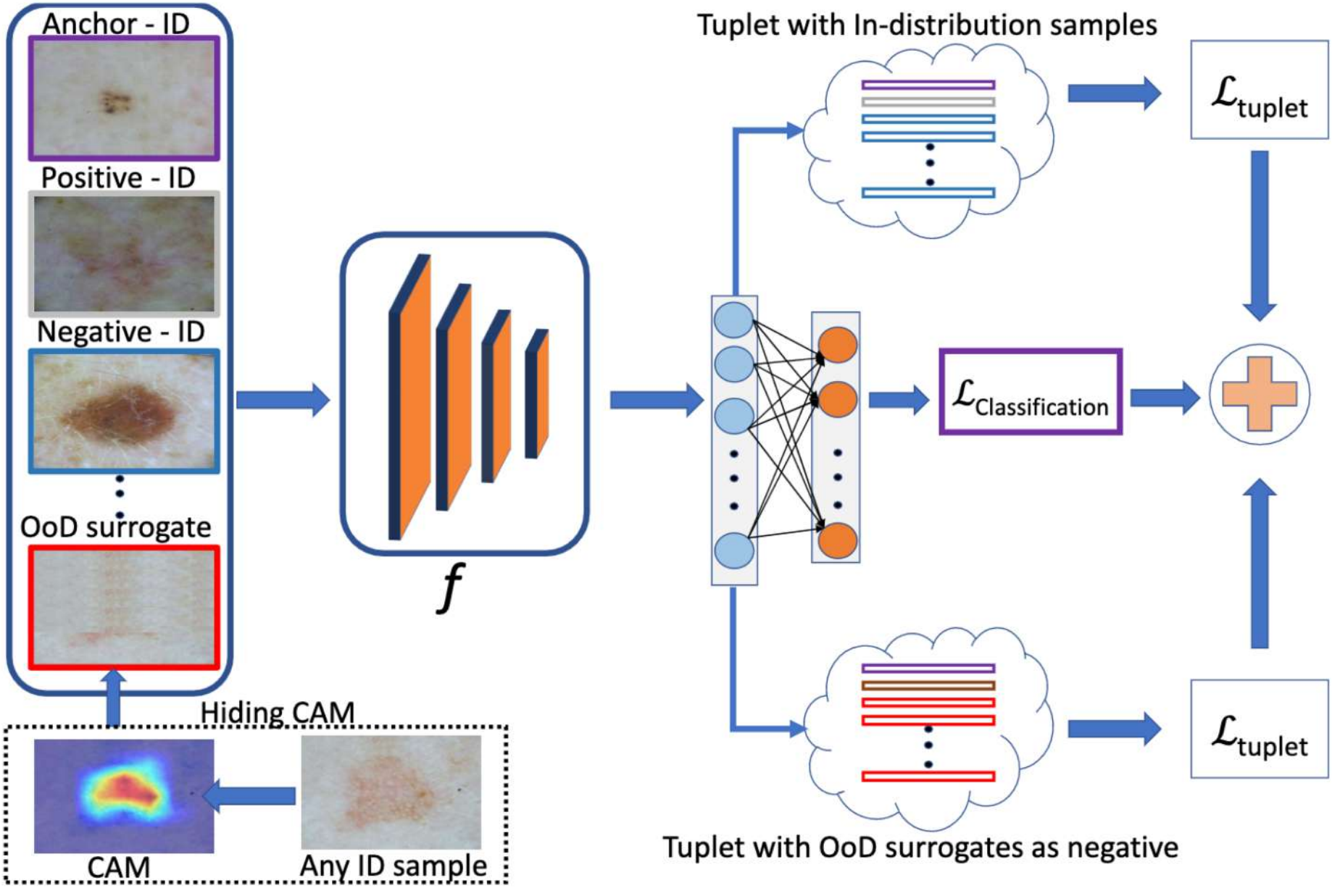
- The Armed Forces Institute of Pathology (AFIP), Rawalpindi, reports skin cancer to be **the fourth (6.68% of all cancers)** and second (5.63 of all cancers) leading cause of cancer in males and females respectively in Northern Pakistan

Out of Distribuition Detection in Skin Cancer Images



Skin Cancer Detection

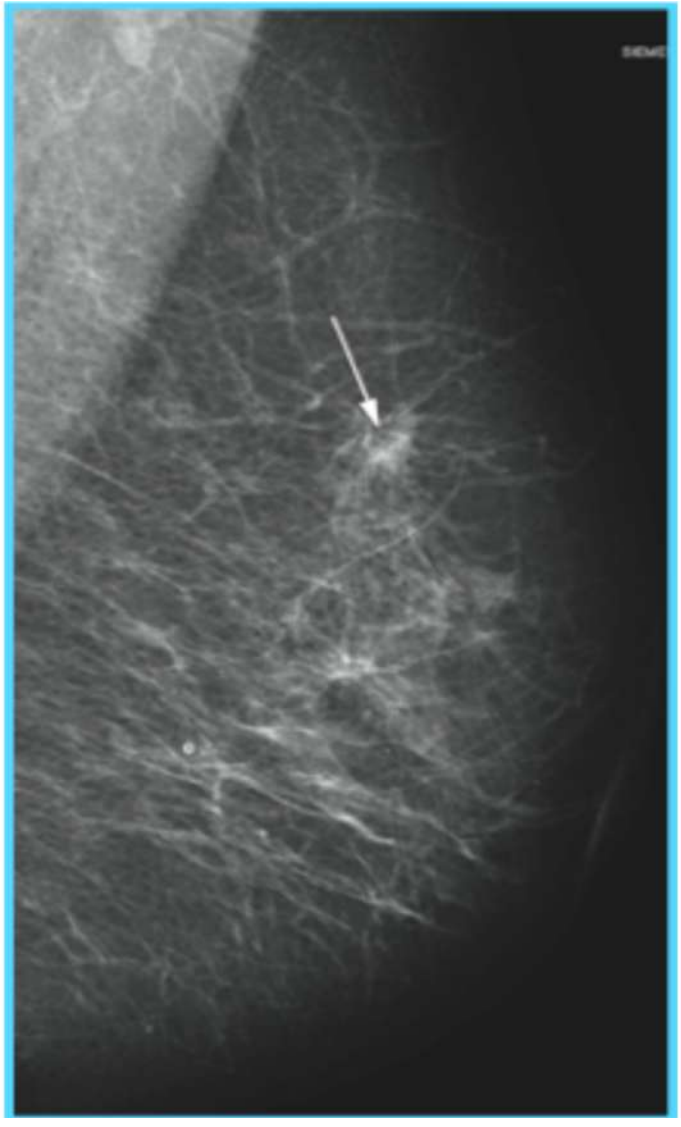


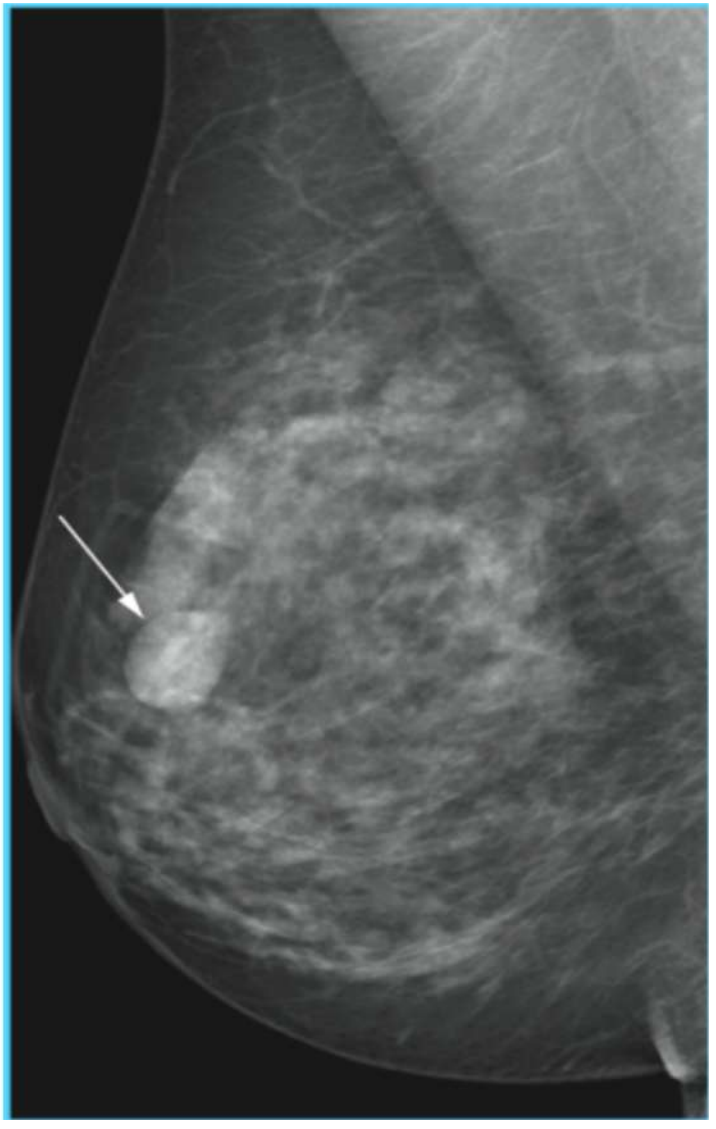

Anchor - ID
Positive - ID
Negative - ID
OoD surrogate
Hiding CAM
CAM
Any ID sample
f
Tuplet with In-distribution samples
$\mathcal{L}_{\text{tuplet}}$
$\mathcal{L}_{\text{Classification}}$
$\mathcal{L}_{\text{tuplet}}$
Tuplet with OoD surrogates as negative

Breast Cancer

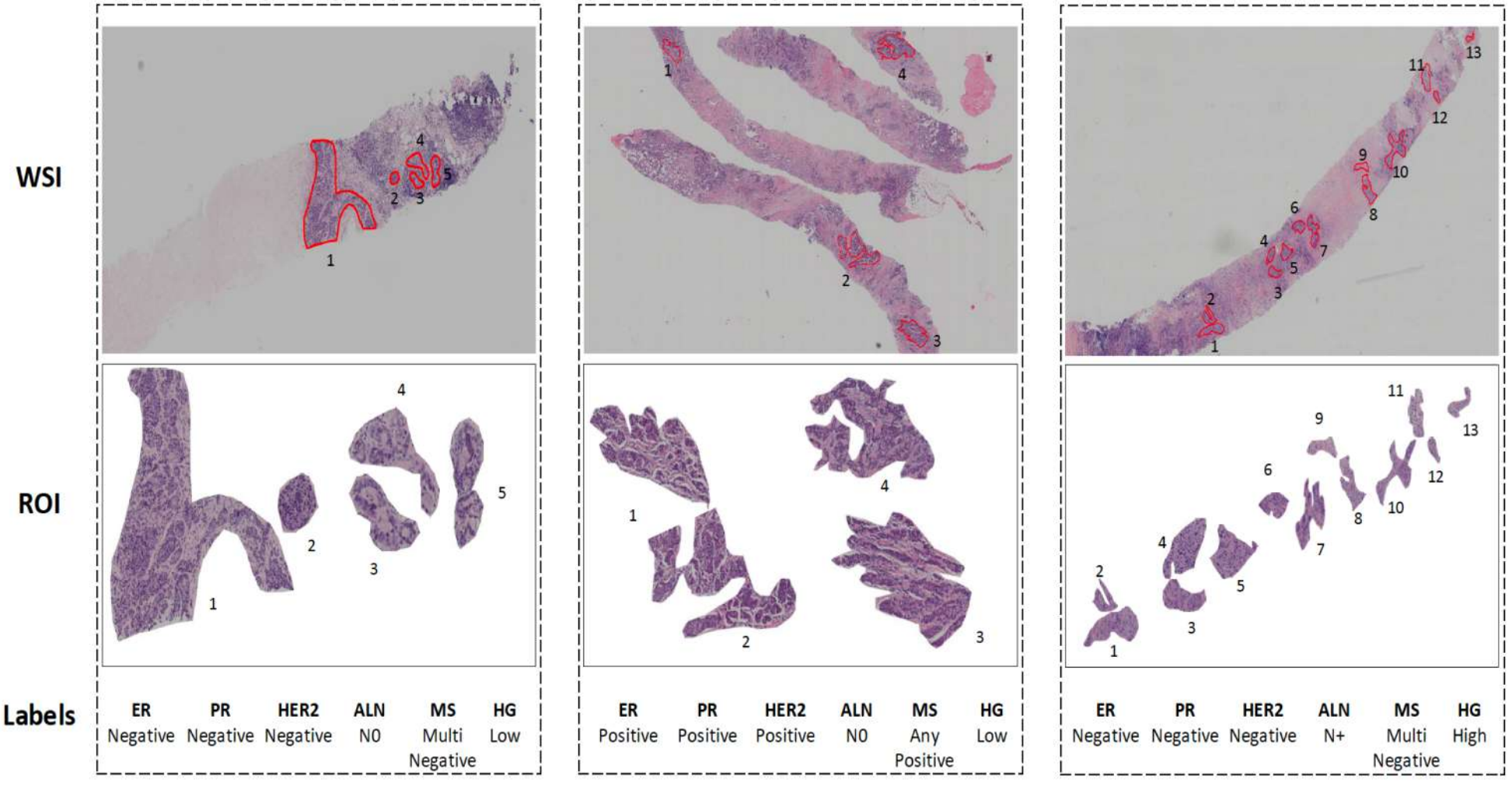
- Breast cancer rate in Pakistan is highest in Asia
- Almost 99 percent breast cancer patients, if diagnosed at an early stage, are treatable.
- In Pakistan, as many as 36,000 cases of breast cancer among women were reported in 2020 out of which 13,000 women died. “One in every nine women in Pakistan is likely to suffer from breast cancer in her lifetime,”.
- Every woman after the age of 40 years must get her mammography scan done. “Late diagnosis of breast cancers reduces chances of recovery from 1 to 10 percent,”

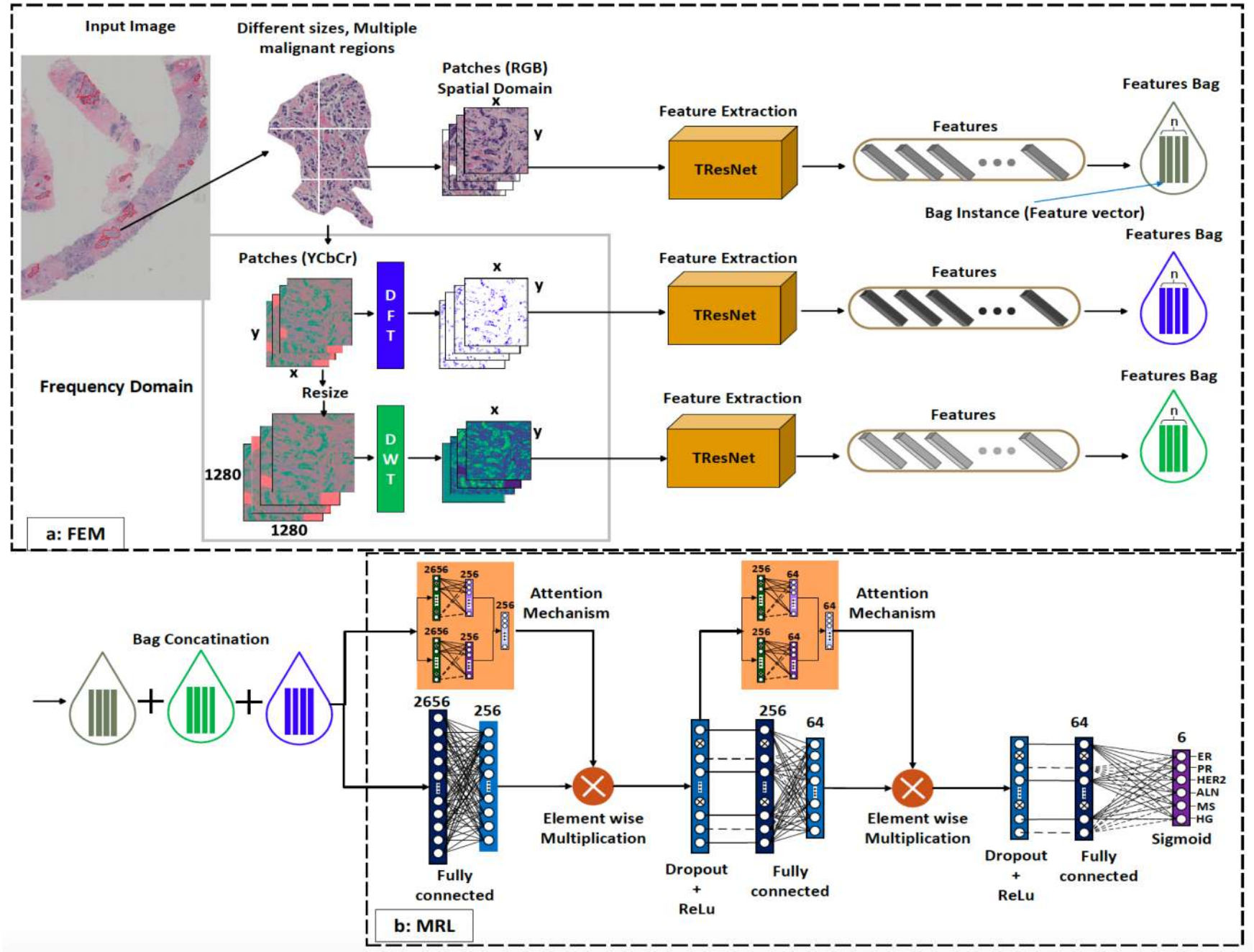
Benign





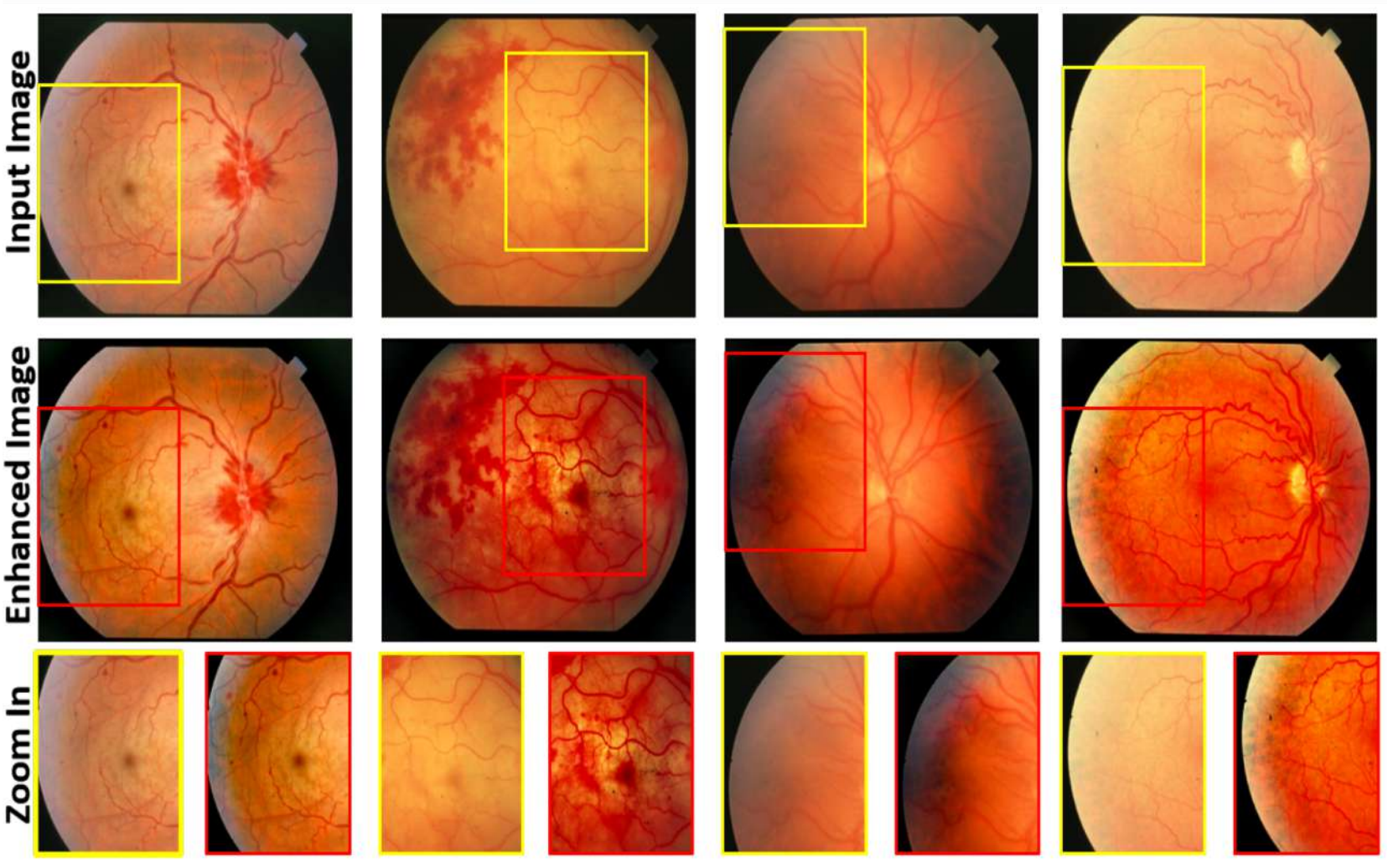
Malignant


WSI
ROI
Labels
ER PR HER2 ALN MS HG
Negative Negative Negative N0 Multi Negative Low
ER PR HER2 ALN MS HG
Positive Positive Positive N0 Any Positive Low
ER PR HER2 ALN MS HG
Negative Negative Negative N+ Multi Negative High



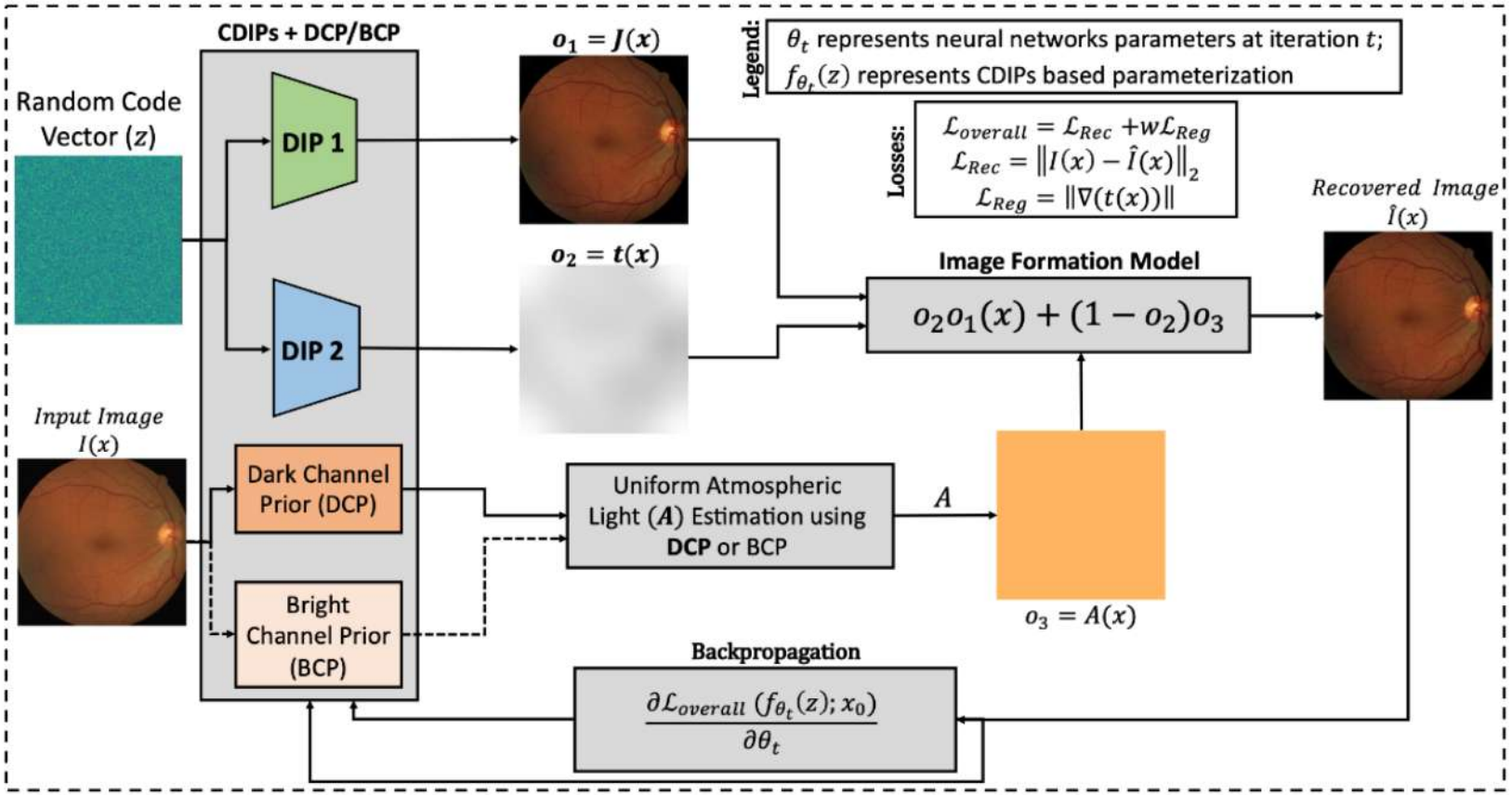
Under review in Biomedical Signal Processing 2023

Retinal Image Enhancement

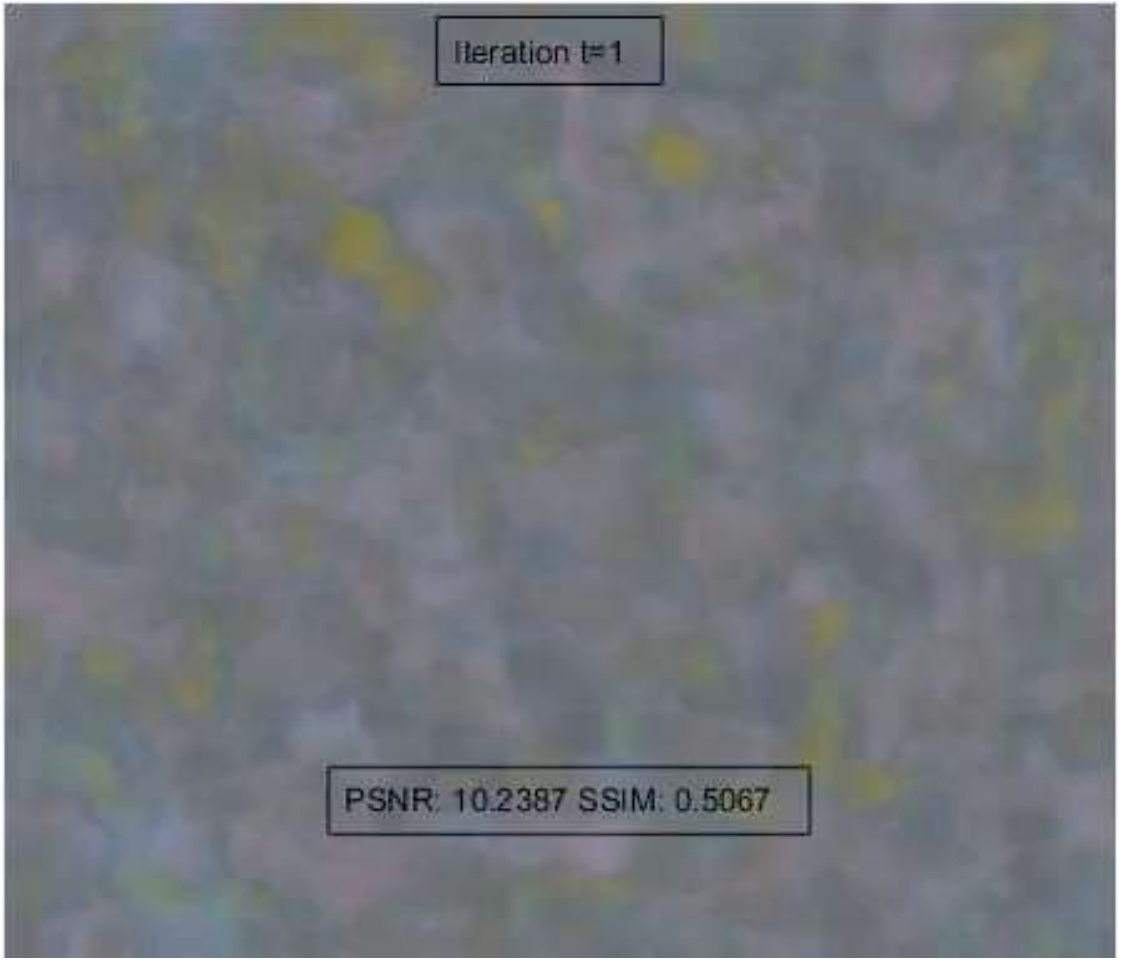


Published in MICCAI 2020 and Computers in Biology and Medicine 2022

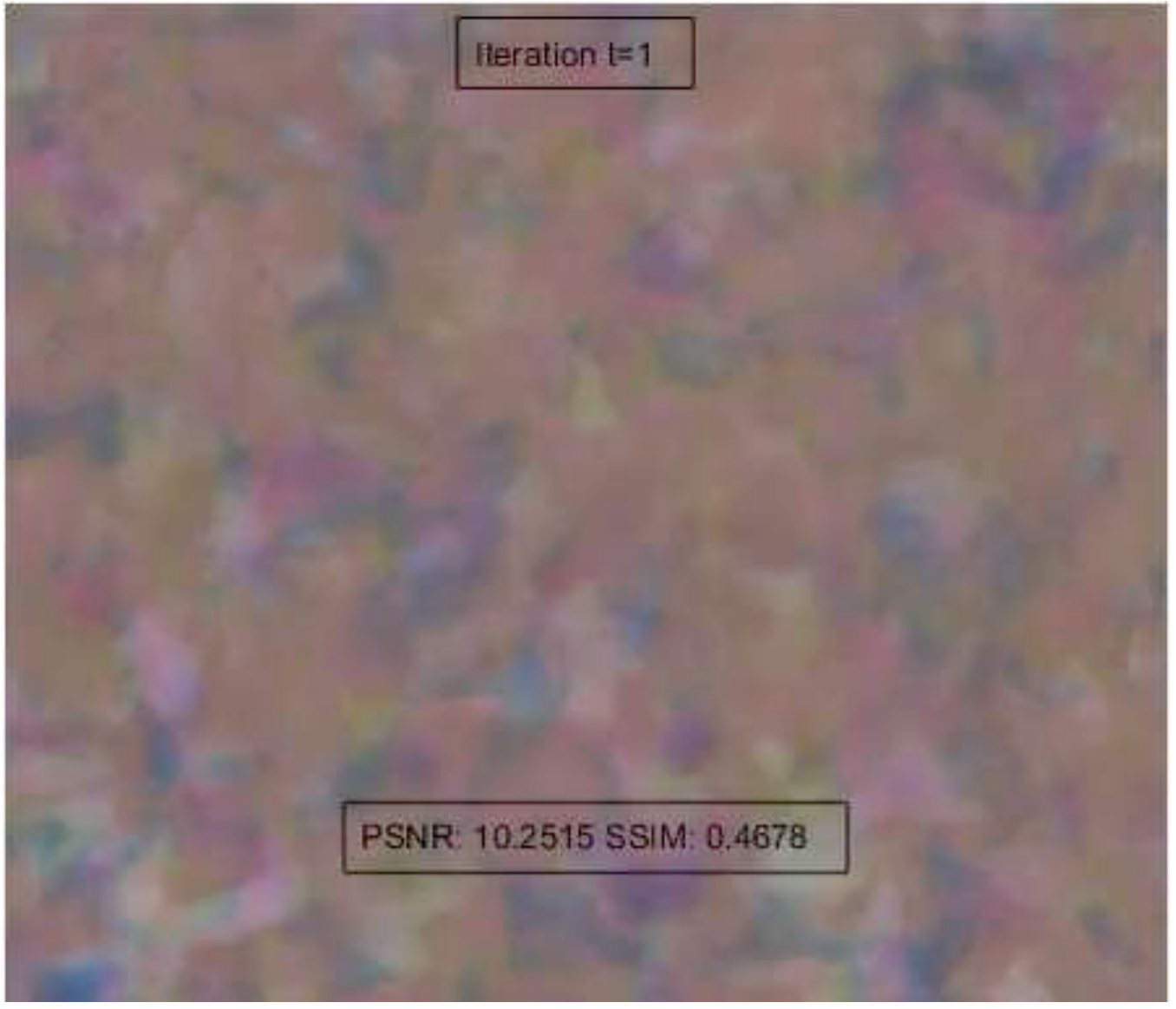
https://link.springer.com/chapter/10.1007/978-3-030-59722-1_61


CDIPs + DCP/BCP
Random Code Vector (z)
DIP 1
DIP 2
$o_1 = J(x)$
$o_2 = t(x)$
Legend:
θ_t represents neural networks parameters at iteration t;
$f_{\theta_t}(z)$ represents CDIPs based parameterization
Losses:
$\mathcal{L}_{overall} = \mathcal{L}_{Rec} + w\mathcal{L}_{Reg}$
$\mathcal{L}_{Rec} = \|I(x) - \hat{I}(x)\|_2$
$\mathcal{L}_{Reg} = \|\nabla(t(x))\|$
Recovered Image $\hat{I}(x)$
Image Formation Model
$o_2 o_1(x) + (1 - o_2) o_3$
Input Image $I(x)$
Dark Channel Prior (DCP)
Bright Channel Prior (BCP)
Uniform Atmospheric Light (A) Estimation using DCP or BCP
A
$o_3 = A(x)$
Backpropagation
$\frac{\partial \mathcal{L}_{overall}(f_{\theta_t}(z); x_0)}{\partial \theta_t}$

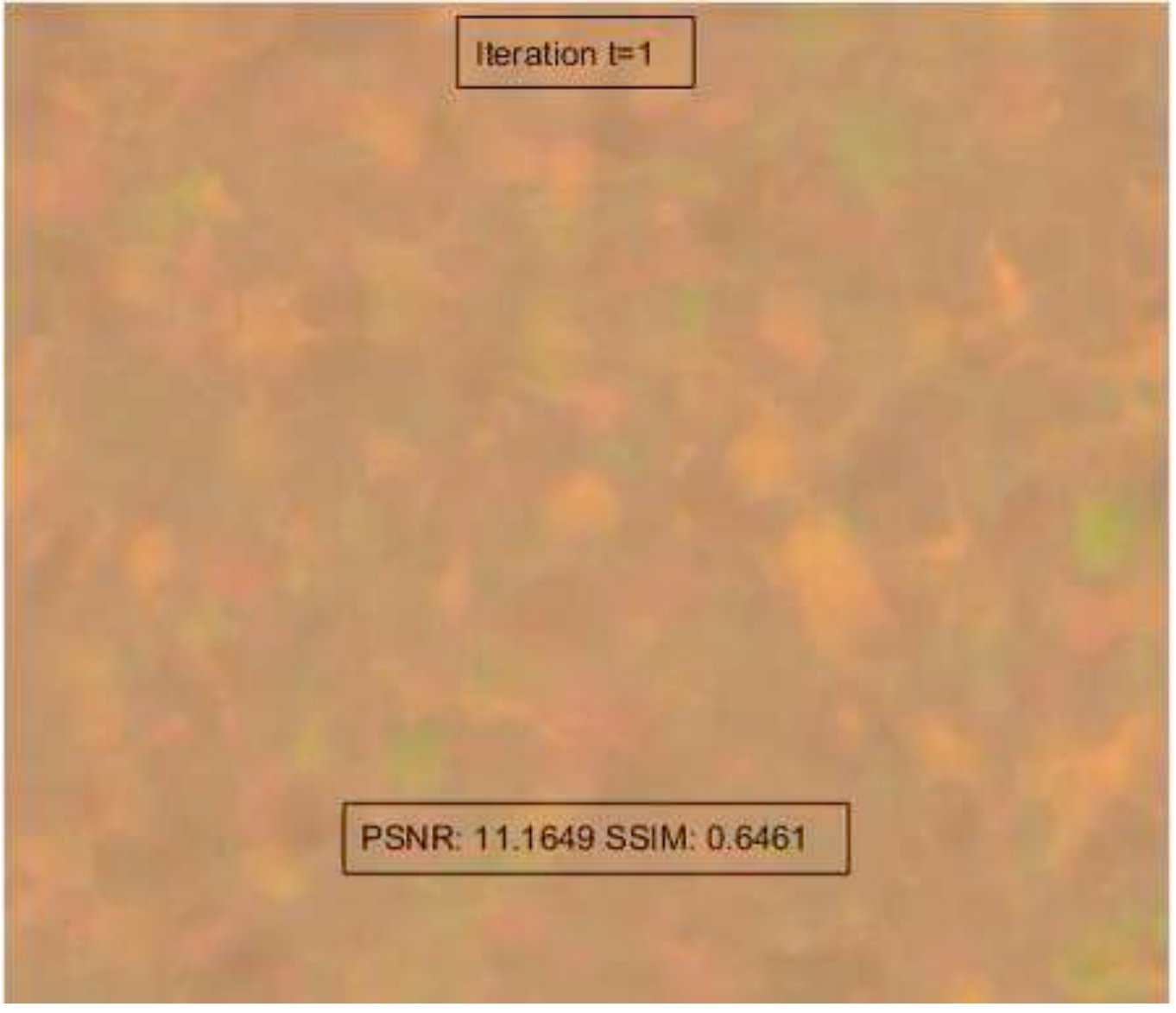
Retinal Image Enhancement



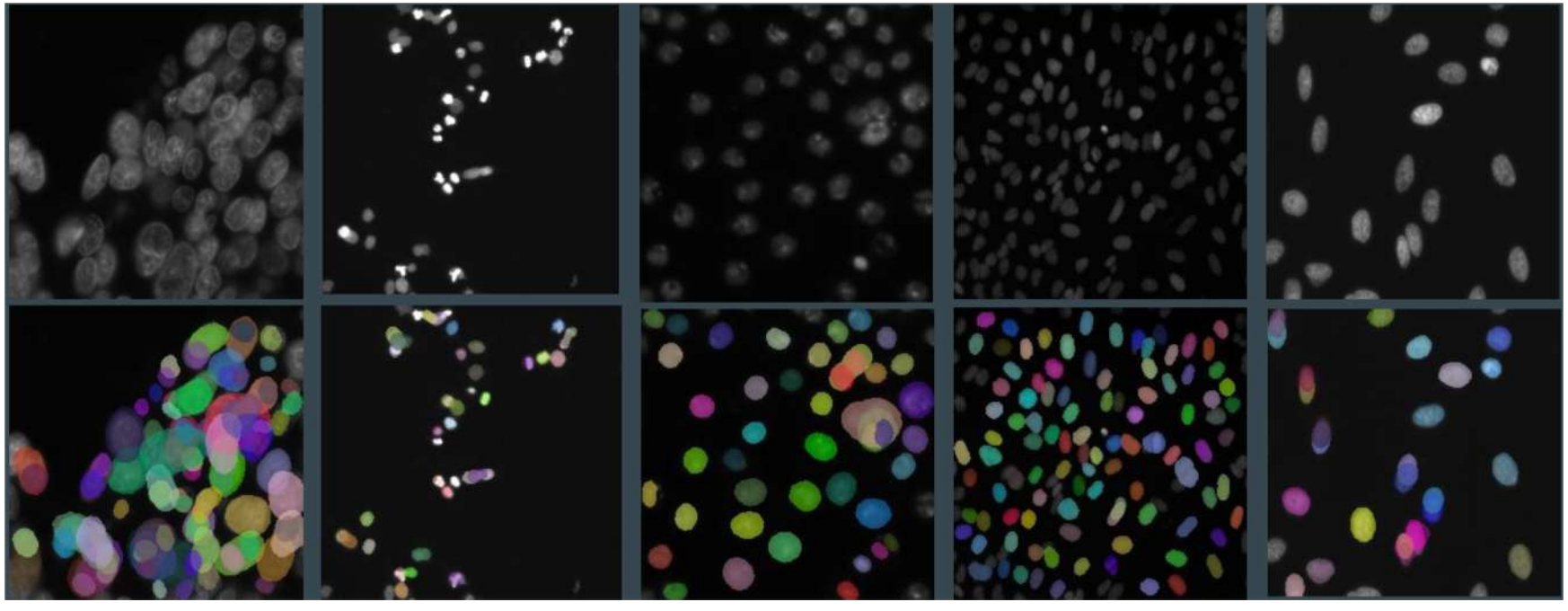
Retinal Image Enhancement

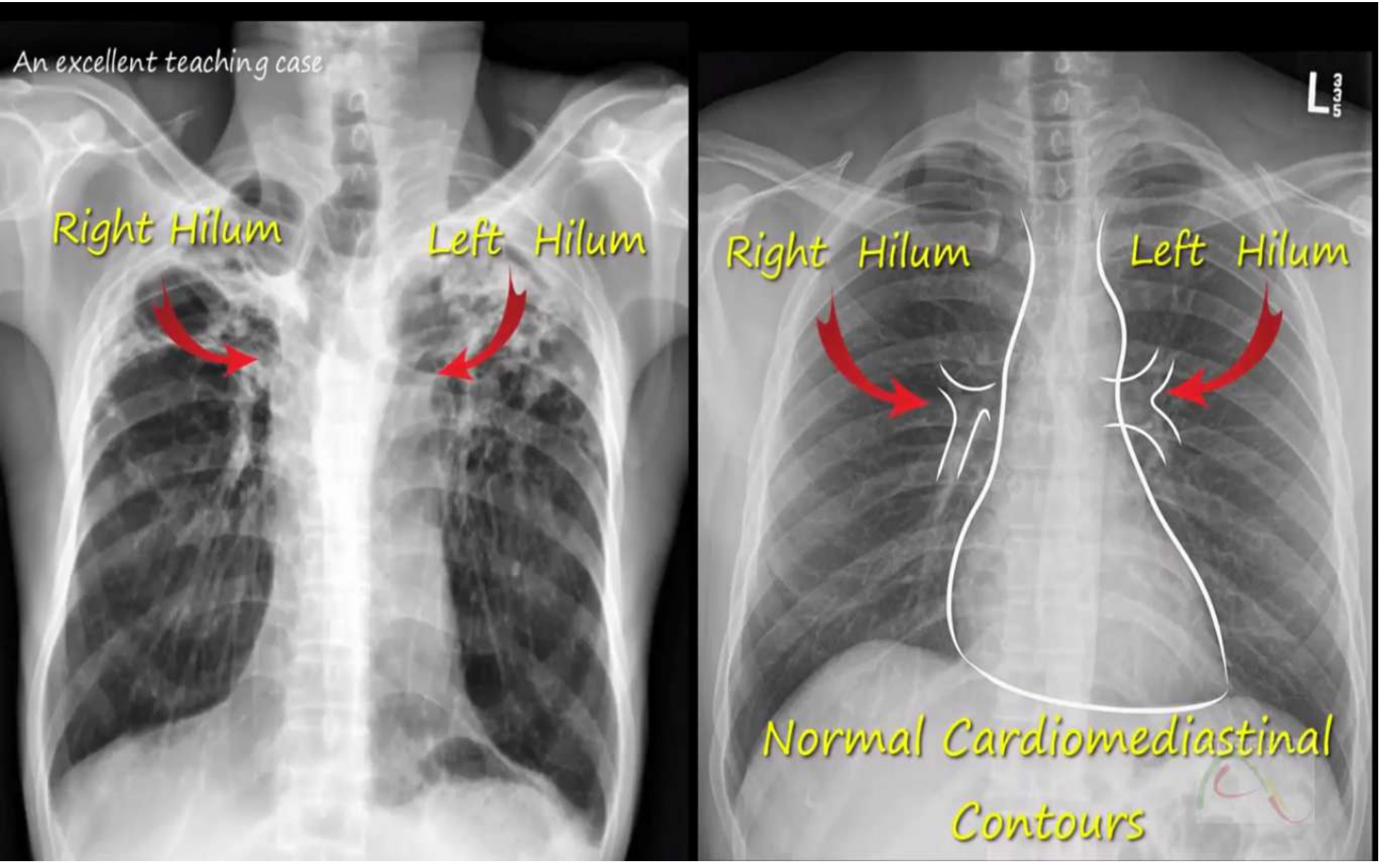


Retinal Image Enhancement



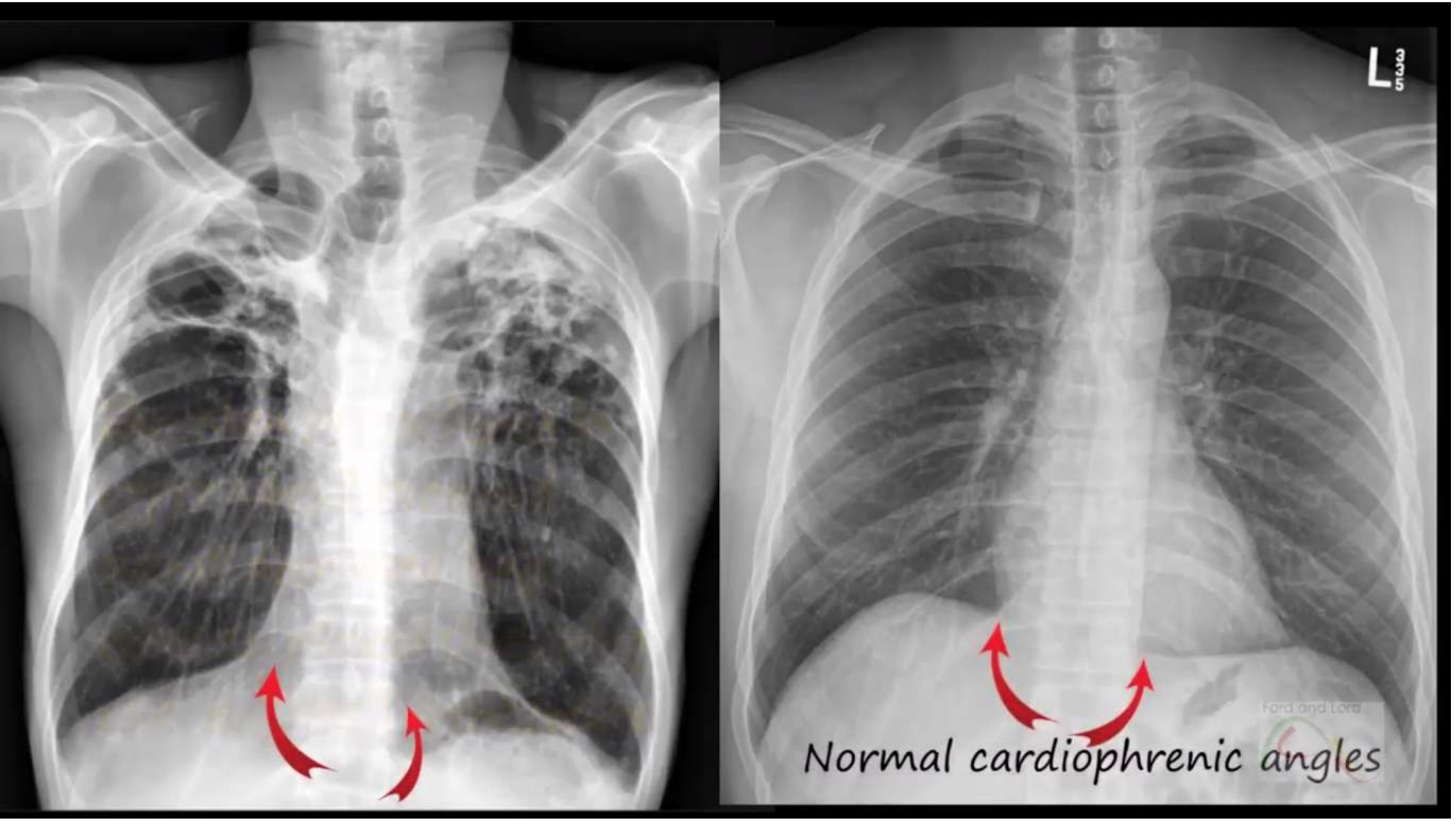
Microscopic Cell Counting

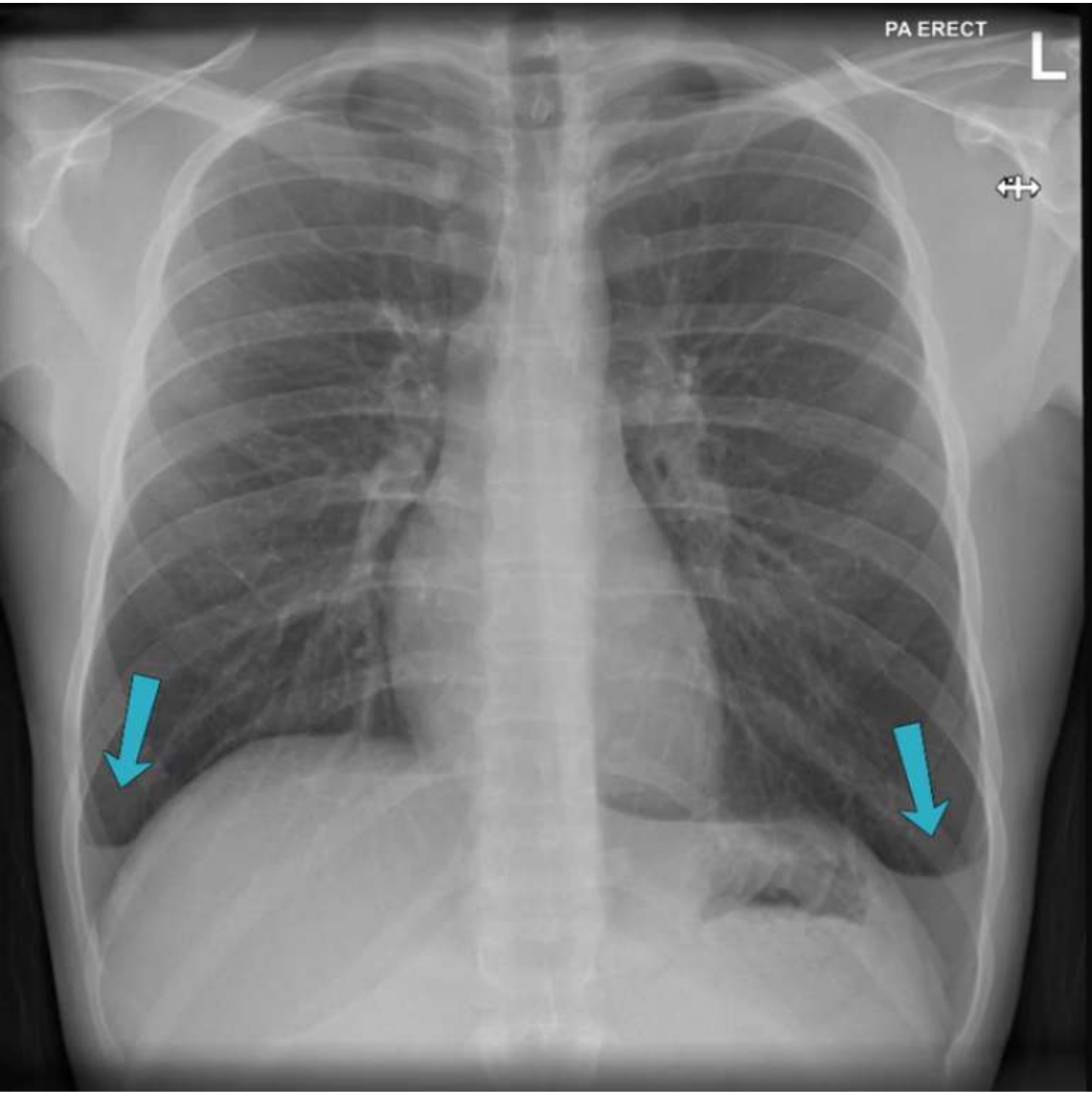



An excellent teaching case
Right Hilum
Left Hilum
Right Hilum
Left Hilum
Normal Cardiomediastinal
Contours

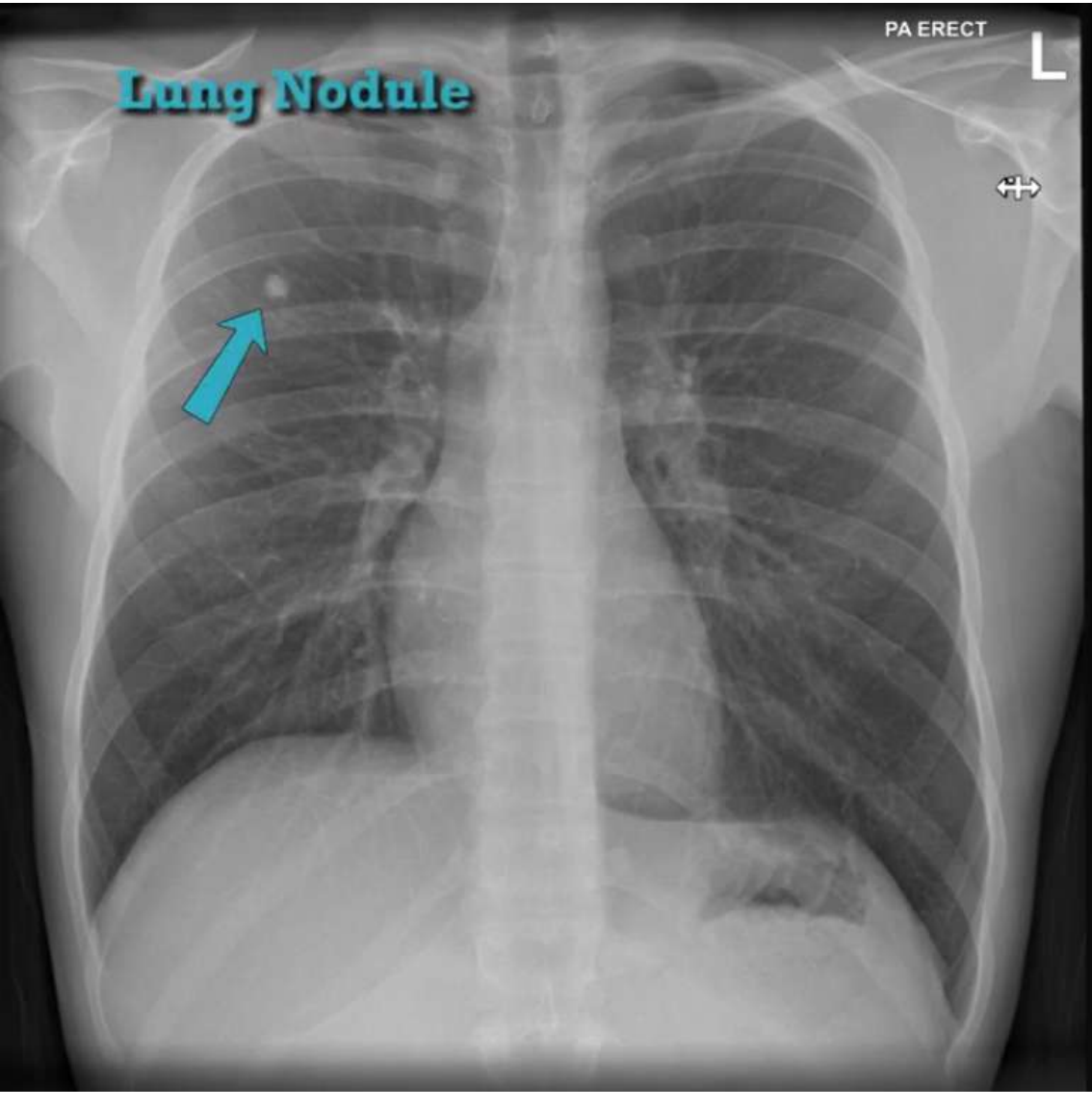

An excellent teaching case
Decreased lung volumes


Normal chest x-ray
L


Normal cardiophrenic angles

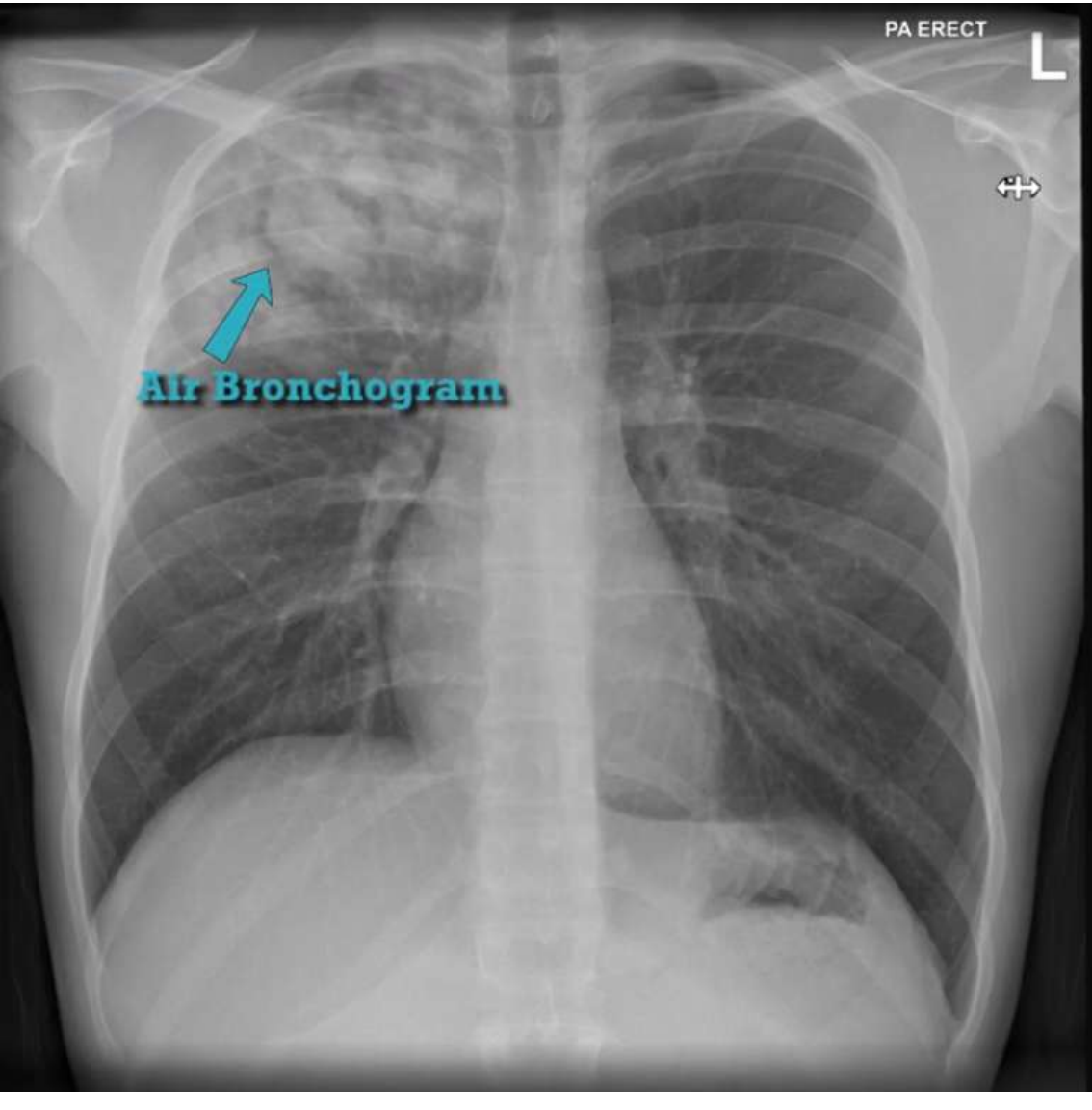

PA ERECT
L

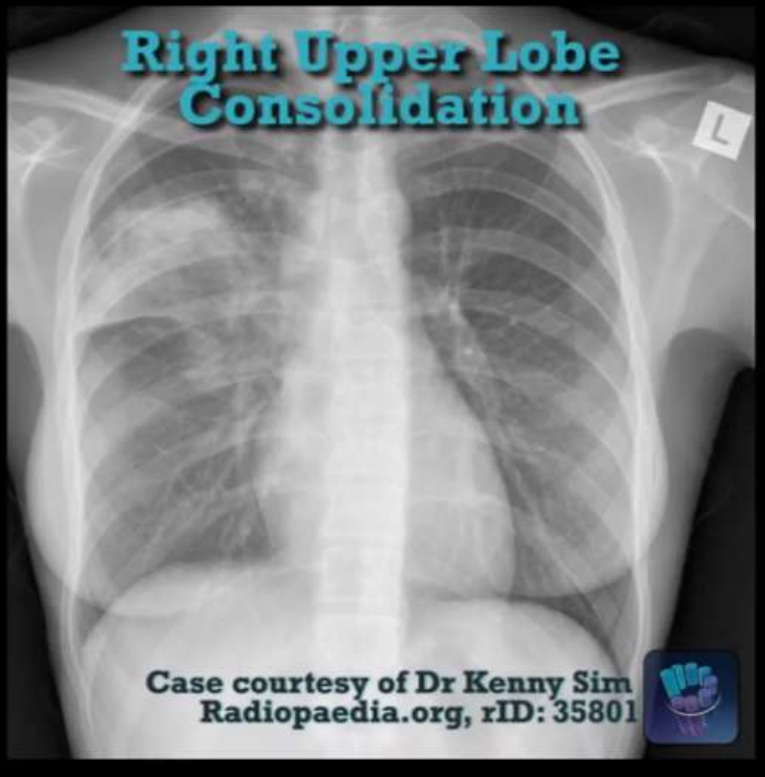

Right Pleural Effusion
Case courtesy of A.Prof Frank Gaillard
Radiopaedia.org, rID: 25595



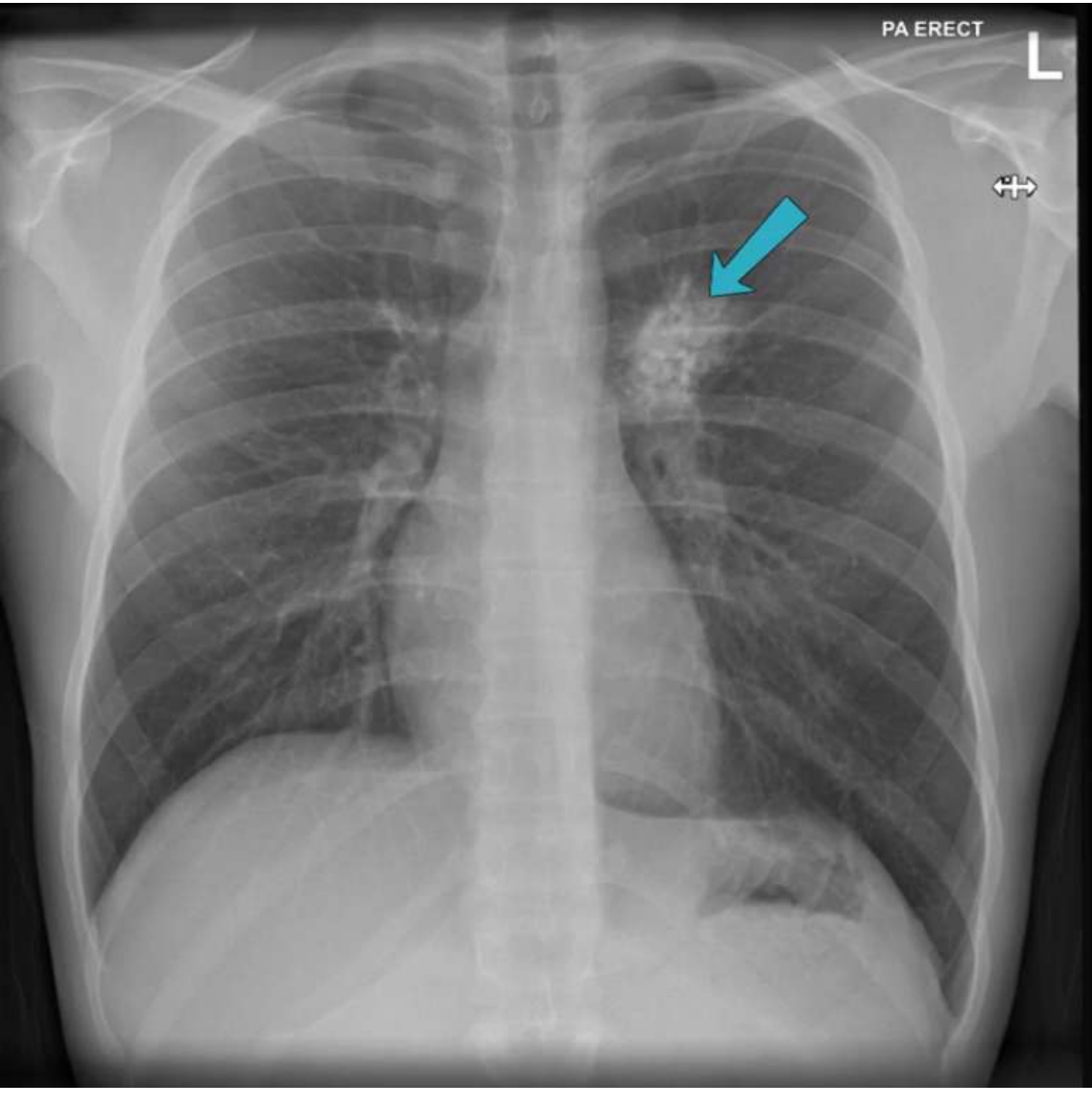


Look for abnormal lesions such as metastases or lung nodules.



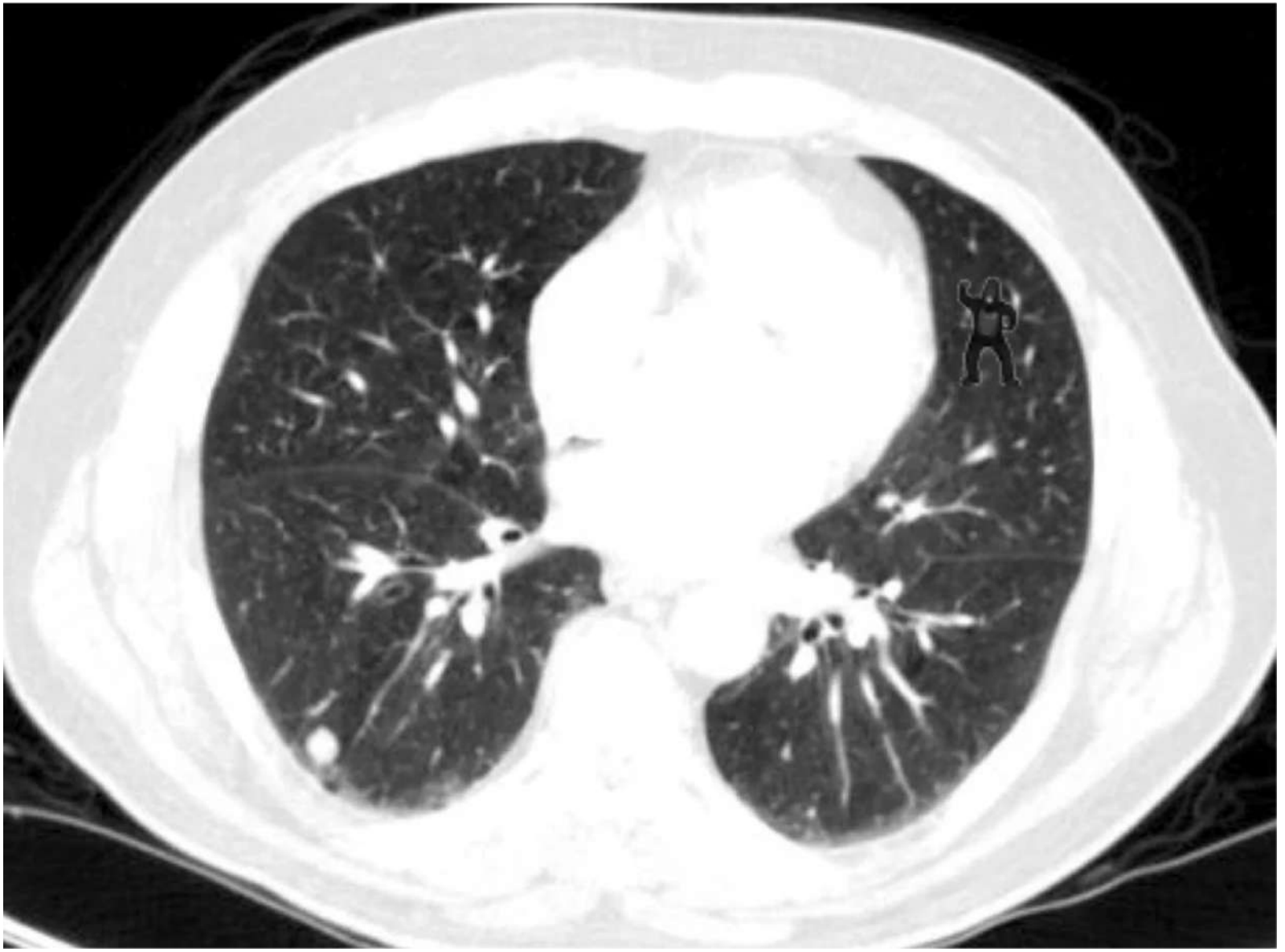


Consolidation or airspace opacification is another common finding and is confirmed by the presence of air bronchograms.

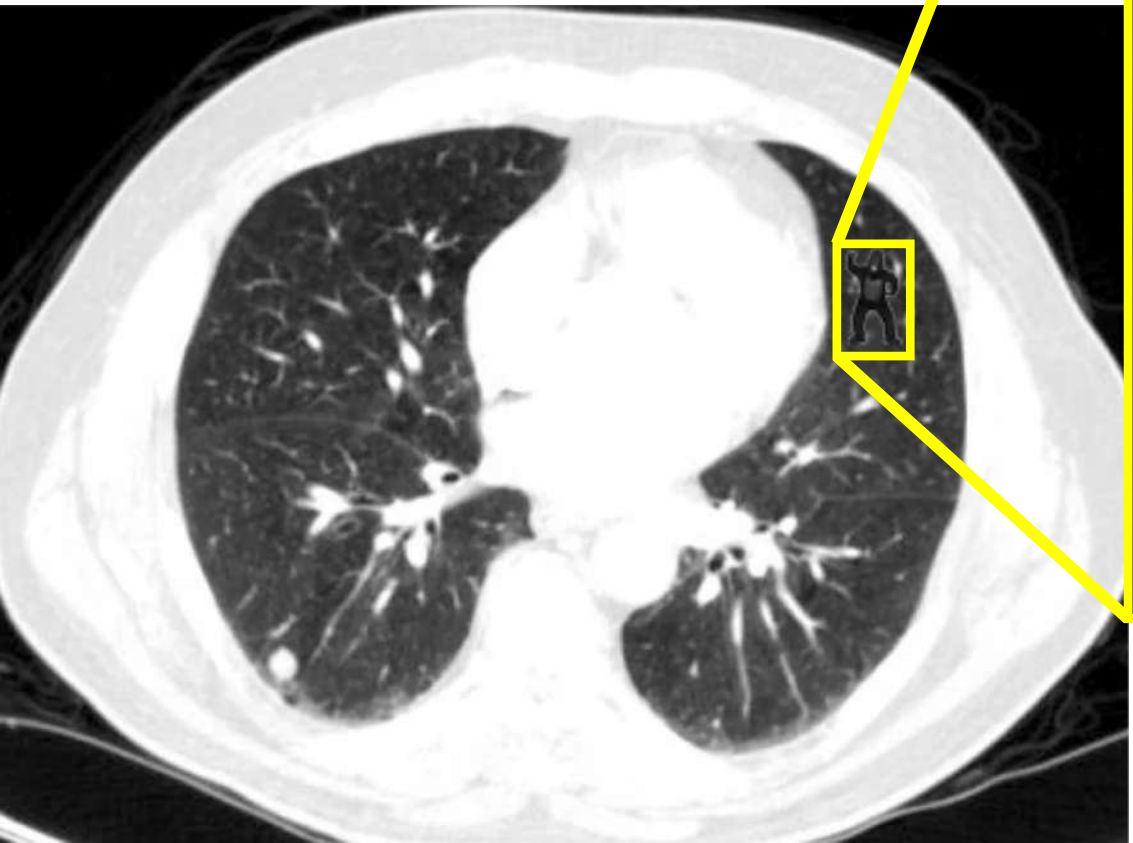




Particular attention should be paid to the hilar regions as malignancy or enlarged lymph nodes can be overlooked in this area.



Cancer Detection in the Lungs MRI





83% of radiologists did not see the gorilla

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3964612/

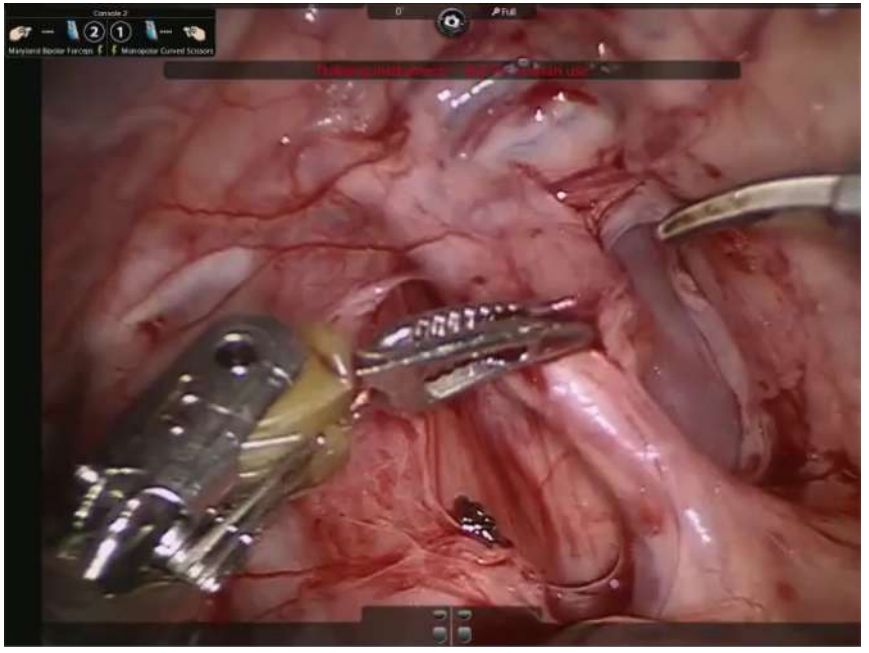
What is Surgery ?

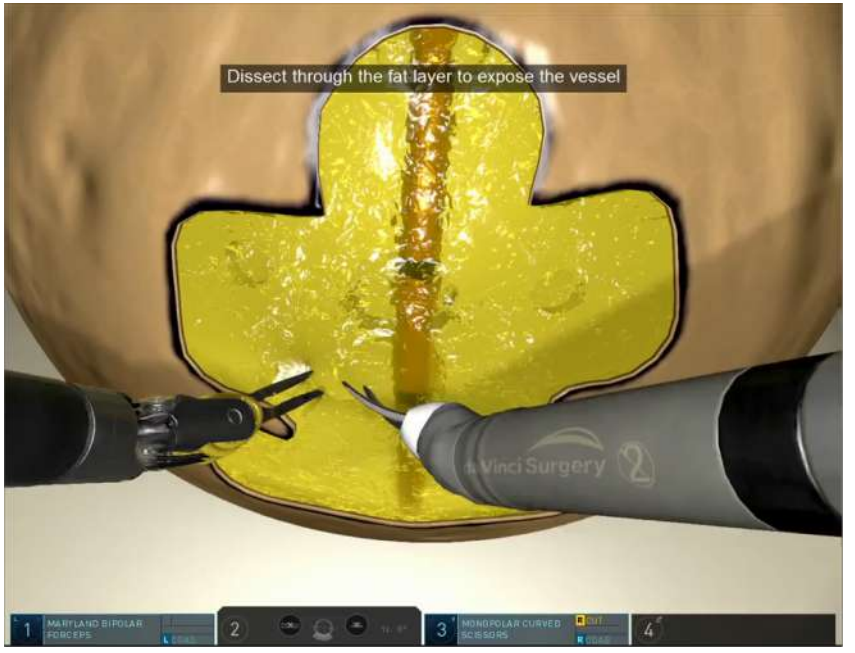
Surgery treats diseases and injuries by removal, repair or adjustment of organs. Patient needs surgery due to the following common reasons:



- Diagnosis
- Remove or repair diseased tissues or organs
- Reposition structure
- Redirect blood vessels
- Transplant tissue or whole organs etc.

Dissection



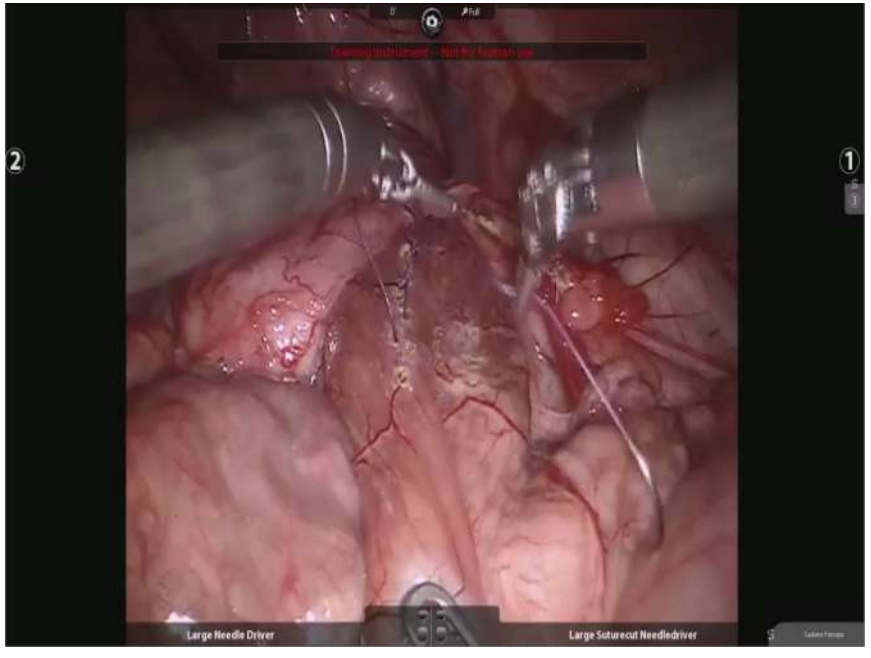


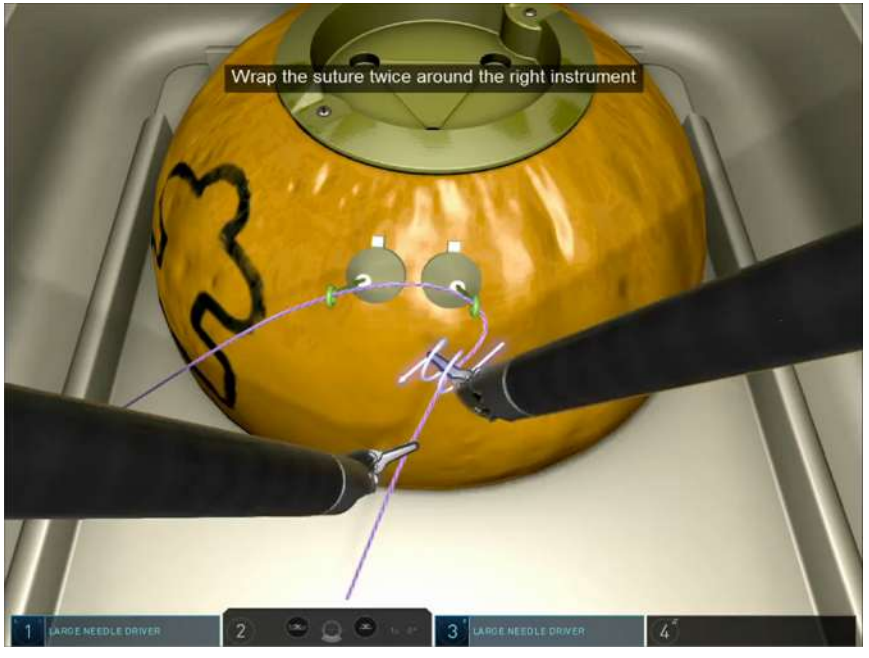
Dissection Actual Surgery Video

Dissection VR Video

https://www.synapse.org/#!Synapse:syn22083820

Knot Tying



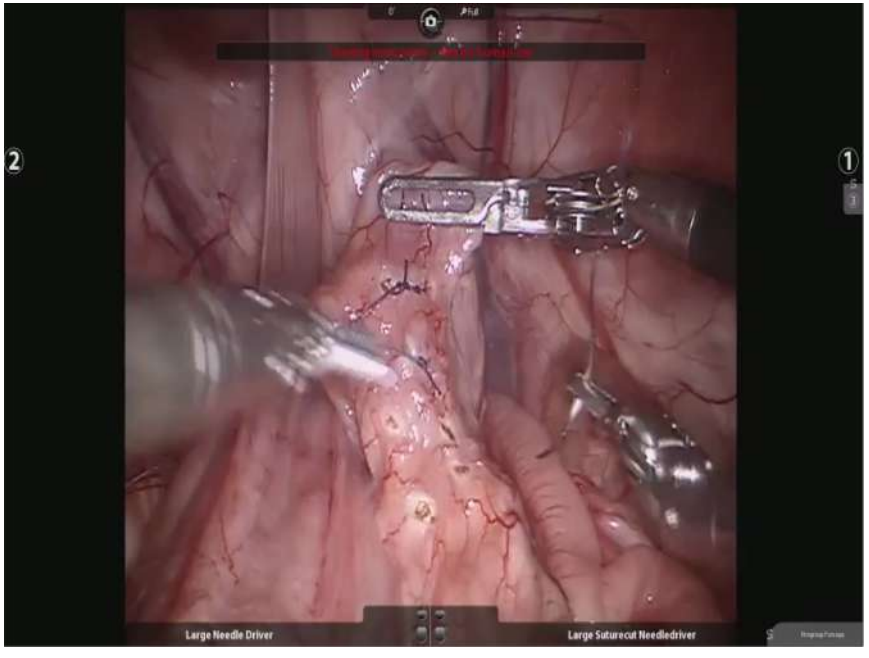


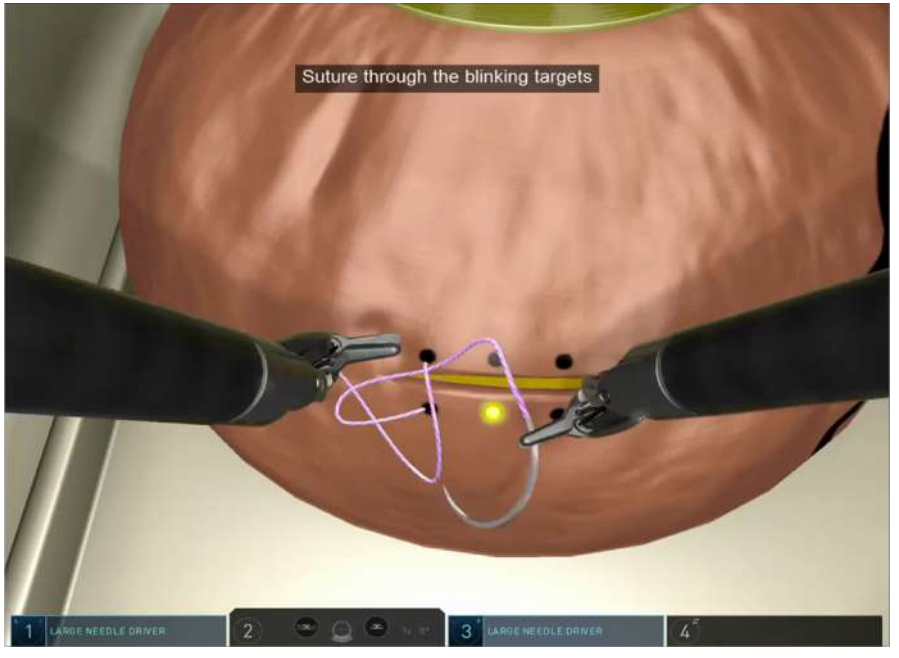
Knot Tying Actual Surgery Video

Knot Tying VR Video

https://www.synapse.org/#!Synapse:syn22083820

Needle Driving





Needle Driving Actual Surgery Video Needle Driving VR Video

https://www.synapse.org/#!Synapse:syn22083820

Surveillance Videos

Datasets

1) Real-world Anomaly Detection Dataset

- Large-scale benchmark dataset
  - 1,900 real-world surveillance videos of 128 hours
  - 15 times more videos than existing datasets.
  - Average video length is over four times

13 real world anomalies



Abuse



Arrest



Arson



Explosion


Robbery


Arrest



Assault



Burglary



Stealing



Fighting



Road Accident



Shooting

Anomaly	Number of videos
Burglary	100
Fighting	50
Road Accidents	150
Robbery	150
Shooting	50
Shoplifting	50
Stealing	100
Abuse	50
Arrest	50
Arson	50
Assault	50
Explosion	50
Vandalism	50
Normal	950

US police testing AI that learns to spot crimes in CCTV footage

TECHNOLOGY 14 August 2018

By Richard Kemeny



Just like humans, machines can be trained to spot illicit activity
MediaProduction/Getty

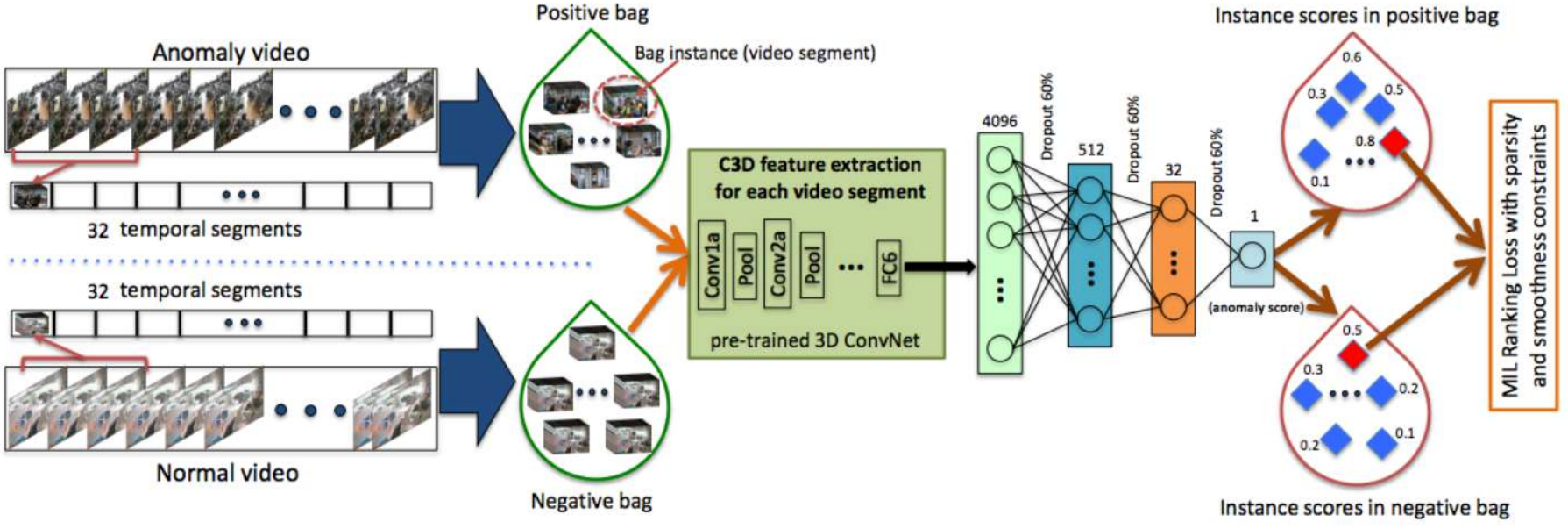
SURVEILLANCE cameras are already ubiquitous, but you still need trained guards to spot crime. Keeping that level of attention up can be hard, even for the most focused individual.

Now police in Orlando, Florida, have been testing a system that automatically scans CCTV looking for potentially illicit activity.

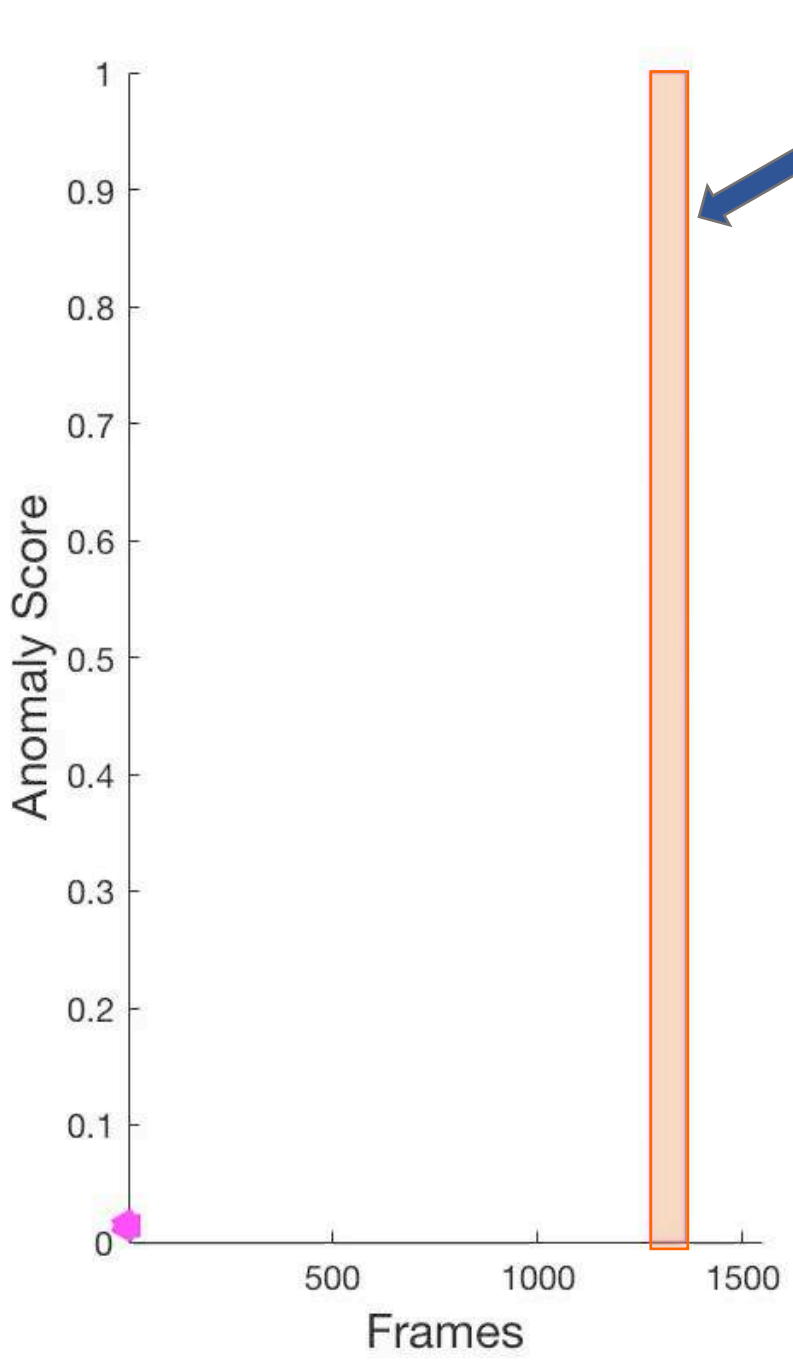
Previous AIs have been trained to spot specific activities, such as violence. Yet as crime comes in many forms, these systems are inherently limited. Waqas Sultani at the Information Technology University in Pakistan and his colleagues tried to incorporate a ...

Comparison with other datasets

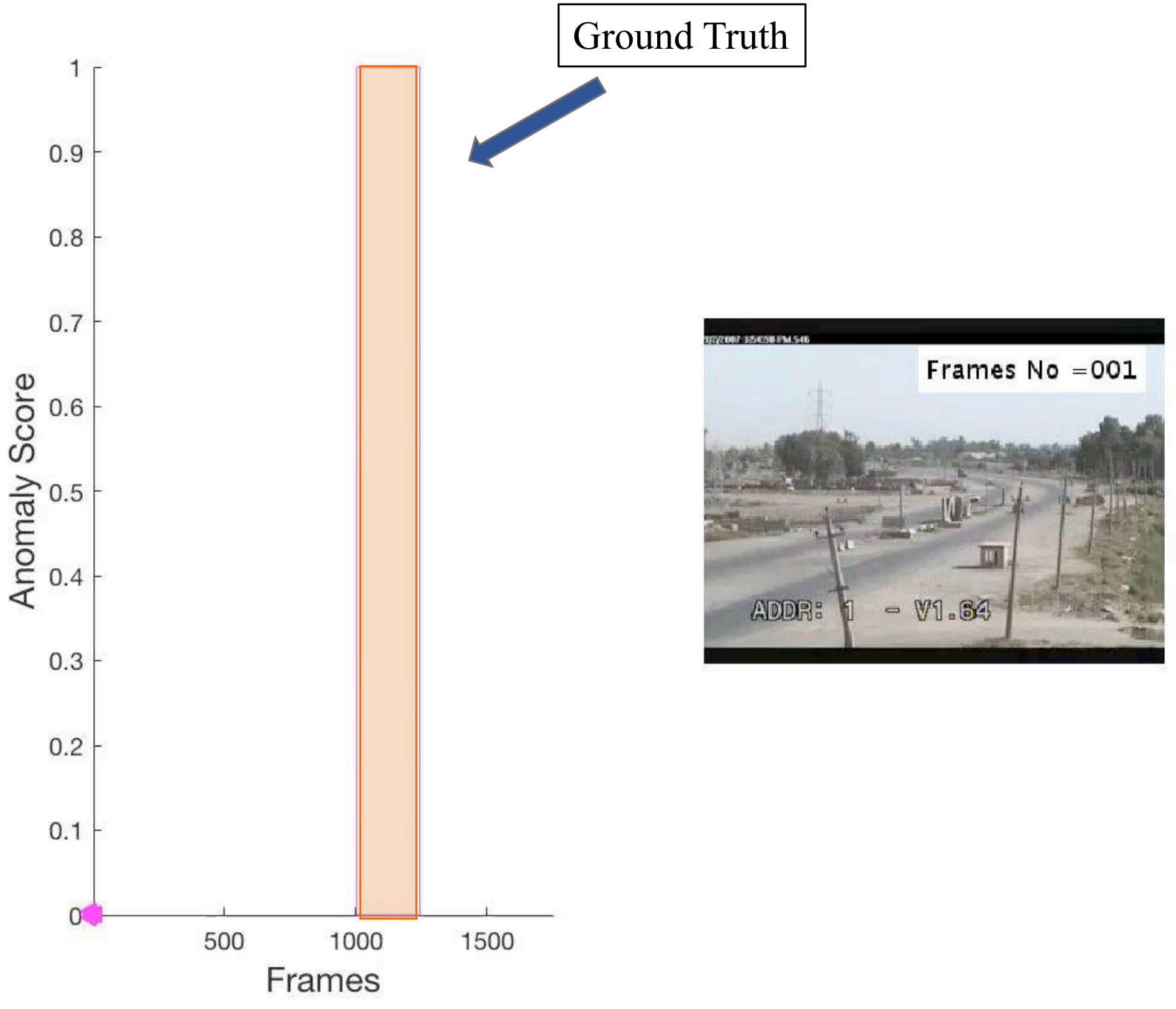
	# of videos	Average # of frames	Dataset length	Example anomalies
UCSD Ped1 [27]	70	201	5 min	Bikers, small carts, walking across walkways
UCSD Ped2 [27]	28	163	5 min	Bikers, small carts, walking across walkways
Subway Entrance [3]	1	121,749	1.5 hours	Wrong direction, No payment
Subwa Exit [3]	1	64,901	1.5 hours	Wrong direction, No payment
Avenue [28]	37	839	30 min	Run, throw, new object
UMN [2]	5	1290	5 min	Run
BOSS [1]	12	4052	27 min	Harass, disease, panic
Abnormal Crowd [31]	31	1408	24 min	Panic, fight, congestion, obstacle, neutral
Ours	**1900**	**7247**	**128 hours**	**Abuse, arrest, arson, assault, accident, burglary, fighting, robbery**

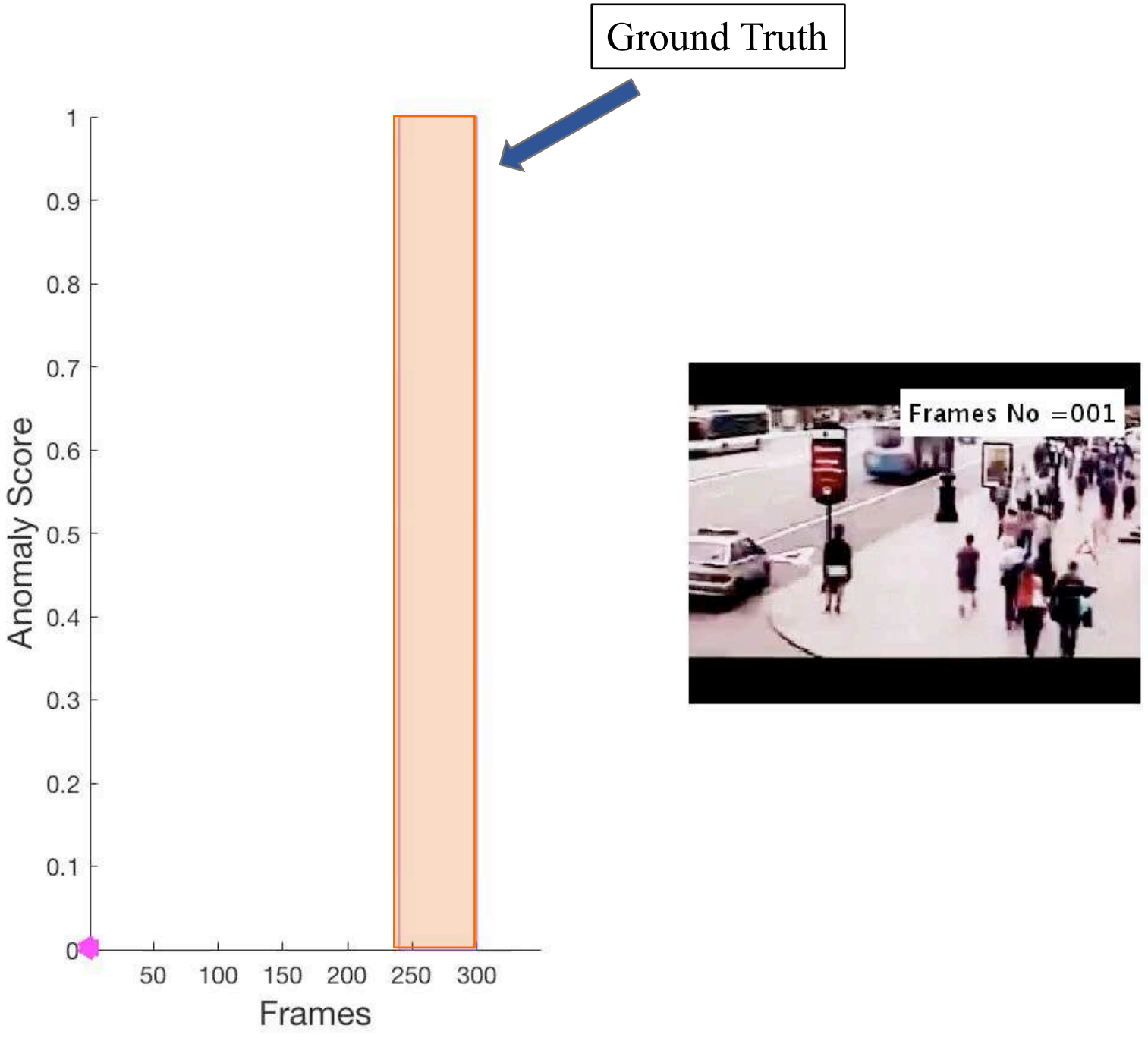

Anomaly video
32 temporal segments
32 temporal segments
Normal video
Positive bag
Bag instance (video segment)
Negative bag
C3D feature extraction for each video segment
Conv1a
Pool
Conv2a
Pool
FC6
pre-trained 3D ConvNet
4096
Dropout 60%
512
Dropout 60%
32
Dropout 60%
1
(anomaly score)
Instance scores in positive bag
0.6
0.3
0.5
0.8
0.1
0.5
0.3
0.2
0.2
0.1
Instance scores in negative bag
MIL Ranking Loss with sparsity and smoothness constraints

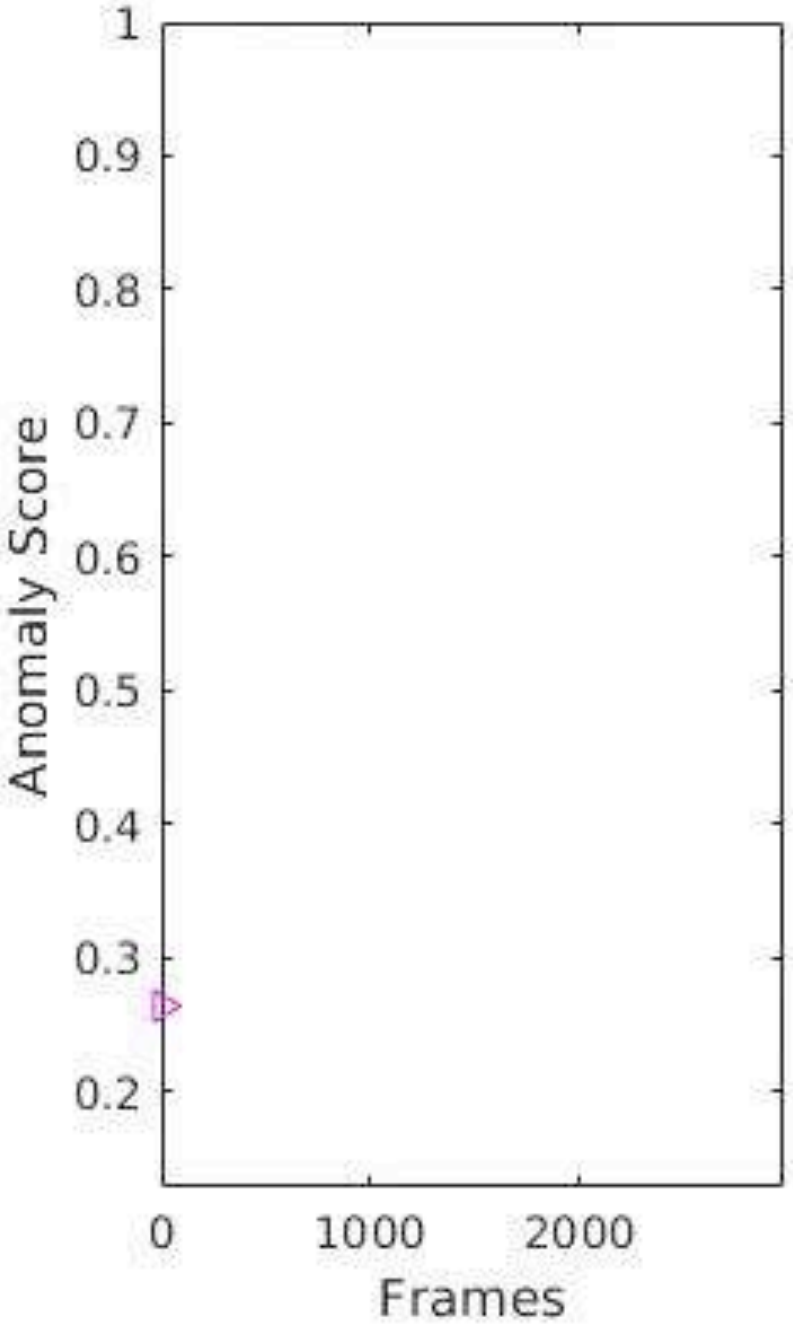
Anomaly Detection Examples of Our Method


Ground Truth
Anomaly Score
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
500
1000
1500
Frames


1:31:51 PM
Frames No =001


Ground Truth
Anomaly Score
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
500
1000
1500
Frames
Frames No =001
ADDR: 1 - V1.64


Ground Truth
Anomaly Score
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
50
100
150
200
250
300
Frames
Frames No =001


1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
Anomaly Score
0
1000
2000
Frames

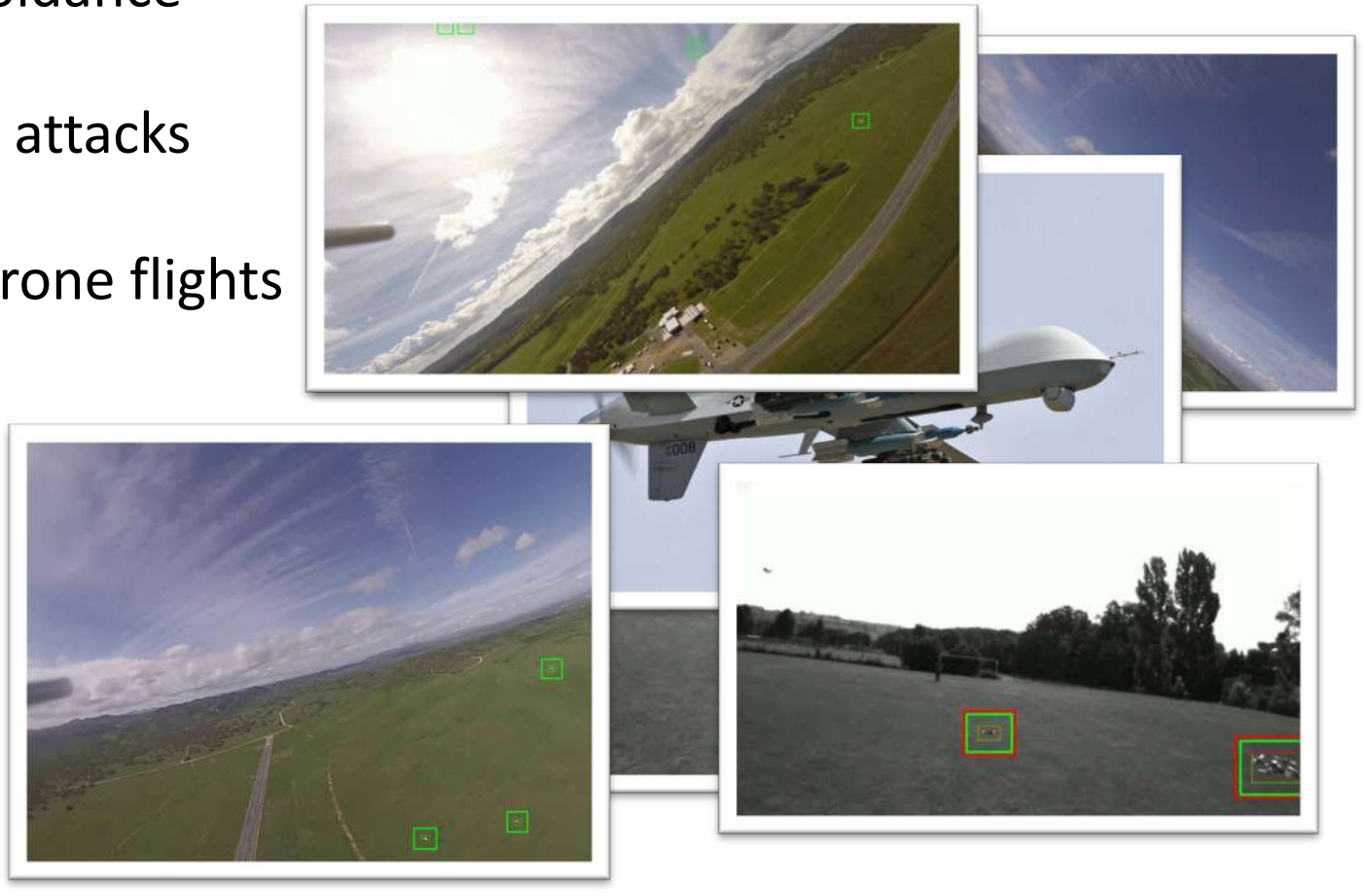

Frame No: = 002
大街上強搶小孩，搶死人！
在中國大陸真要非常小心哦！

UAV Visual Analysis

Motivation

Important to detect drones for:

- Collision Avoidance
- Avoid drone attacks
- Safe multi-drone flights



Problem Statement

Drone detection from other flying drones.

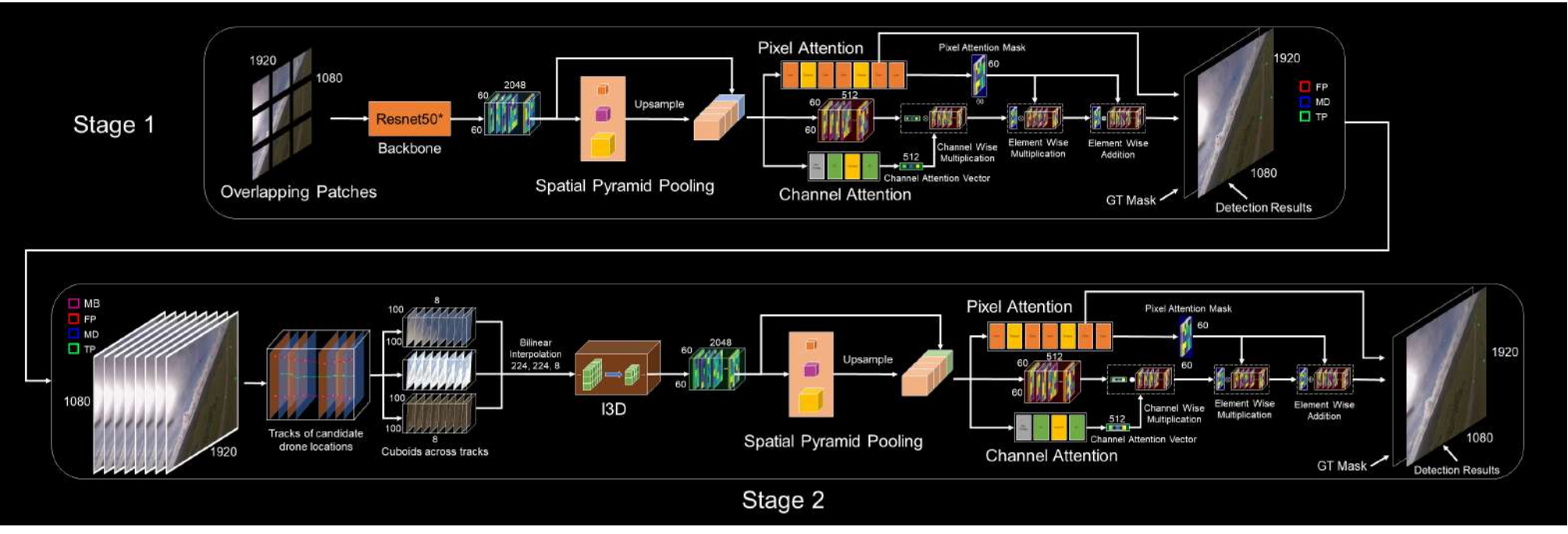
Challenges

- Erratic movement of the source and target drones
- Small size. The average drone size is 0.05% of frame size.
- Arbitrary shape
- Large intensity variations and occlusion

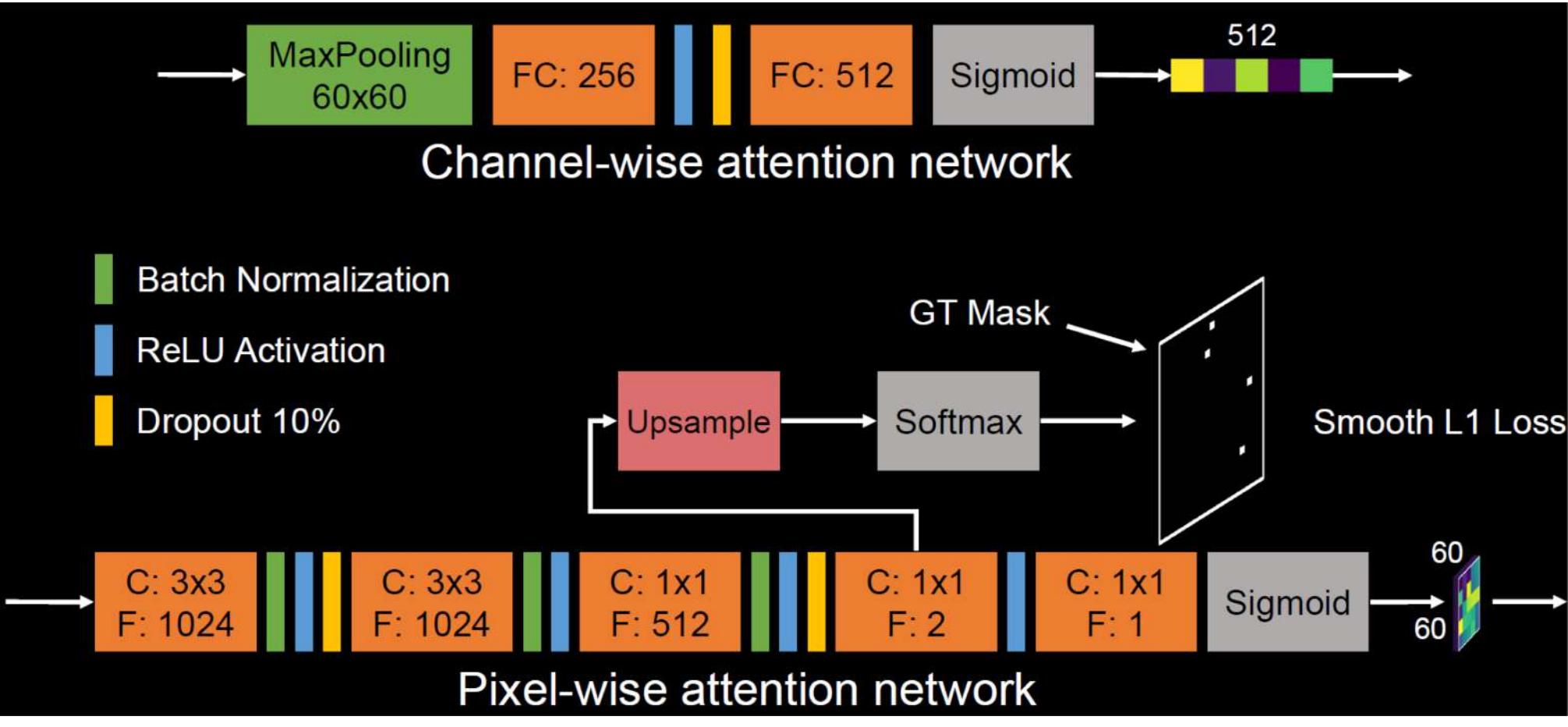





Proposed Approach



Attention networks



Experimental Results



Ground Truth Proposed detections

Bigger boxes are shown for better visualization






Ground Truth Proposed detections

Bigger boxes are shown for better visualization






Ground Truth Proposed detections

Bigger boxes are shown for better visualization





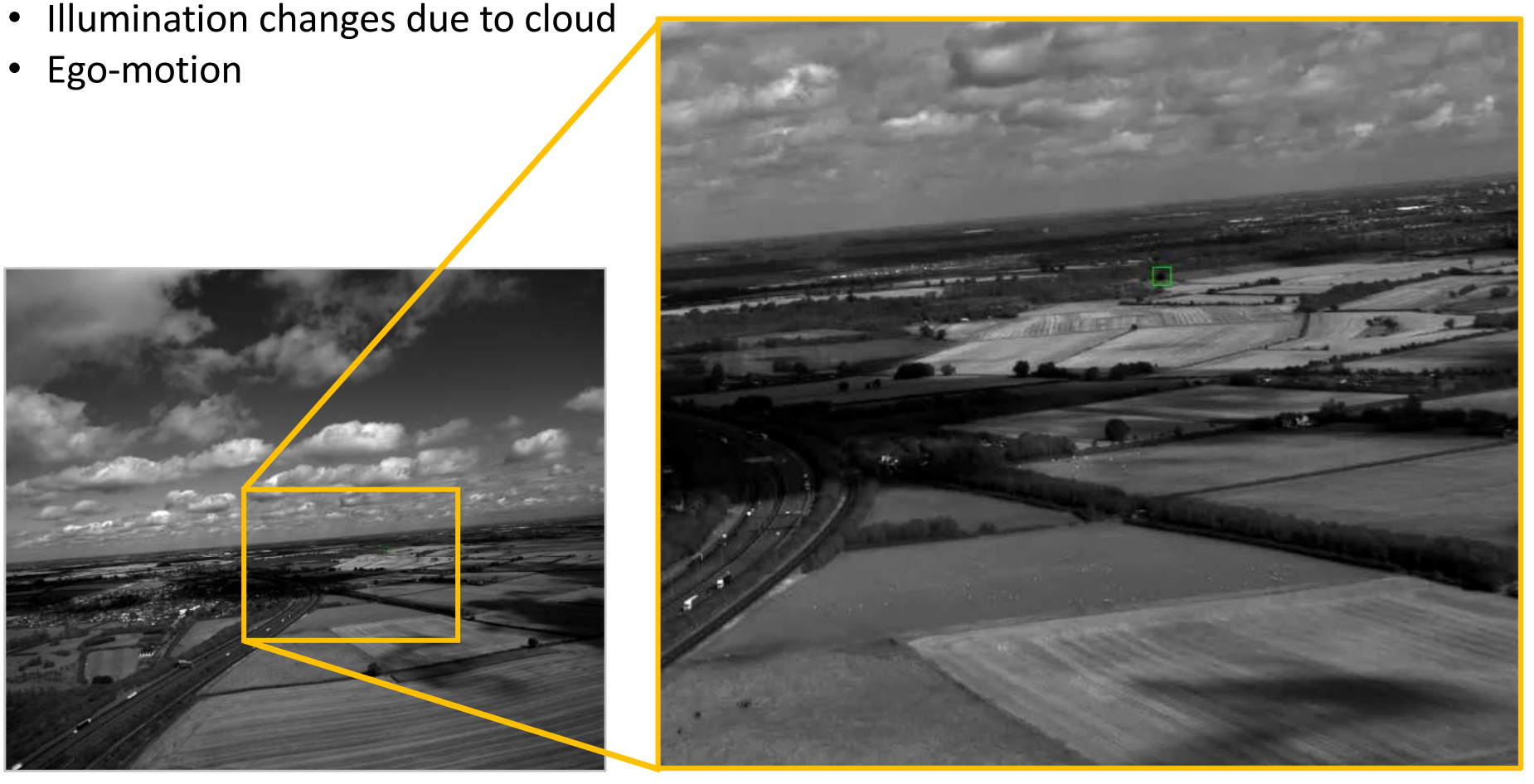

Ground Truth Proposed detections

Bigger boxes are shown for better visualization

- Urban Area
- Diverse background objects: building roof, ground objects



- Moving ground vehicles
- Illumination changes due to cloud
- Ego-motion



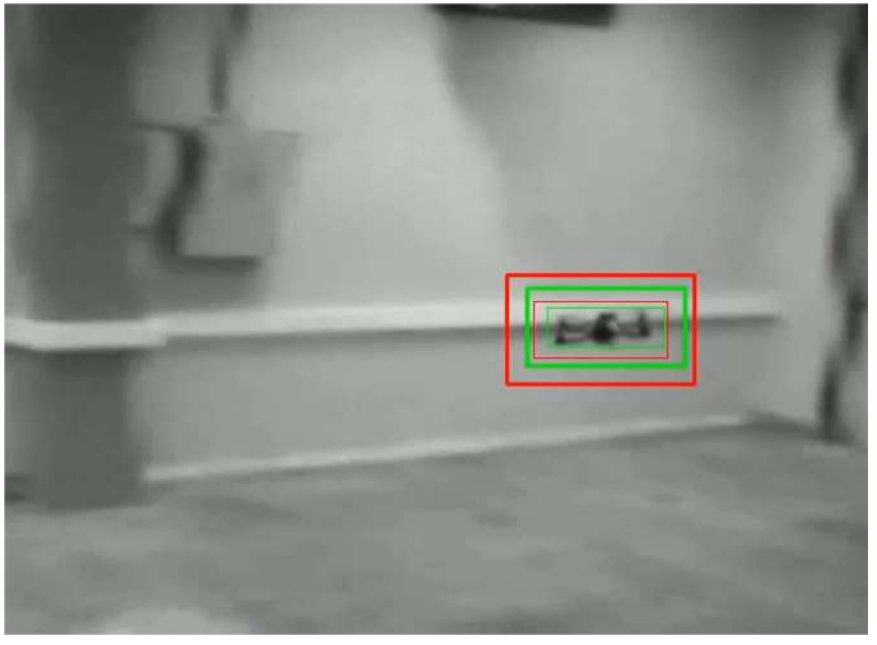
- Low visibility
- High range detection (>1 km)
- Stable detection while moving close to camera

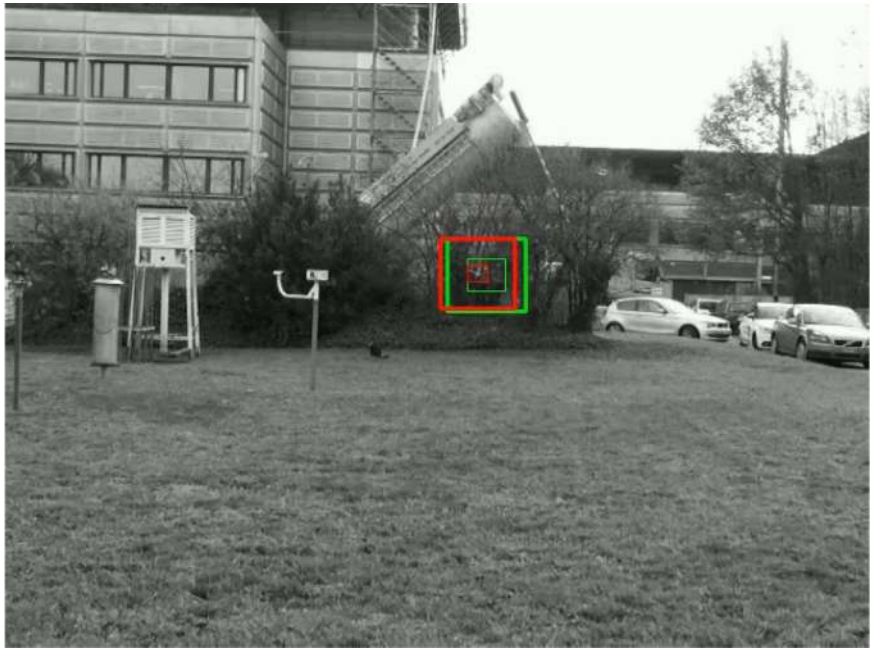
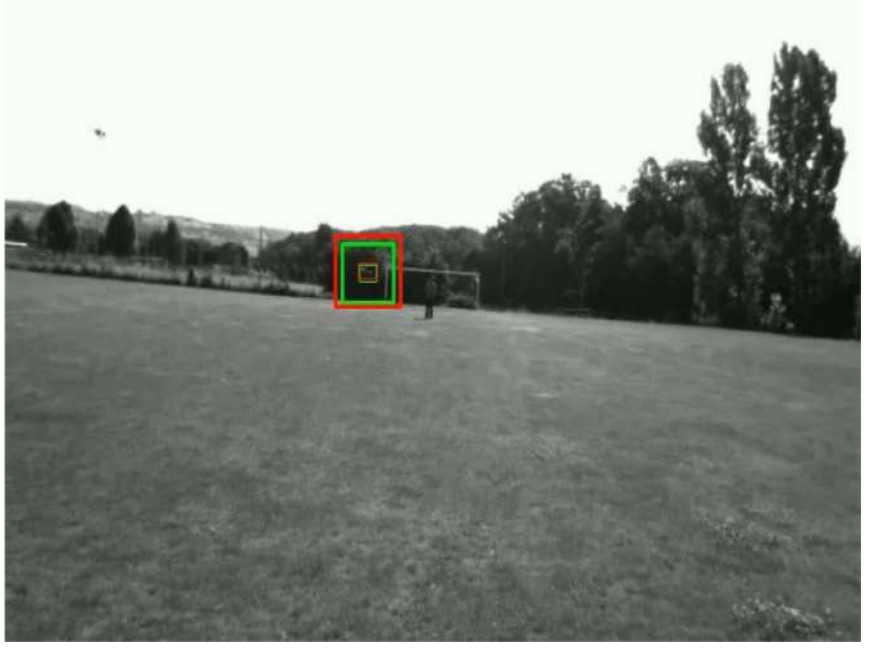
Predictions

Ground-Truth



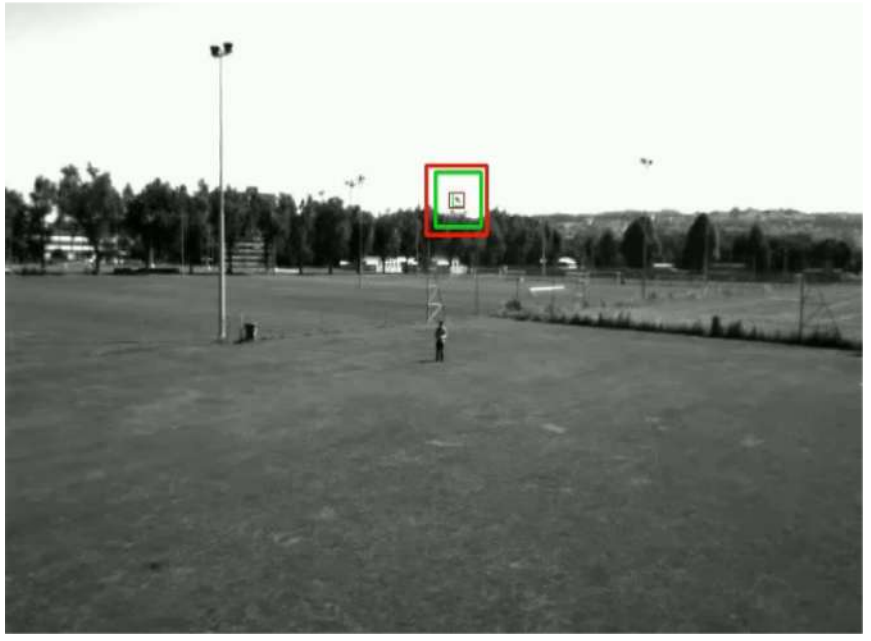
FL-Drones Dataset



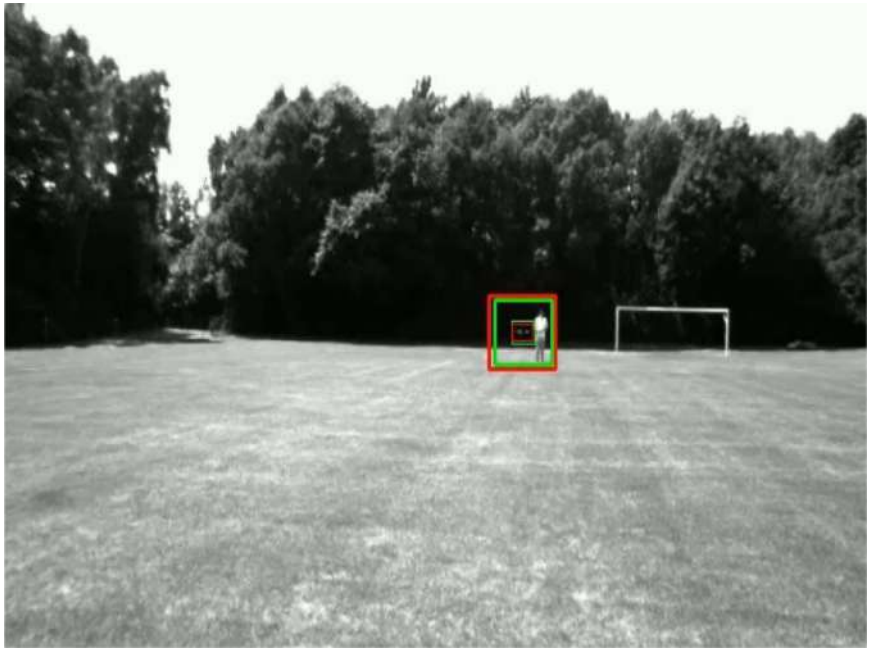



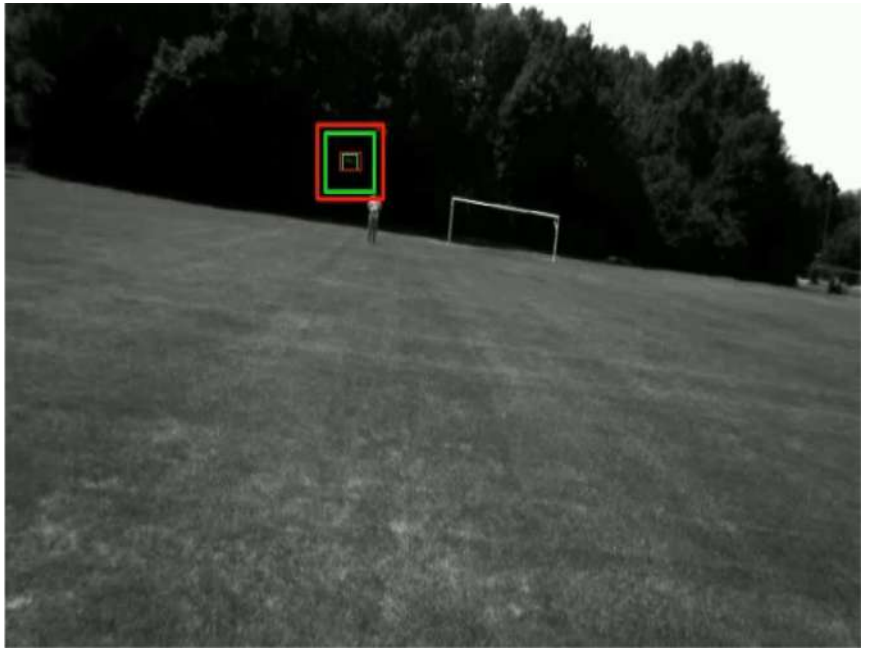
Ground Truth Proposed detections

Bigger boxes are shown for better visualization





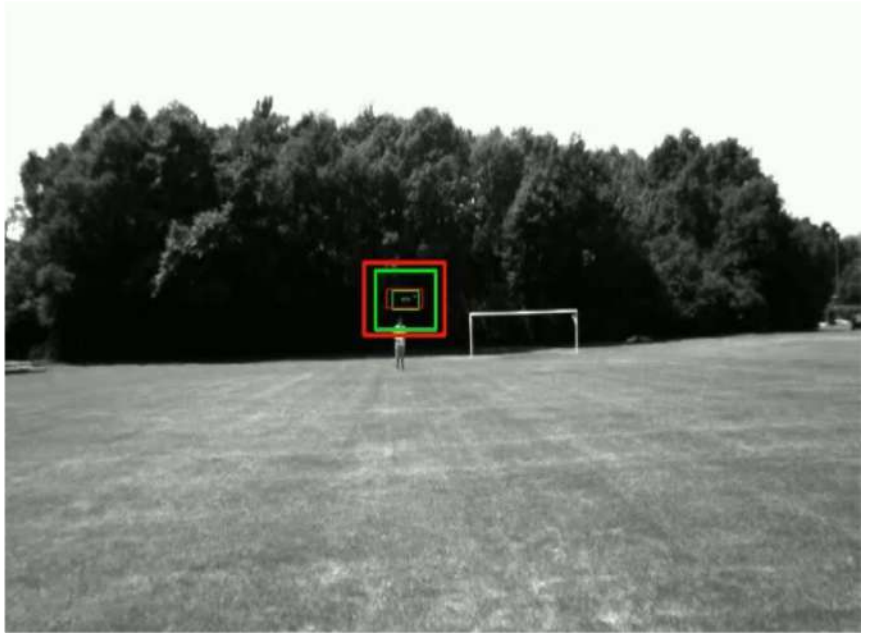


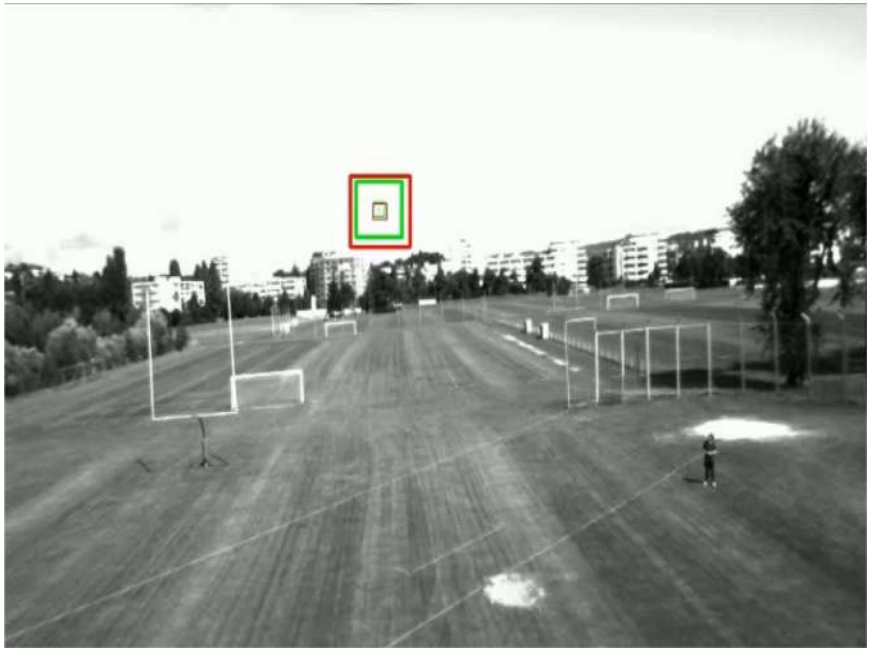


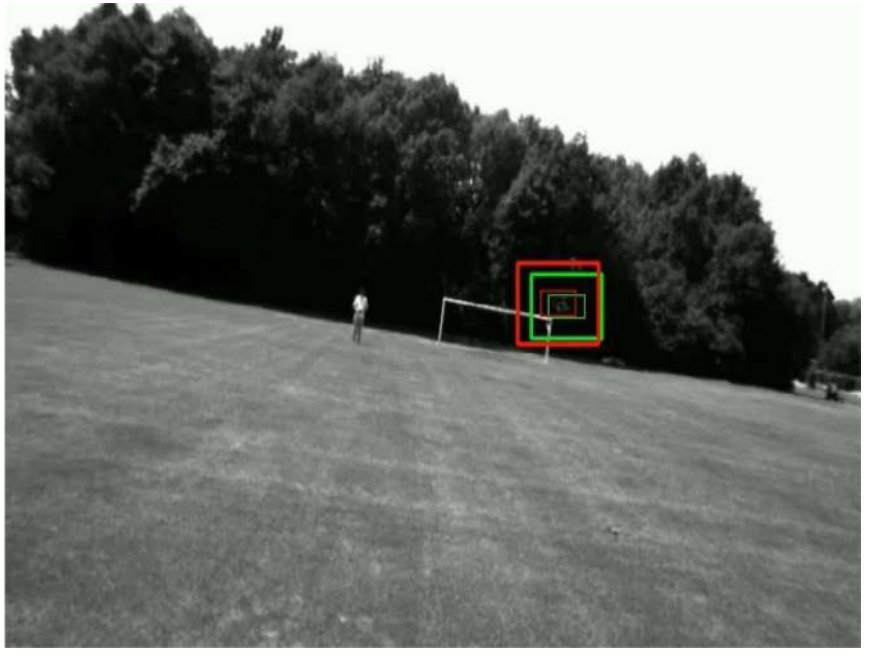
Ground Truth Proposed detections

Bigger boxes are shown for better visualization









Ground Truth Proposed detections

Bigger boxes are shown for better visualization

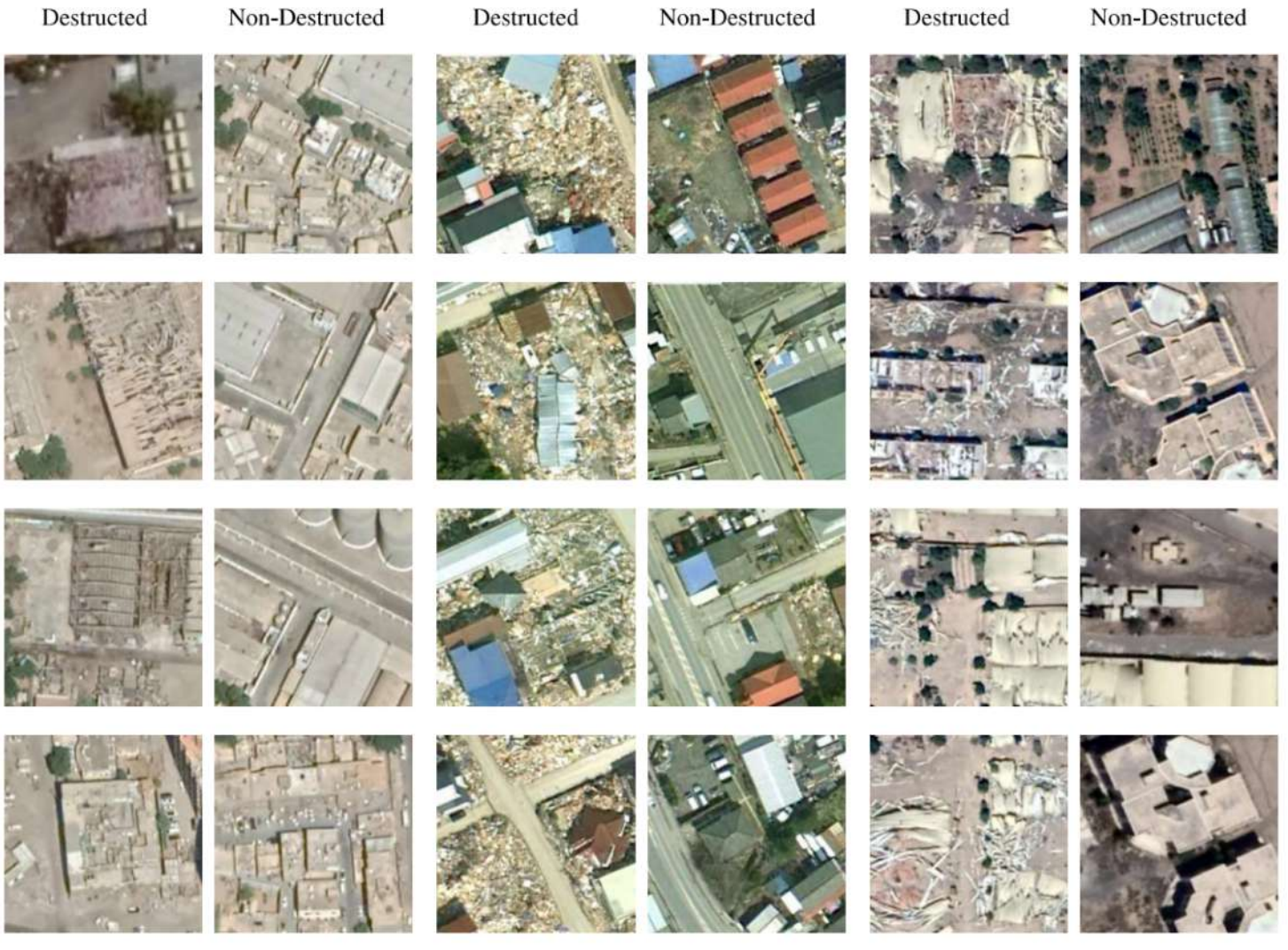


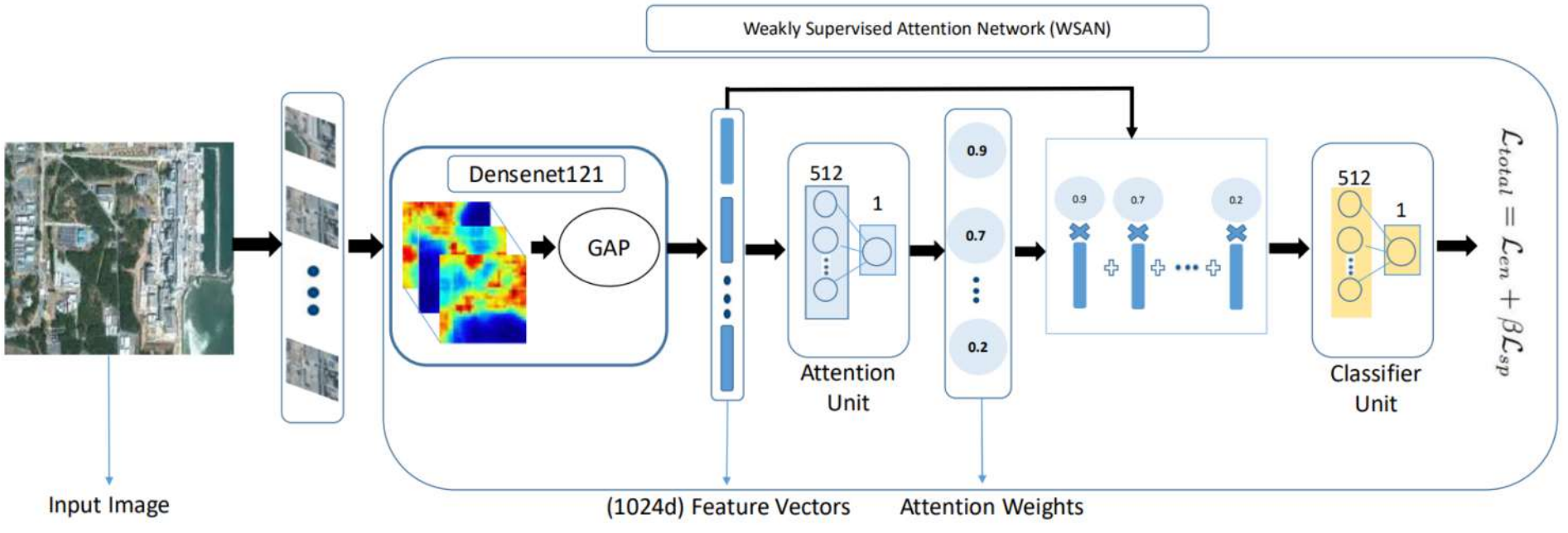
Ground Truth Proposed detections

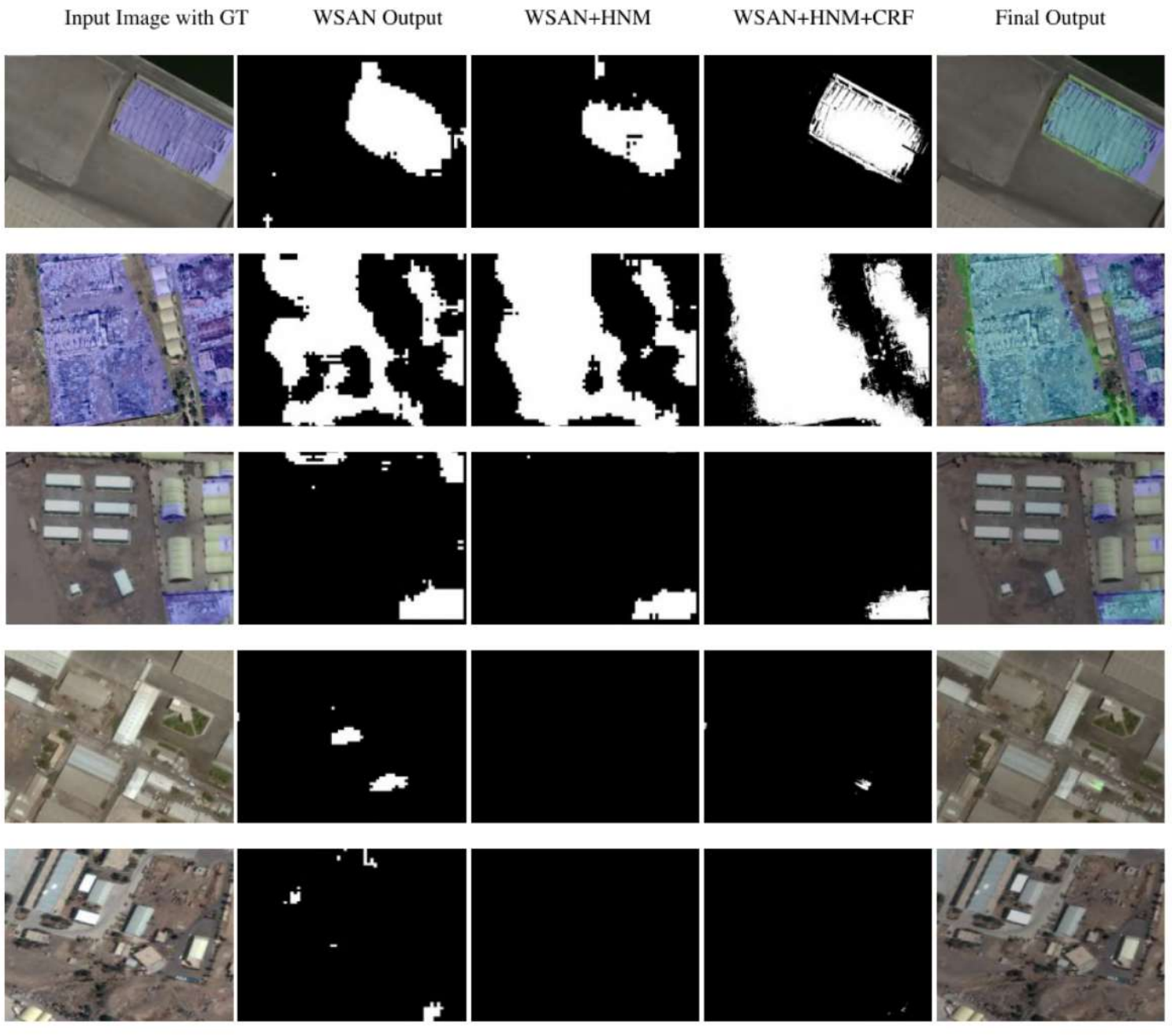
Bigger boxes are shown for better visualization

Remote Sensing

Detecting Destruction from Sky

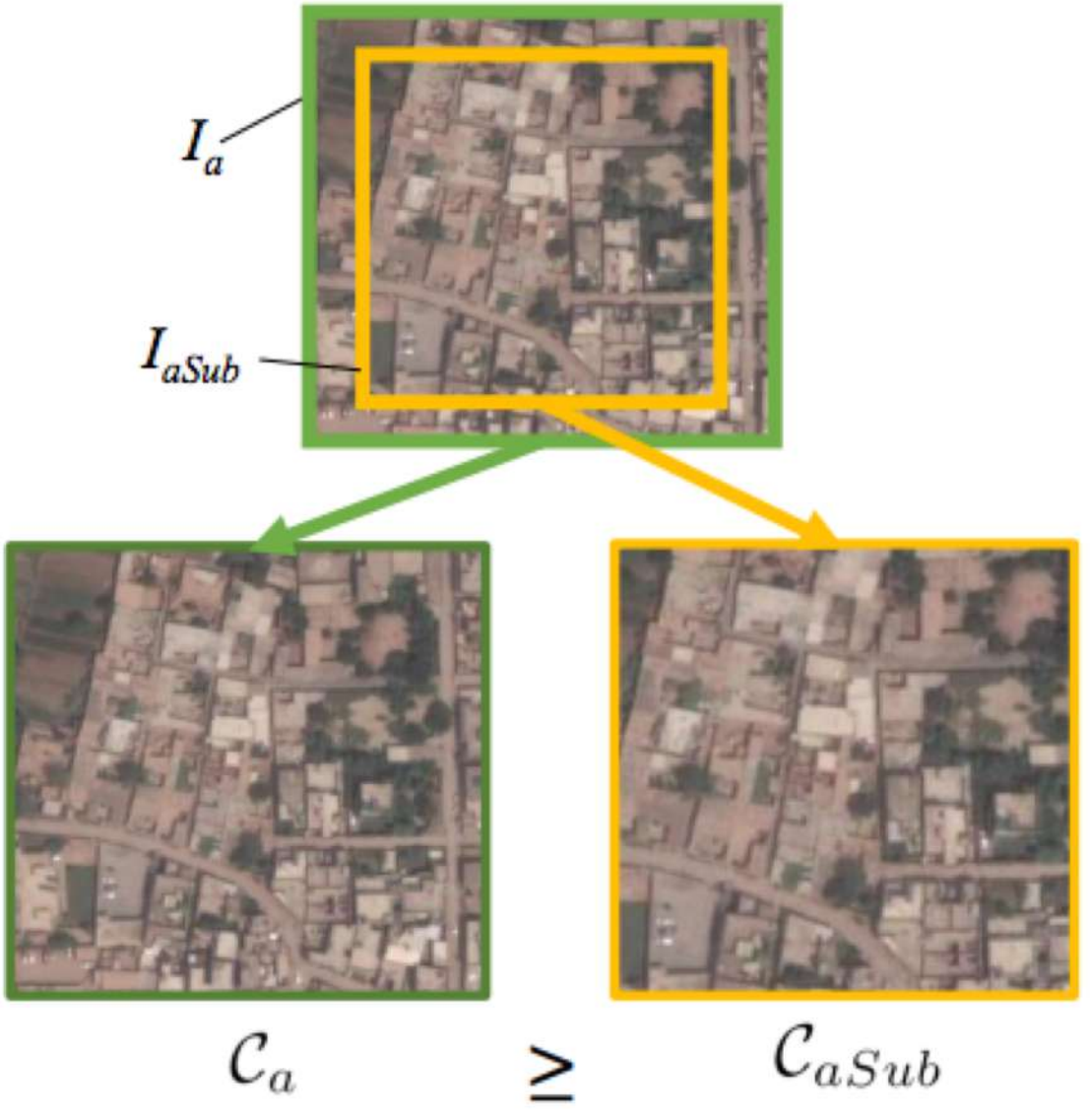


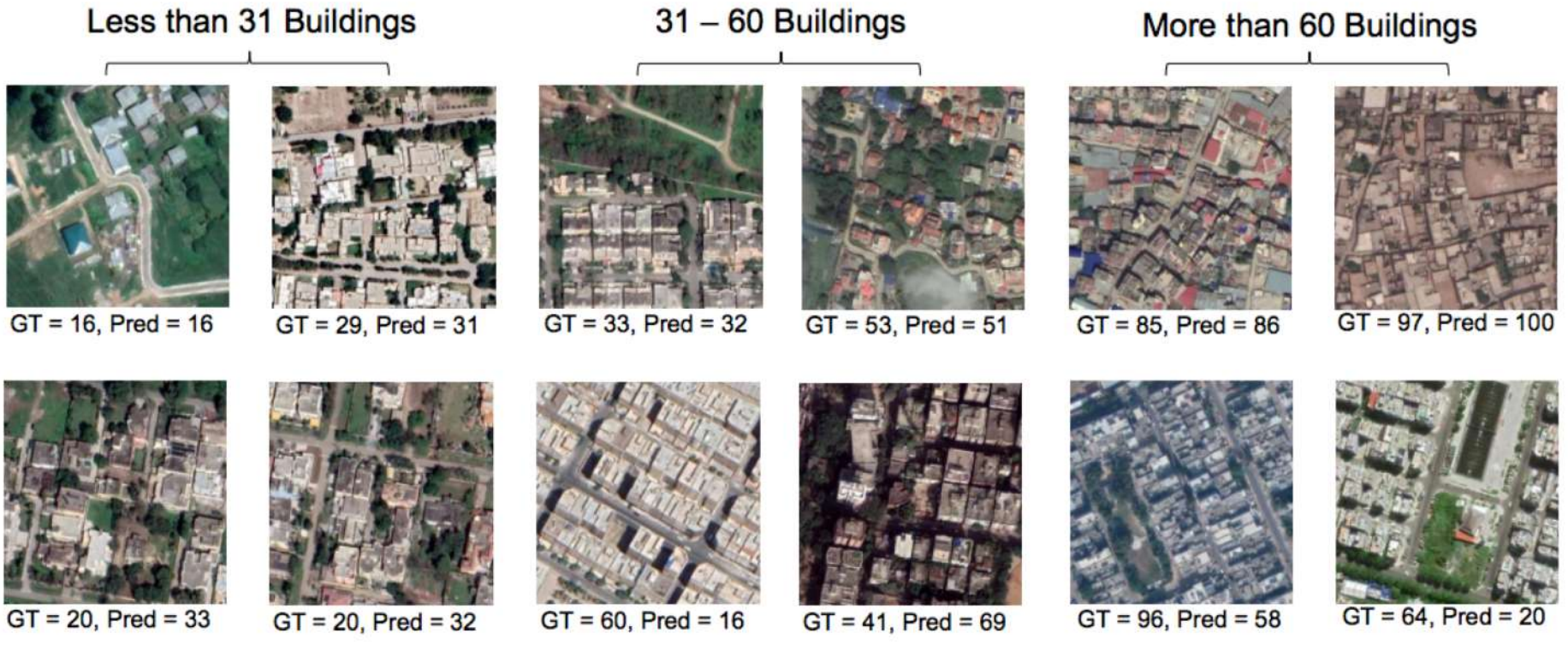

Weakly Supervised Attention Network (WSAN)
Densenet121
GAP
512
1
Attention
Unit
0.9
0.7
0.2
512
1
Classifier
Unit
$\mathcal{L}_{total} = \mathcal{L}_{en} + \beta\mathcal{L}_{sp}$
Input Image
(1024d) Feature Vectors
Attention Weights


Input Image with GT
WSAN Output
WSAN+HNM
WSAN+HNM+CRF
Final Output

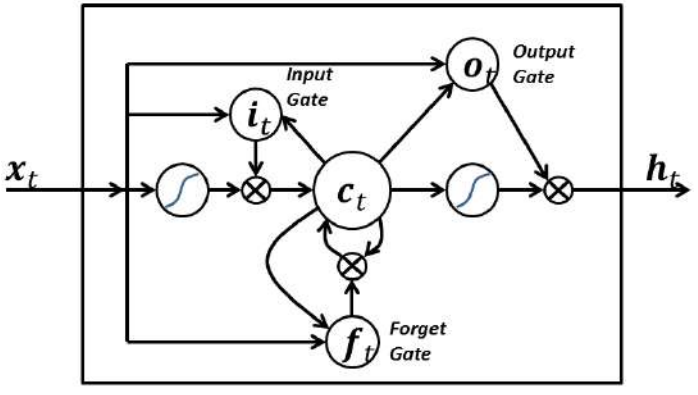
Cross-Region Building Counting in Satellite Imagery using Counting Consistency

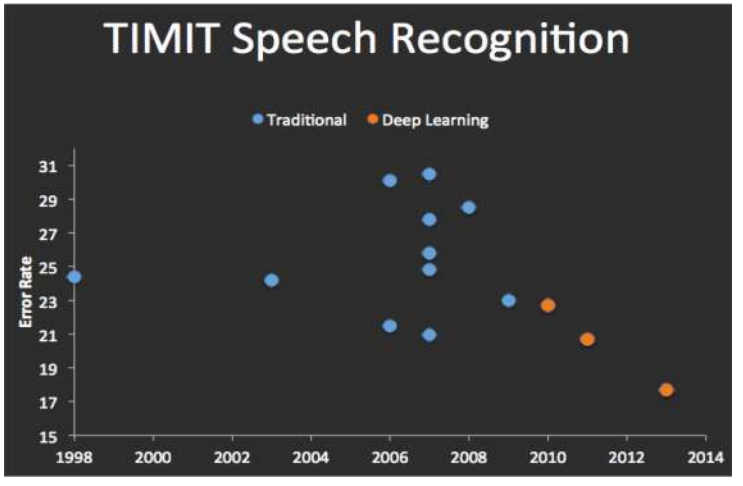

Sloped Roofs
Flat Roofs

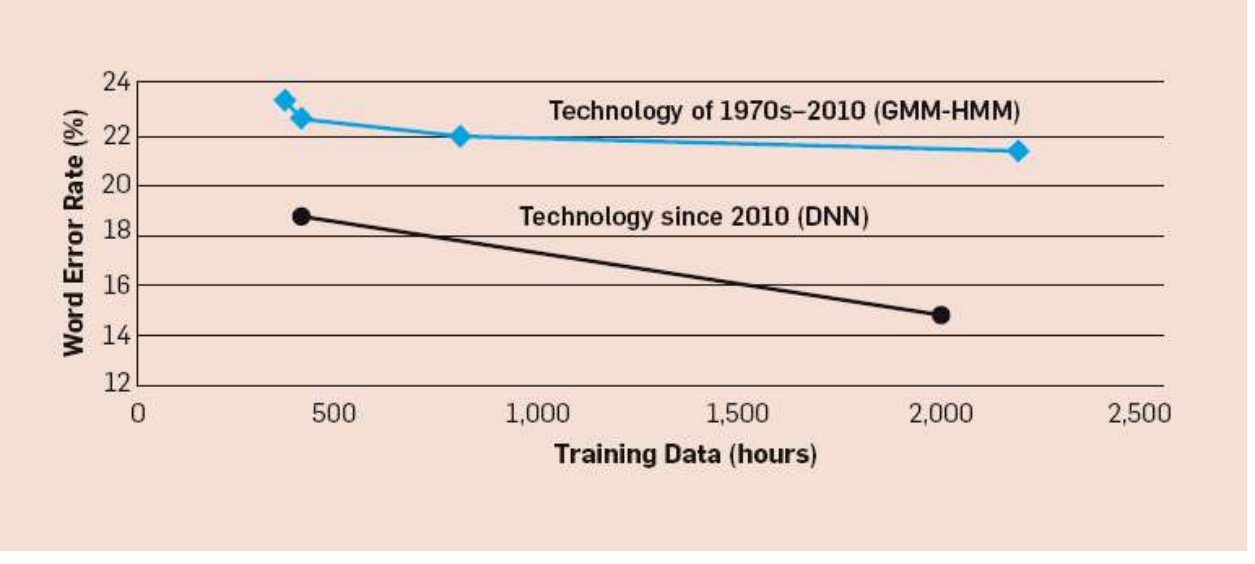

I_a
I_{aSub}
C_a
$\geq$
C_{aSub}


Less than 31 Buildings
31 – 60 Buildings
More than 60 Buildings
GT = 16, Pred = 16
GT = 29, Pred = 31
GT = 33, Pred = 32
GT = 53, Pred = 51
GT = 85, Pred = 86
GT = 97, Pred = 100
GT = 20, Pred = 33
GT = 20, Pred = 32
GT = 60, Pred = 16
GT = 41, Pred = 69
GT = 96, Pred = 58
GT = 64, Pred = 20

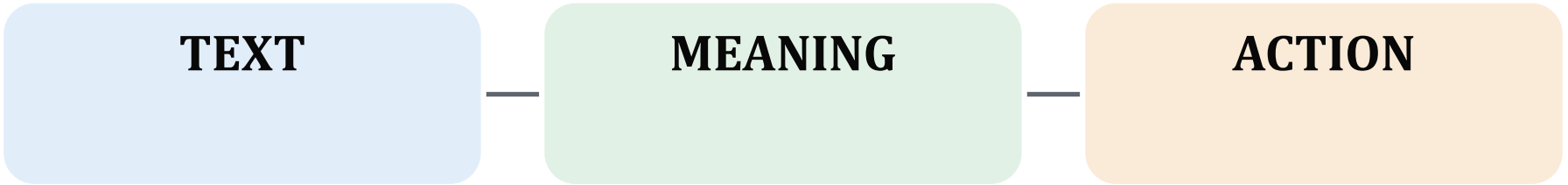
Recurrent Nets: LSTMs (1997):





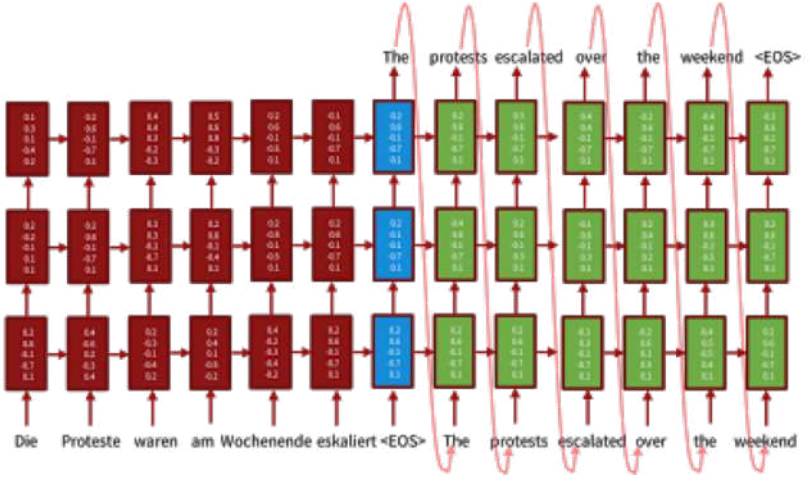


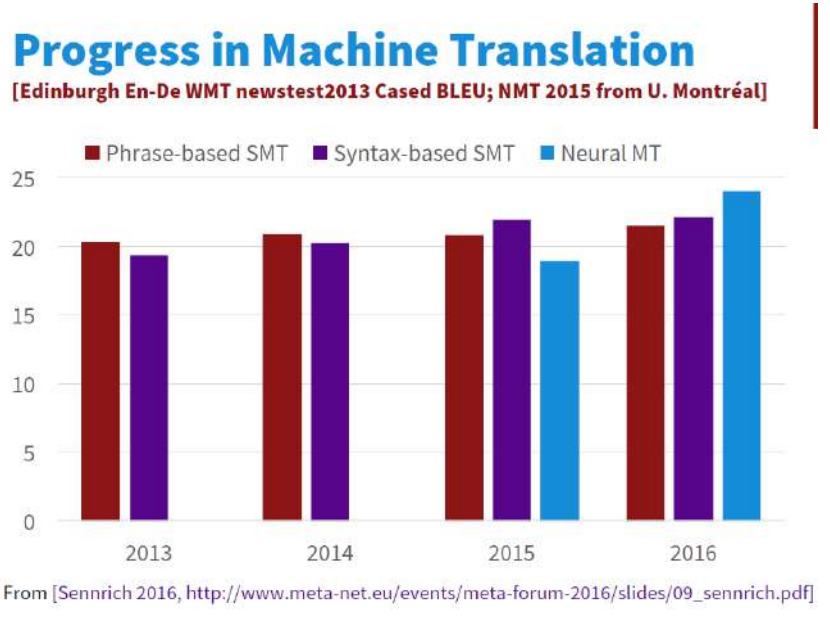
Deep Learning Meets Language



How can a computer understand, classify, translate, answer questions, and generate human language?

Sequence-to-sequence models with LSTMs and attention:





Source Luong, Cho, Manning ACL Tutorial 2016.

Where Do We Use NLP?

Search	Chatbots	Translation	Email	Reviews
Relevant results	Answer questions	English → Urdu	Spam detection	Sentiment

Summarization	Question Answering	Generation	Healthcare	Legal
Long → short	Find answers	Write text	Clinical info	Document search

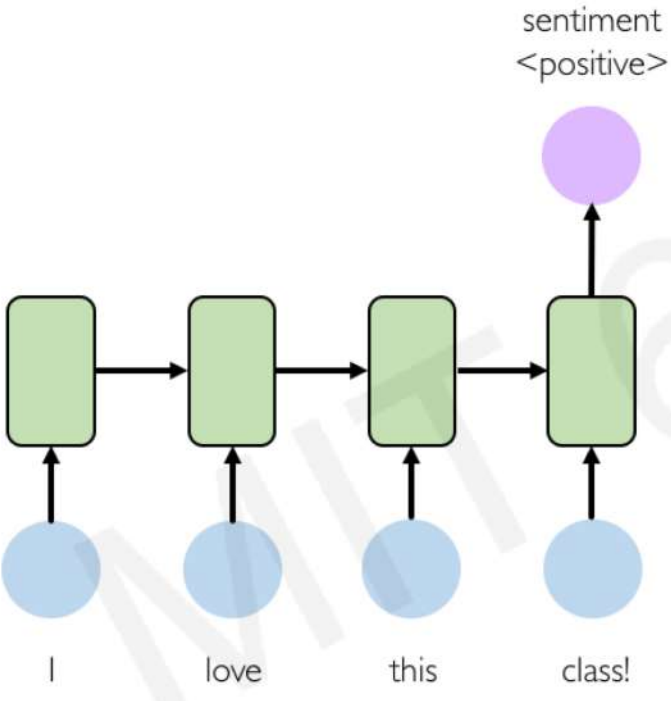
Application 1 — Sentiment Analysis



- Product reviews
- Customer feedback
- Social media monitoring

Input → Text **Output → Sentiment**

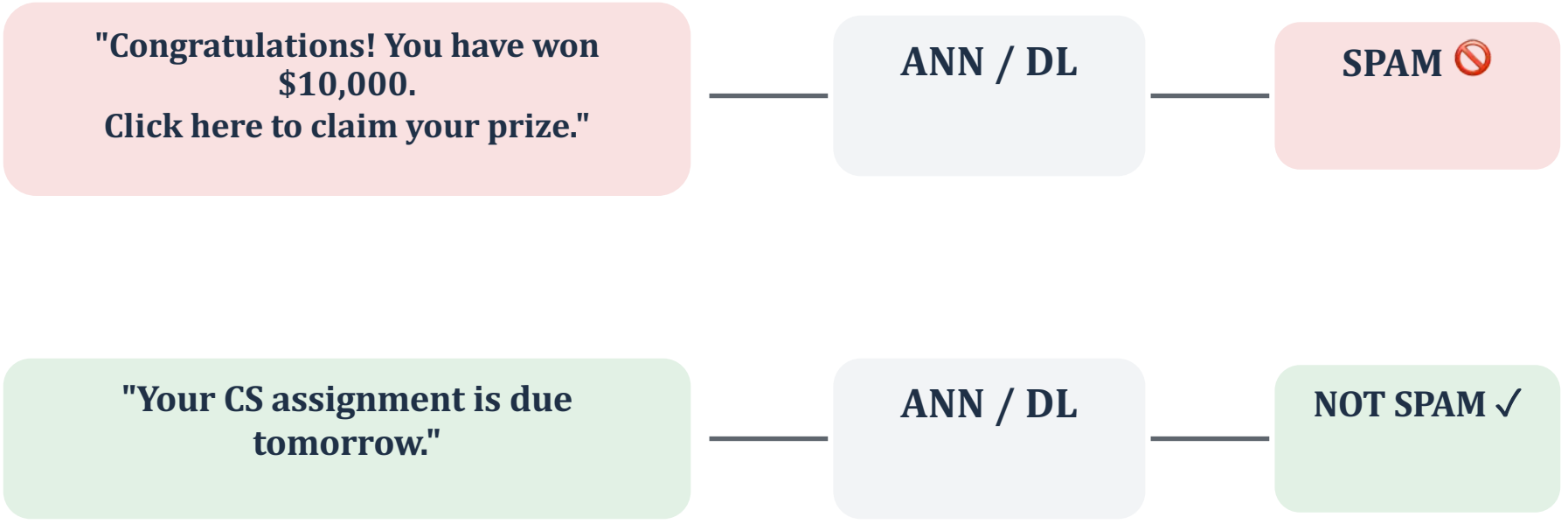
Example Task: Sentiment Classification



Tweet sentiment classification



Application 2 — Spam Detection



Application 3 — Machine Translation



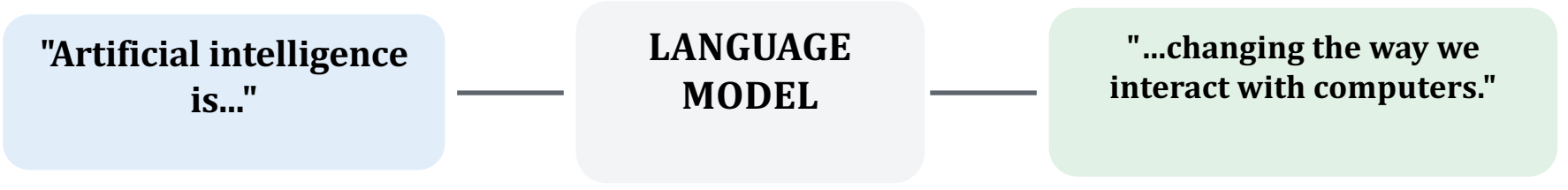
English → Urdu | English → Chinese | English → French

Application 4 — Question Answering

Pakistan's national animal is the markhor. It is a wild goat found in mountainous regions.



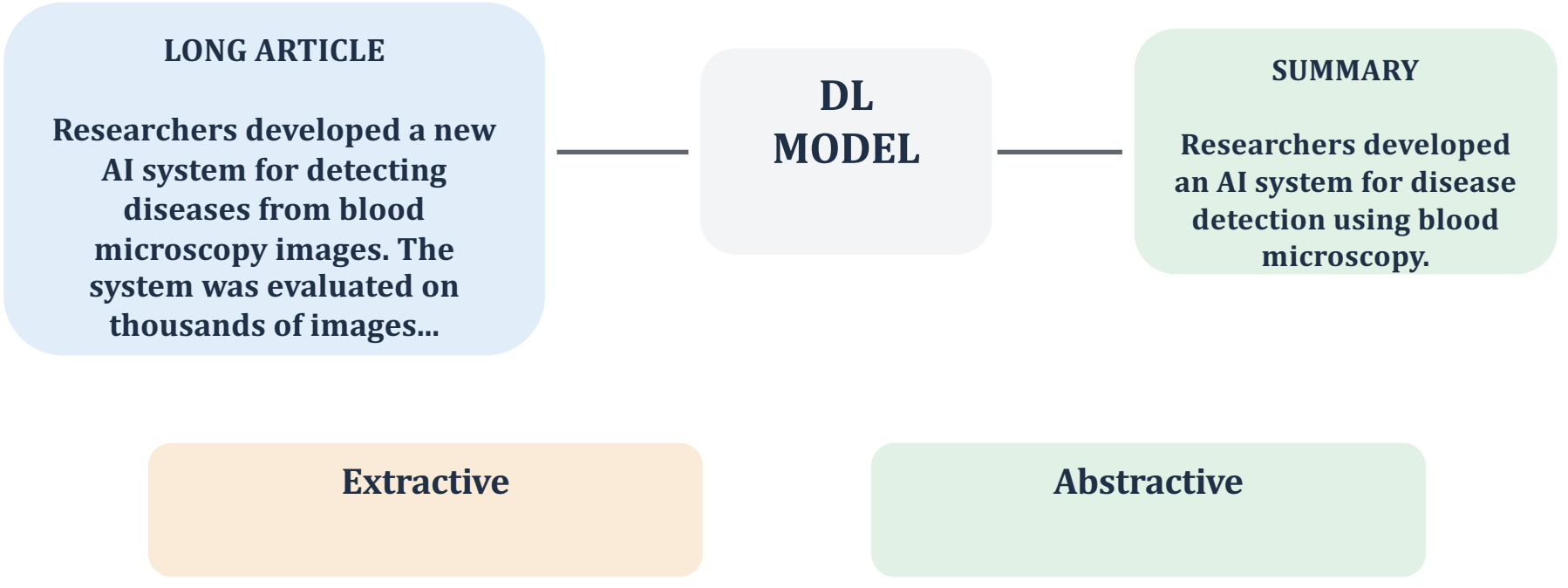
Application 5 — Text Generation



- Complete a sentence
- Write an email
- Generate a story
- Answer a prompt

Modern example: Large Language Models

Application 6 — Text Summarization



Application 7 — Information Extraction

Named Entity Recognition (NER)

"Dr. Ali joined ITU in 2024 and works in Lahore."

Dr. Ali	ITU	2024	Lahore
PERSON	ORGANIZATION	DATE	LOCATION

Useful in search, healthcare, legal & news applications

Application 8 — Intent Classification

"I want to cancel my flight."	———	CANCEL_FLIGHT
"Where is my order?"	———	ORDER_TRACKING
"I forgot my password."	———	PASSWORD_RESET
"What is your refund policy?"	———	REFUND_POLICY

One Sentence — Many Deep Learning Tasks

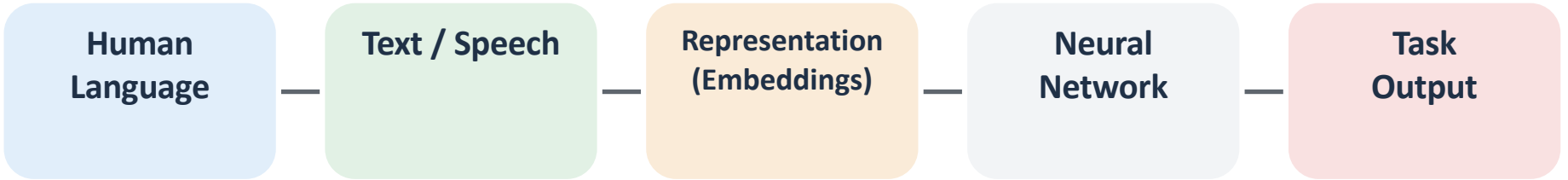
"I loved the new iPhone, but its battery died after only three hours."

Sentiment	Entity	Topic	QA
Mixed / negative	iPhone → Product	Smartphones	Battery died

Summary	Translation	Generation	Intent
Poor battery life	Urdu / Chinese	Write a review	Product complaint

How Does Deep Learning Do All This?

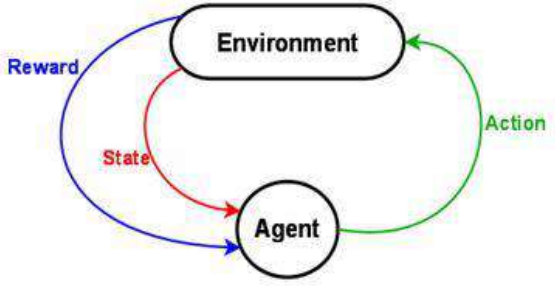
From language to representations to tasks

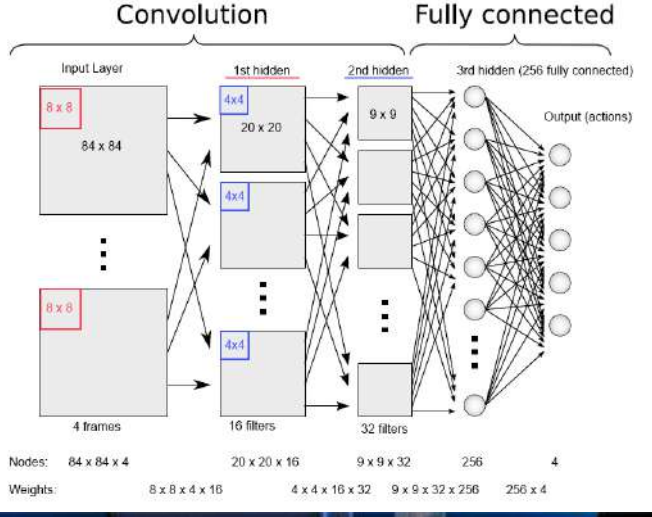




Next question: How does a neural network convert words into numbers?

In 2013, Deep Mind's arcade player beat human expert on six Atari Games. Acquired by Google in 2014,.





In 2016, Deep Mind's

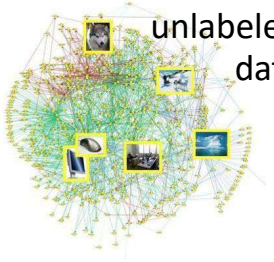
alphaGo defeats former world champion Lee Sedol

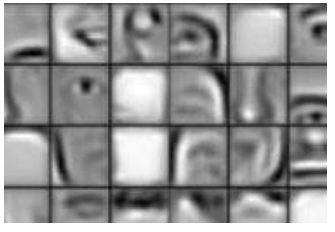


Speech Applications and DL

- Speech recognition / Speech-to-Text
- Speaker identification
- Speaker verification
- Emotion recognition
- Keyword spotting
- Audio event detection
- Music classification
- Sound/source separation
- Speech enhancement / noise removal
- Text-to-Speech Voice assistants Healthcare applications using sounds

Unsupervised learning





representation

what does that mean?

generative modeling:







GANs
VAEs
pixel RNN, etc.

Src: Sergey Levine

What is Clustering?

- Organizing data into classes such that:
  - High intra-class similarity
  - Low inter-class similarity

- Finding the class labels and the number of classes directly from the data (as opposed to *classification tasks*)

Why do we cluster?

- **Summarizing data**
  - Look at large amounts of data
  - Patch-based compression or denoising
  - Represent a large continuous vector with the cluster number

- **Counting**
  - Histograms of texture, color, SIFT vectors

- **Segmentation**
  - Separate the image into different regions

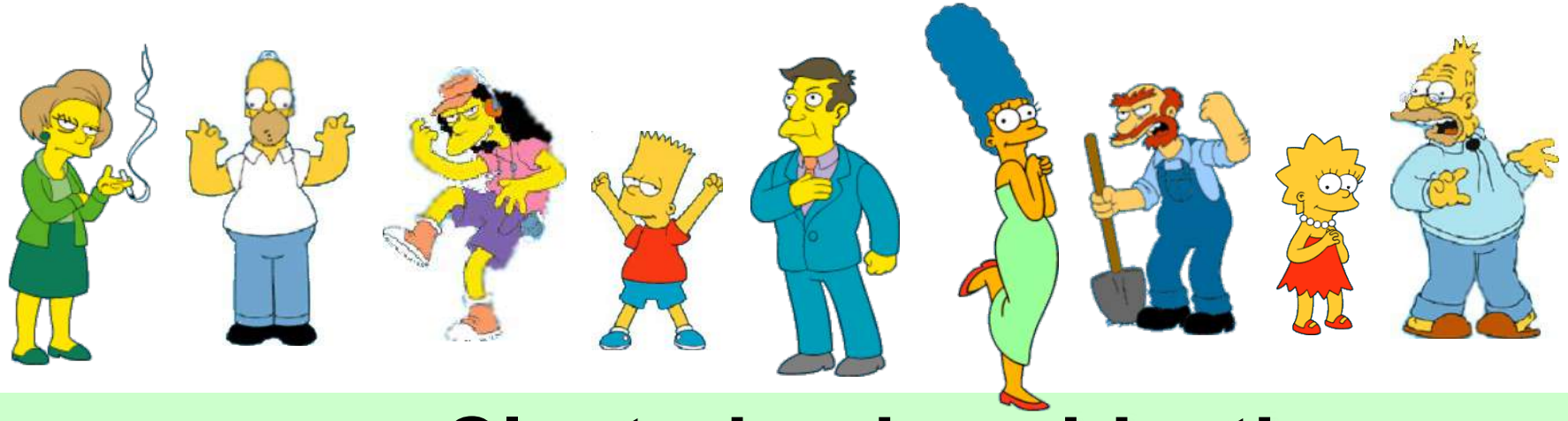
- **Prediction**
  - Images in the same cluster may have the same labels

Slide: Derek Hoiem

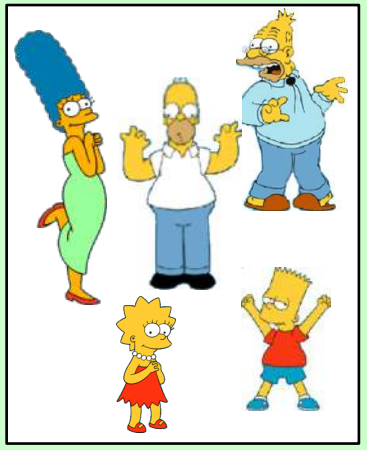
What is a natural grouping?



What is a natural grouping?



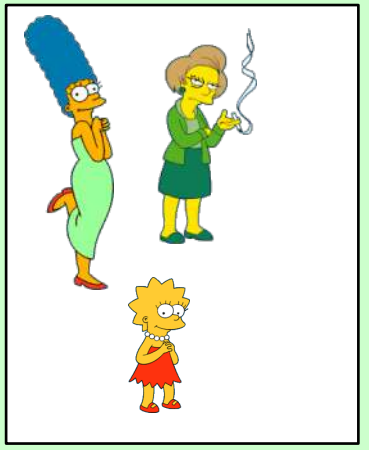
Clustering is subjective



Simpson's Family



School Employees



Females



Males

What is similarity ?

Cluster by features

- **Color**
- **Intensity**
- **Location**
- **Texture**
- **….**

Distance metrics









0.23

Peter Piotr







3









342.7

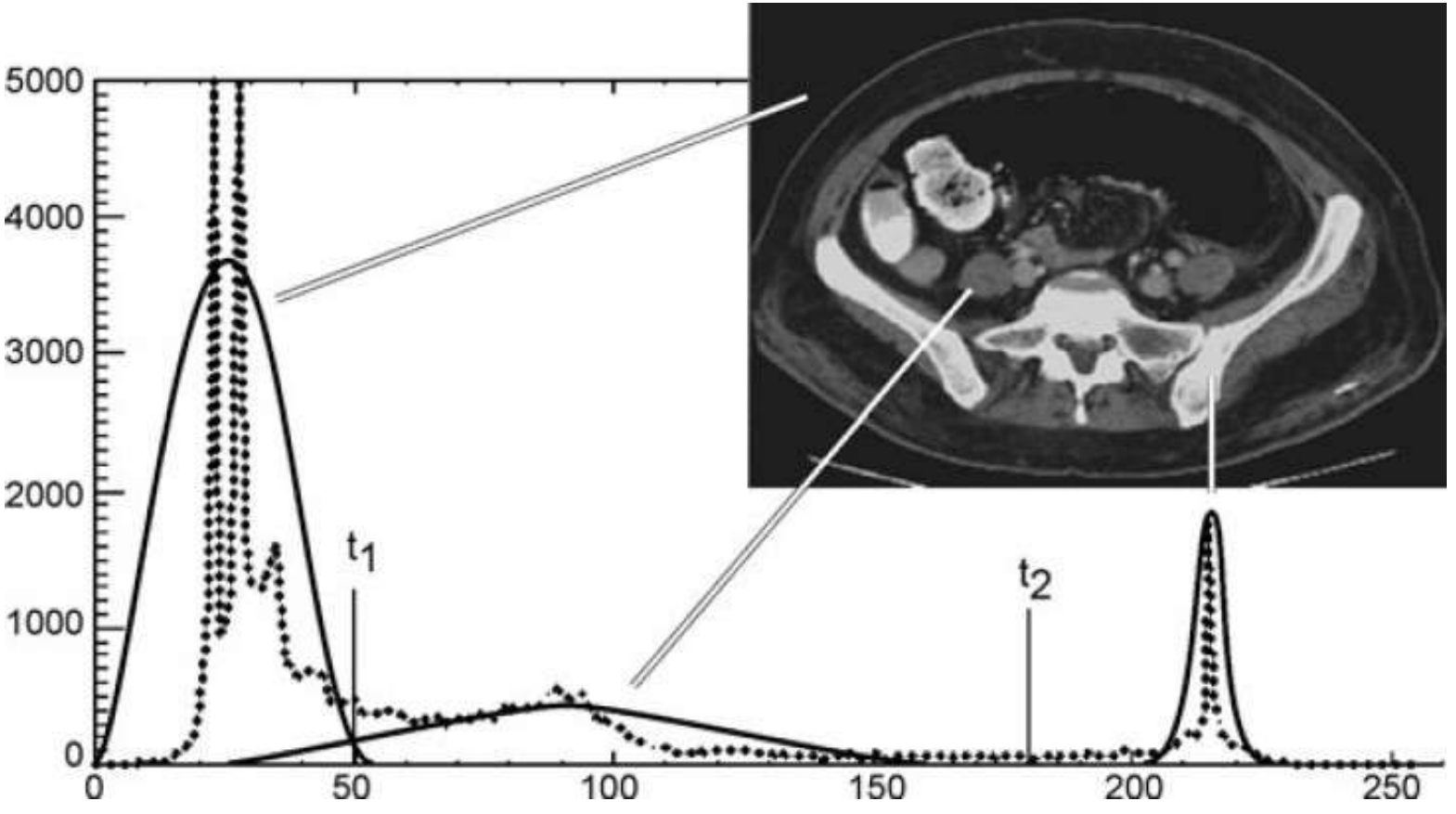
- **Color, rotation, scale/size invariance**
- **Local Distance , Global Distance**

- **Clustering algorithms** essentially perform the same function as **classifier methods** without the use of training data.

- Thus, they are termed **unsupervised methods.**

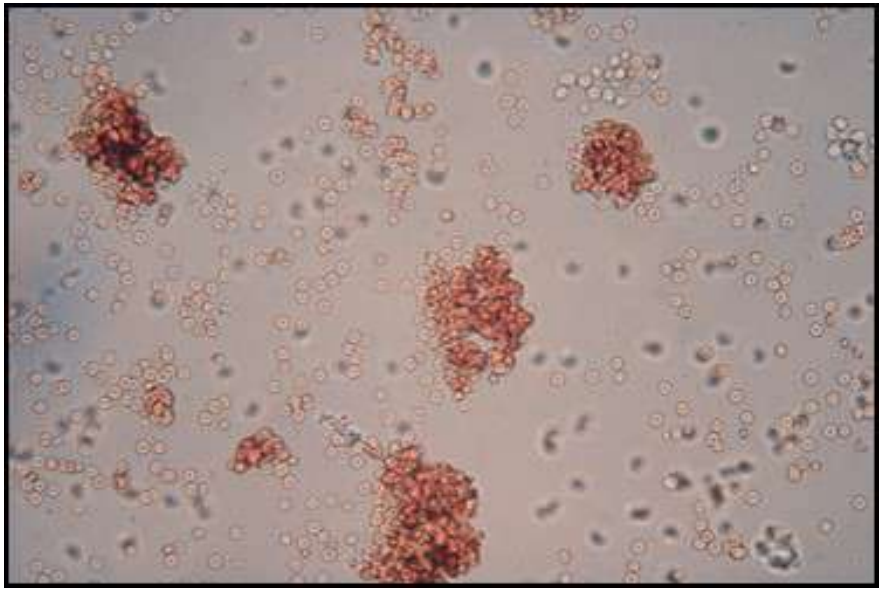
- In order to compensate for the lack of training data, clustering methods iterate between segmenting the image and characterizing the properties of the each class.

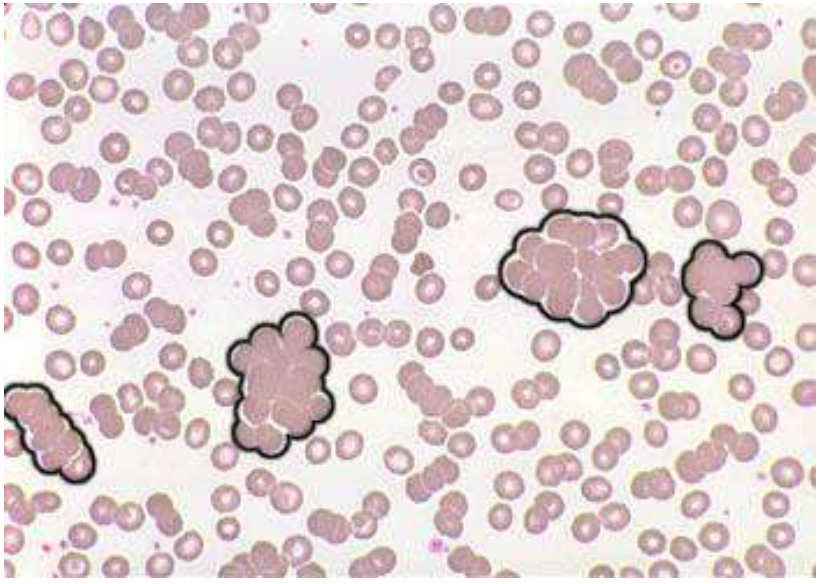
Motivation in Biomedical Image Segmentation

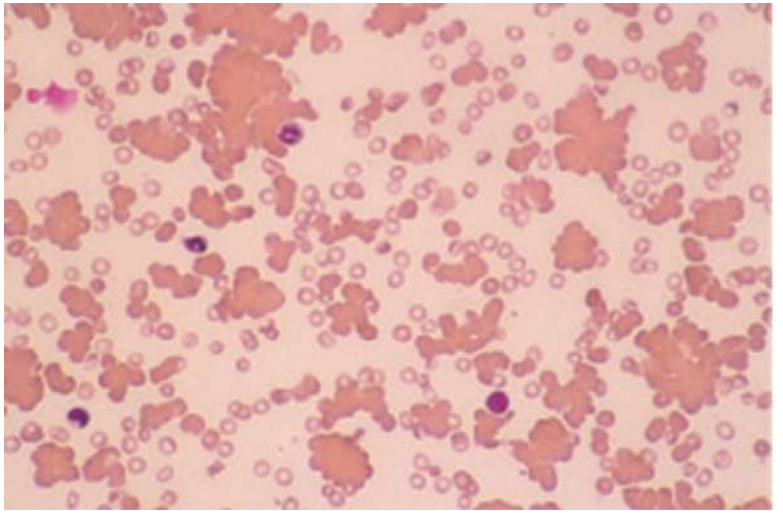


Segmenting the image based on threshold

Autoagglutination







Rouleaux formation

